

8

7

6

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2

1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

http://shop61976717.taobao.com

SCHEM, EVT, MLB, K21

04/18/11

REV

ECN

DESCRIPTION OF REVISION

CK APPD

DATE

2011-04-18

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(02/16/2010)

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(K09_MLB)

05/15/2010

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06/15/2010

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01/10/2011

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PCB Rule Definitions

K21_CONSTRAINTS

04/06/2011

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-8870	1	SCHEM, MLB, K21	SCH	CRITICAL	
020-3023	1	PCBF, MLB, K21	PCB	CRITICAL	

DRAWING

TITLE=MLB

ABBREV=DRAWING

LAST MODIFIED=Rev A/11 01/10/2011

PRODUCT SAFETY REQUIREMENTS:

PCB,UL RECOGNIZED, MIN. 130-C TEMP RATING AND V-0 FLAME RATING PER UL 796 & UL 94

PCB TO BE SILK-SCREENED WITH UL/CUL RECOGNITION MARK, MANUFACTURER'S UL FILE

NUMBER, UL PCB MATERIAL DESIGNATION, 130-C TEMP RATING AND V-0 FLAME RATING

DRAWING TITLE

SCHEM, MOCKUP, MLB, K21

Apple Inc.

051-8870

3.13.0

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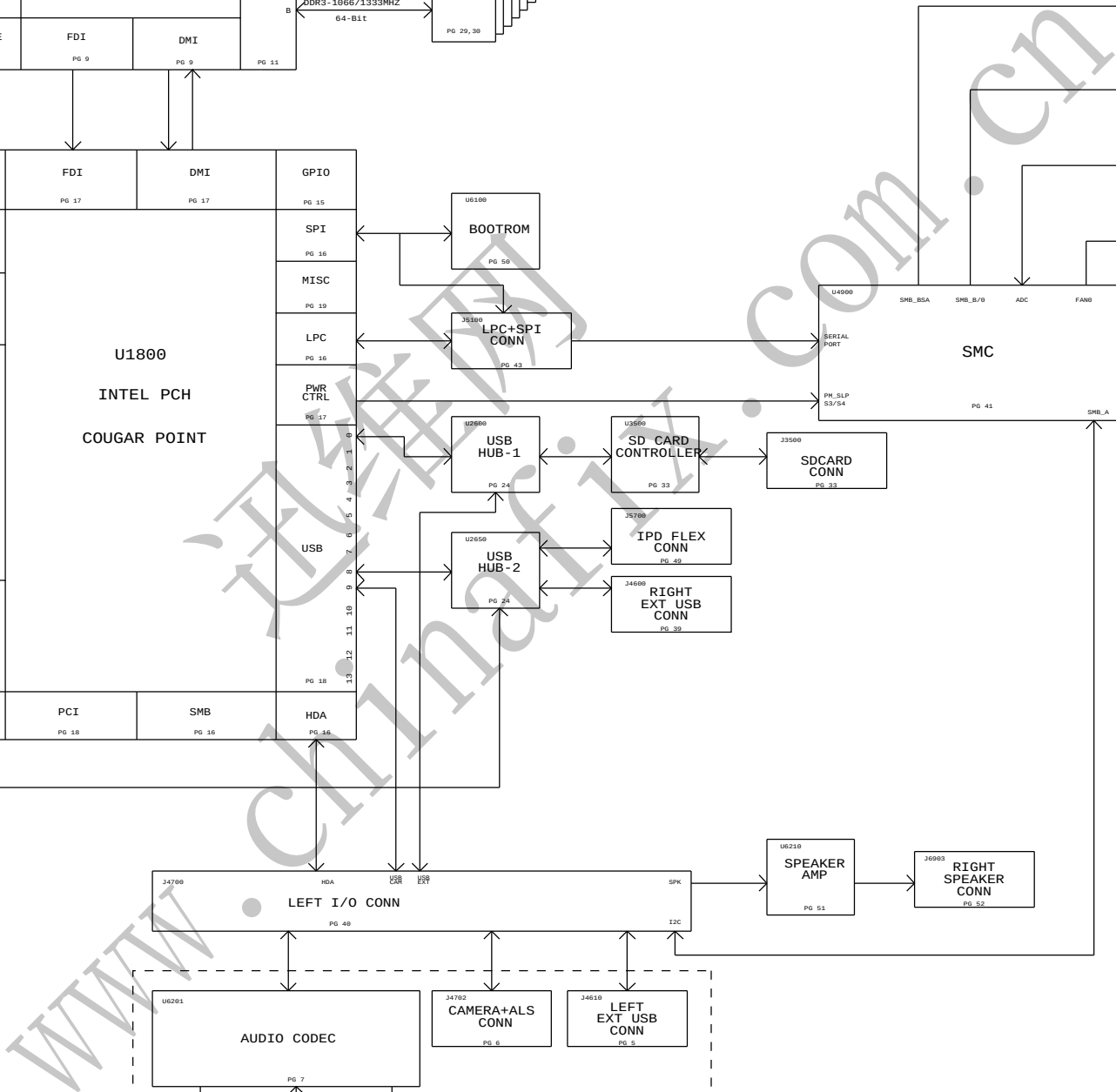
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
085-2684	K211 MLB DEVELOPMENT BOM	K21_DEVEL-ENG
607-B041	CMN PTS, PCBA, MLB, K21	K21_COMMON
639-2553	PCBA, MLB, 1.8GHZ, HY 2GB, K21	K21_CMNPTS,EEEE:DP1F,CPU:1.8GHZ,DOR3:HYNIX_2GB
639-2554	PCBA, MLB, 1.7GHZ, SA 4GB, K21	K21_CMNPTS,EEEE:DP1G,CPU:1.7GHZ,DOR3:SAMSUNG_4GB
639-2558	PCBA, MLB, 1.8GHZ, EL 4GB, K21	K21_CMNPTS,EEEE:DP1H,CPU:1.8GHZ,DOR3:ELPIDA_4GB
639-2549	PCBA, MLB, 1.7GHZ, EL 4GB, K21	K21_CMNPTS,EEEE:DP1J,CPU:1.7GHZ,DOR3:ELPIDA_4GB
639-2555	PCBA, MLB, 1.8GHZ, HY 4GB, K21	K21_CMNPTS,EEEE:DP1K,CPU:1.8GHZ,DOR3:HYNIX_4GB
639-2557	PCBA, MLB, 1.8GHZ, SA 4GB, K21	K21_CMNPTS,EEEE:DP1L,CPU:1.8GHZ,DOR3:SAMSUNG_4GB
639-2548	PCBA, MLB, 1.7GHZ, HY 2GB, K21	K21_CMNPTS,EEEE:DP1M,CPU:1.7GHZ,DOR3:HYNIX_2GB
639-2550	PCBA, MLB, 1.8GHZ, MI 2GB, K21	K21_CMNPTS,EEEE:DP1N,CPU:1.8GHZ,DOR3:MICRON_2GB
639-2551	PCBA, MLB, 1.7GHZ, HY 4GB, K21	K21_CMNPTS,EEEE:DP1P,CPU:1.7GHZ,DOR3:HYNIX_4GB
639-2552	PCBA, MLB, 1.7GHZ, SA 2GB, K21	K21_CMNPTS,EEEE:DP1Q,CPU:1.7GHZ,DOR3:SAMSUNG_2GB
639-2556	PCBA, MLB, 1.8GHZ, SA 2GB, K21	K21_CMNPTS,EEEE:DP1R,CPU:1.8GHZ,DOR3:SAMSUNG_2GB
639-2559	PCBA, MLB, 1.7GHZ, MI 2GB, K21	K21_CMNPTS,EEEE:DP1T,CPU:1.7GHZ,DOR3:MICRON_2GB

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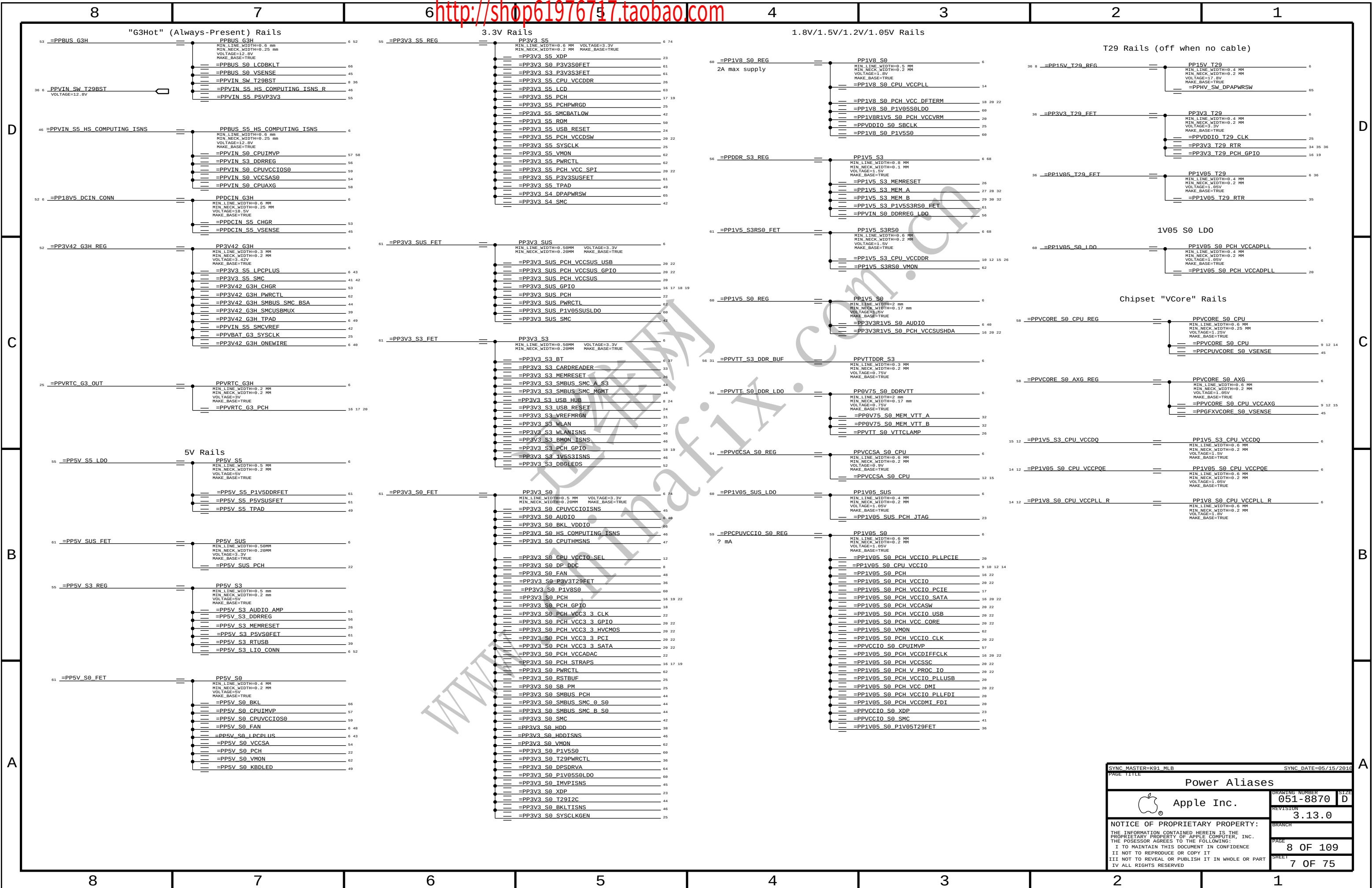
Bar Code Labels / EEEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
825-7563	1	LABEL, L10, K99	[EEEE_DP1F]	CRITICAL	EEEE:DP1F
825-7563	1	LABEL, L10, K99	[EEEE_DP1G]	CRITICAL	EEEE:DP1G
825-7563	1	LABEL, L10, K99	[EEEE_DP1H]	CRITICAL	EEEE:DP1H
825-7563	1	LABEL, L10, K99	[EEEE_DP1J]	CRITICAL	EEEE:DP1J
825-7563	1	LABEL, L10, K99	[EEEE_DP1K]	CRITICAL	EEEE:DP1K
825-7563	1	LABEL, L10, K99	[EEEE_DP1L]	CRITICAL	EEEE:DP1L
825-7563	1	LABEL, L10, K99	[EEEE_DP1M]	CRITICAL	EEEE:DP1M
825-7563	1	LABEL, L10, K99	[EEEE_DP1N]	CRITICAL	EEEE:DP1N
825-7563	1	LABEL, L10, K99	[EEEE_DP1P]	CRITICAL	EEEE:DP1P
825-7563	1	LABEL, L10, K99	[EEEE_DP1Q]	CRITICAL	EEEE:DP1Q
825-7563	1	LABEL, L10, K99	[EEEE_DP1R]	CRITICAL	EEEE:DP1R
825-7563	1	LABEL, L10, K99	[EEEE_DP1T]	CRITICAL	EEEE:DP1T

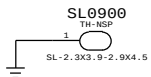
Sub BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM_OPTION
085-2684	1	K21 MLB DEVELOPMENT	DEVEL	CRITICAL	DEVEL_BOM
607-B041	1	CMN PTS,PCBA,MLB,K21	CMNPNTS	CRITICAL	K21_CMNPNTS

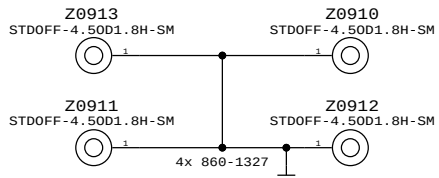
SYNC MASTER-MASTER		SYNC DATE-MASTER	
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	Apple Inc.		DRAWING NUMBER 051-8870
			SIZE D
		REVISION 3.13.0	
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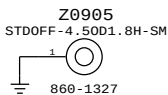
Plated Board Slot



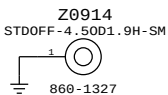
CPU Heat Sink Mounting Bosses



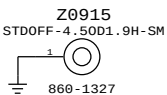
Fan Boss



X21 Boss

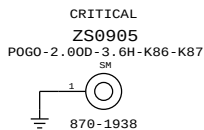


SSD Boss

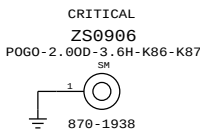


EMI I/O Pogo Pins

DisplayPort Pogo

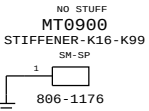


USB/SD Card Pogo

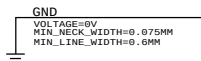


DisplayPort PCB Stiffener

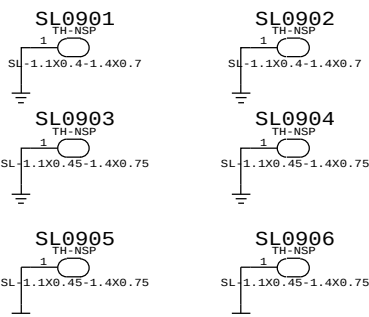
(Provides PCB support for small finger above J9400)



Digital Ground



T29 Can Slots



CPU signals

26	MEMVTT EN	==	DDRVT EN	26	56
70	64	DP EXTA ML C P<3..0>	==	DP IG ML P<3..0>	17
70	64	DP EXTA ML C N<3..0>	==	DP IG ML N<3..0>	17
70	64	DP EXTA AUXCH C P	==	DP IG AUX CH P	17
70	64	DP EXTA AUXCH C N	==	DP IG AUX CH N	17
16	16	PCIE EXCARD D2R N	==	NC PCIE EXCARD D2RN	
16	16	PCIE EXCARD D2R P	==	NC PCIE EXCARD D2RP	
16	16	PCIE EXCARD R2D C N	==	NC PCIE EXCARD R2D CN	
16	16	PCIE EXCARD R2D C P	==	NC PCIE EXCARD R2D CP	
70	16	PCIE CLK100M EXCARD N	==	NC PCIE CLK100M EXCARDN	
70	16	PCIE CLK100M EXCARD P	==	NC PCIE CLK100M EXCARDP	
70	16	PEG CLK100M P	==	NC PEG CLK100MP	
70	16	PEG CLK100M N	==	NC PEG CLK100MN	
68	11	MEM A CLK P<1>	==	TP MEM A CLKP<1>	
68	11	MEM A CLK N<1>	==	TP MEM A CLKN<1>	
68	11	MEM A CKE<1>	==	NC MEM A CKE<1>	
68	11	MEM A CS L<1>	==	NC MEM A CS L<1>	
68	11	MEM A ODT<1>	==	NC MEM A ODT<1>	
68	11	MEM B CLK P<1>	==	TP MEM B CLKP<1>	
68	11	MEM B CLK N<1>	==	TP MEM B CLKN<1>	
68	11	MEM B CKE<1>	==	NC MEM B CKE<1>	
68	11	MEM B CS L<1>	==	NC MEM B CS L<1>	
68	11	MEM B ODT<1>	==	NC MEM B ODT<1>	
68	11	MEM A A<15>	==	TP MEM A A<15>	
68	11	MEM B A<15>	==	TP MEM B A<15>	
TP	PCH_CLKOUT_DPN	==	DP LL_REF_CLK_N	10	67
TP	PCH_CLKOUT_DPP	==	DP LL_REF_CLK_P	10	67

9	=PEG_R2D_C_P<3..0>	67	=PEG_R2D_C_P<3..0>		PCIE_T29_R2D_C_P<3..0>	34	70
9	=PEG_R2D_C_N<3..0>	67	=PEG_R2D_C_N<3..0>		PCIE_T29_R2D_C_N<3..0>	34	70
9	=PEG_D2R_P<3..0>	67	=PEG_D2R_P<3..0>		PCIE_T29_D2R_P<3..0>	34	70
9	=PEG_D2R_N<3..0>	67	=PEG_D2R_N<3..0>		PCIE_T29_D2R_N<3..0>	34	70

T29 DP Ports

17	TP_DP_IG_C_HPD	==	DP_T29SNK0_HPD	34	70
17	TP_DP_IG_C_MLP<3..0>	==	DP_T29SNK0_ML_C_P<3..0>	34	72
17	TP_DP_IG_C_MLN<3..0>	==	DP_T29SNK0_ML_C_N<3..0>	34	72
17	TP_DP_IG_C_AUXP	==	DP_T29SNK0_AUXCH_C_P	34	72
17	TP_DP_IG_C_AUXN	==	DP_T29SNK0_AUXCH_C_N	34	72

LVDS Aliases

6	TP_LVDS_IG_B_CLKP	==	LVDS_IG_B_CLK_P	69
6	TP_LVDS_IG_B_CLKN	==	LVDS_IG_B_CLK_N	69
6	NC_LVDS_IG_B_DATAP<0..3>	==	LVDS_IG_B_DATA_P<0..3>	69
6	NC_LVDS_IG_B_DATA<0..3>	==	LVDS_IG_B_DATA_N<0..3>	69
6	NC_LVDS_IG_A_DATAP<3>	==	LVDS_IG_A_DATA_P<3>	69
6	NC_LVDS_IG_A_DATA<3>	==	LVDS_IG_A_DATA_N<3>	69
66	LCD_BKLT_PWM	==	LVDS_IG_BKL_PWM	17
63	LCD_IG_PWR_EN	==	LVDS_IG_PANEL_PWR	17
66	LCD_BKLT_EN	==	LVDS_IG_BKL_ON	17

SATA Aliases

Unused SATA ODD Signals

69	16	NEW	SATA_ODD_R2D_C_P	==	NC_SATA_ODD_R2DCP	
69	16	NEW	SATA_ODD_R2D_C_N	==	NC_SATA_ODD_R2DCN	
69	16	NEW	SATA_ODD_D2R_P	==	NC_SATA_ODD_D2RP	
69	16	NEW	SATA_ODD_D2R_N	==	NC_SATA_ODD_D2RN	

Unused PG00D signal

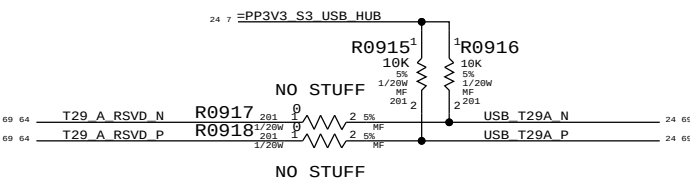
TP_P1V5S3RS0_RAMP_DONE	==	P1V5S3RS0_RAMP_DONE	61
TP_DDRREG_PG00D	==	DDRREG_PG00D	56

T29 JTAG

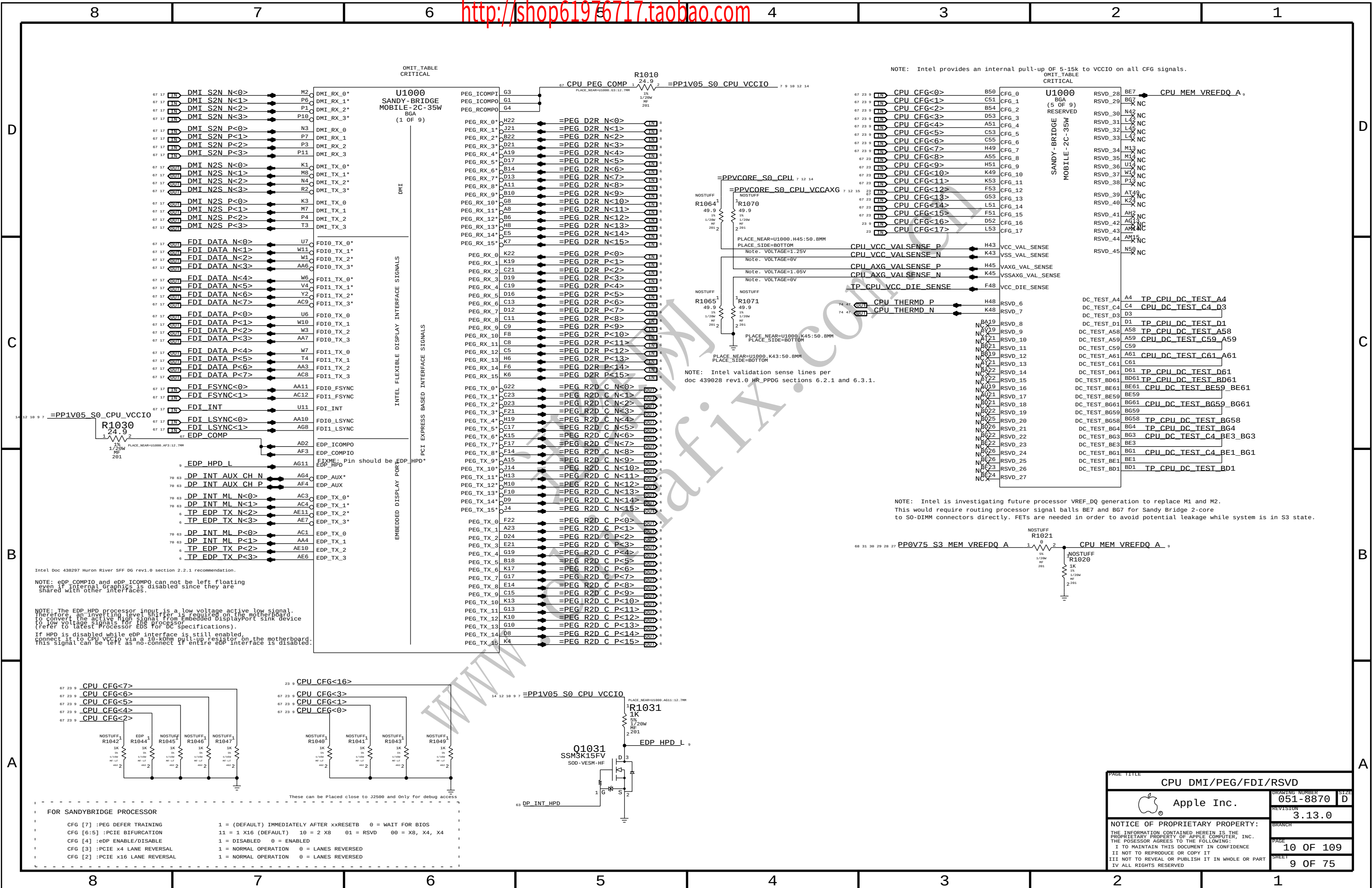
23	19	NEW	JTAG_ISP_TCK	==	JTAG_T29_TCK_R	34
19	19	NEW	JTAG_ISP_TDI	==	JTAG_T29_TDI	34
19	19	NEW	JTAG_ISP_TDO	==	JTAG_T29_TDO	34

T29 Aliases

Unused USB ports

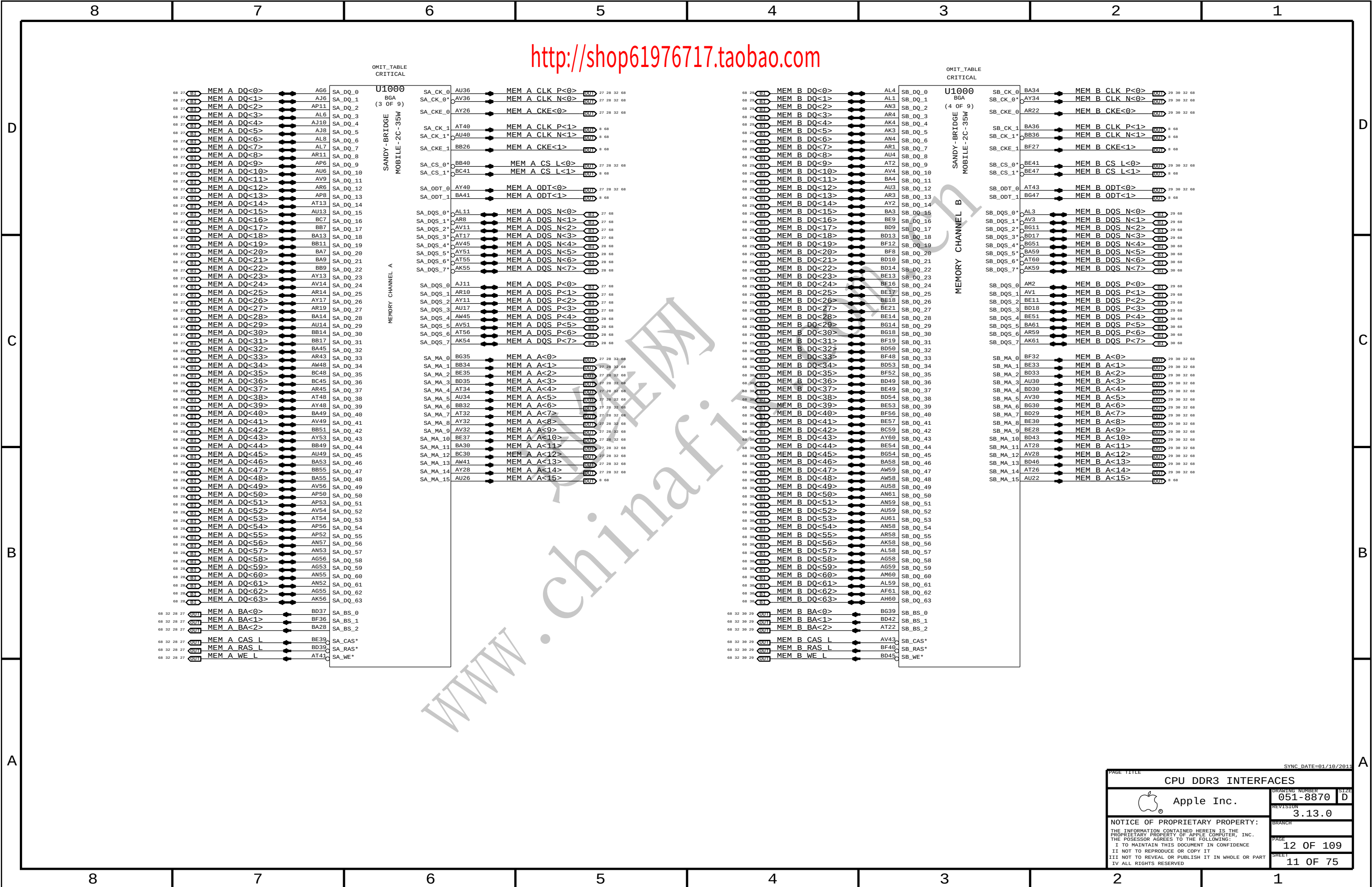


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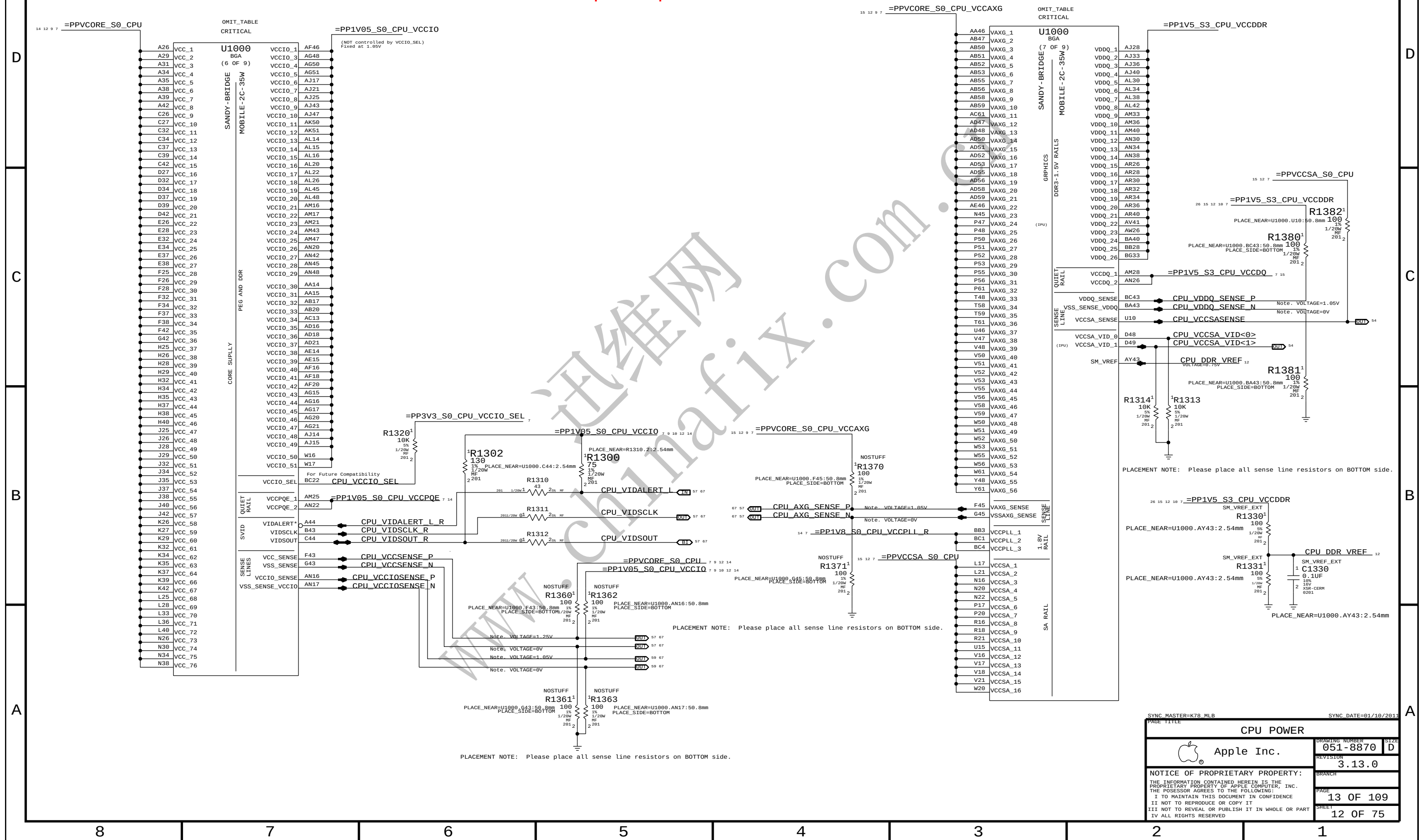




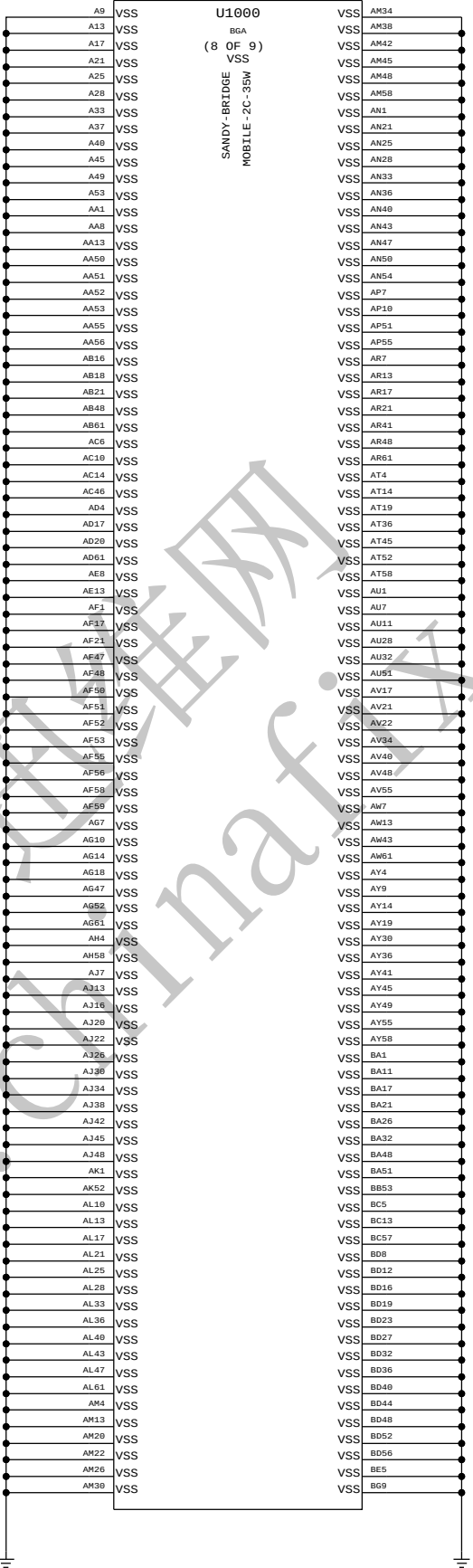
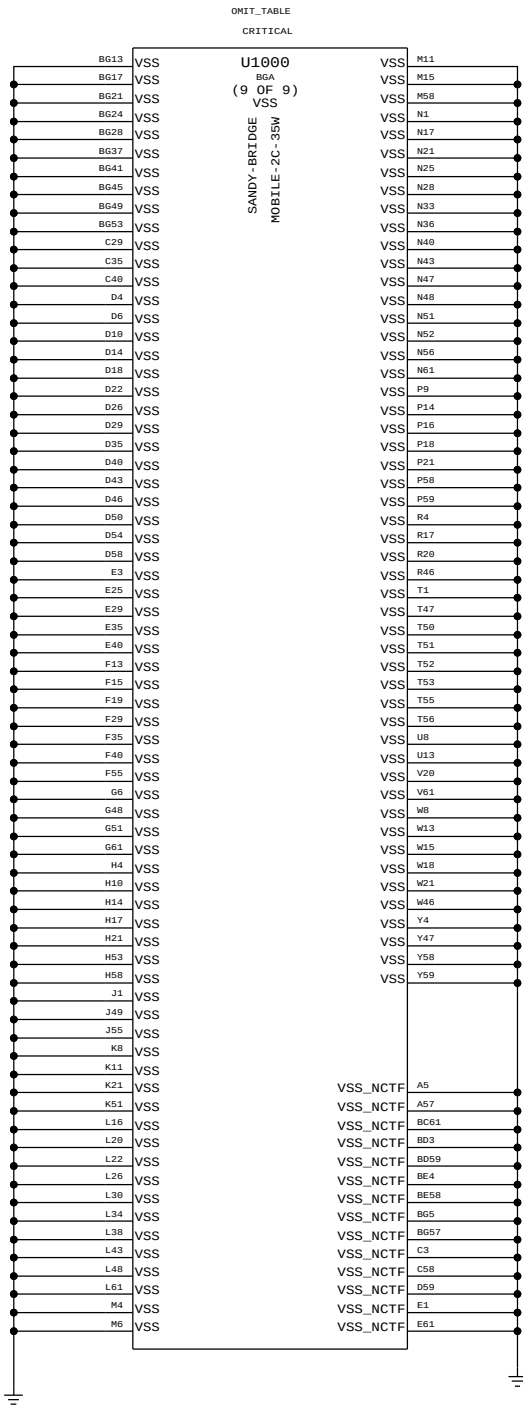
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BRANCH		PAGE		
		11 OF 109		
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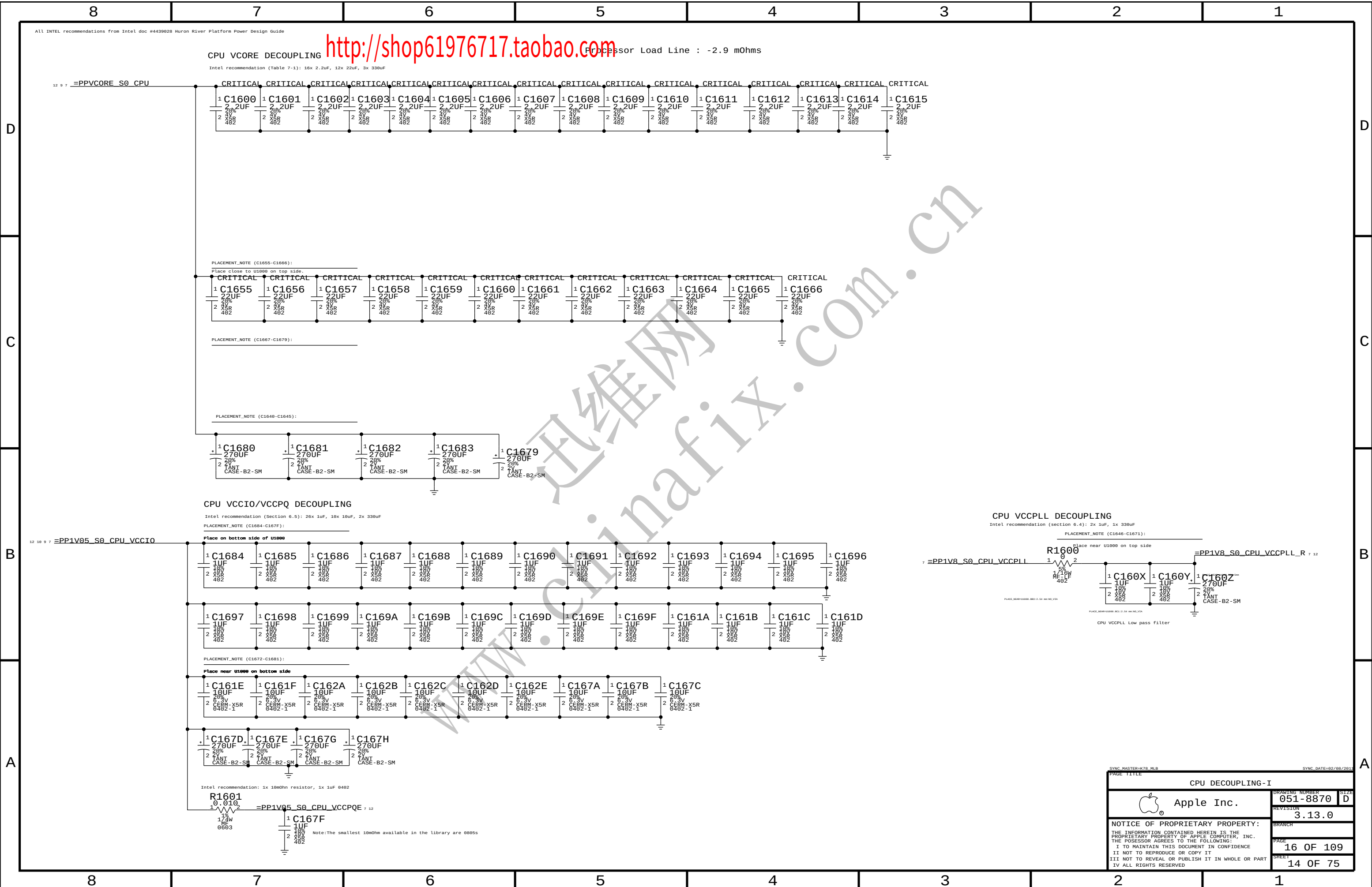


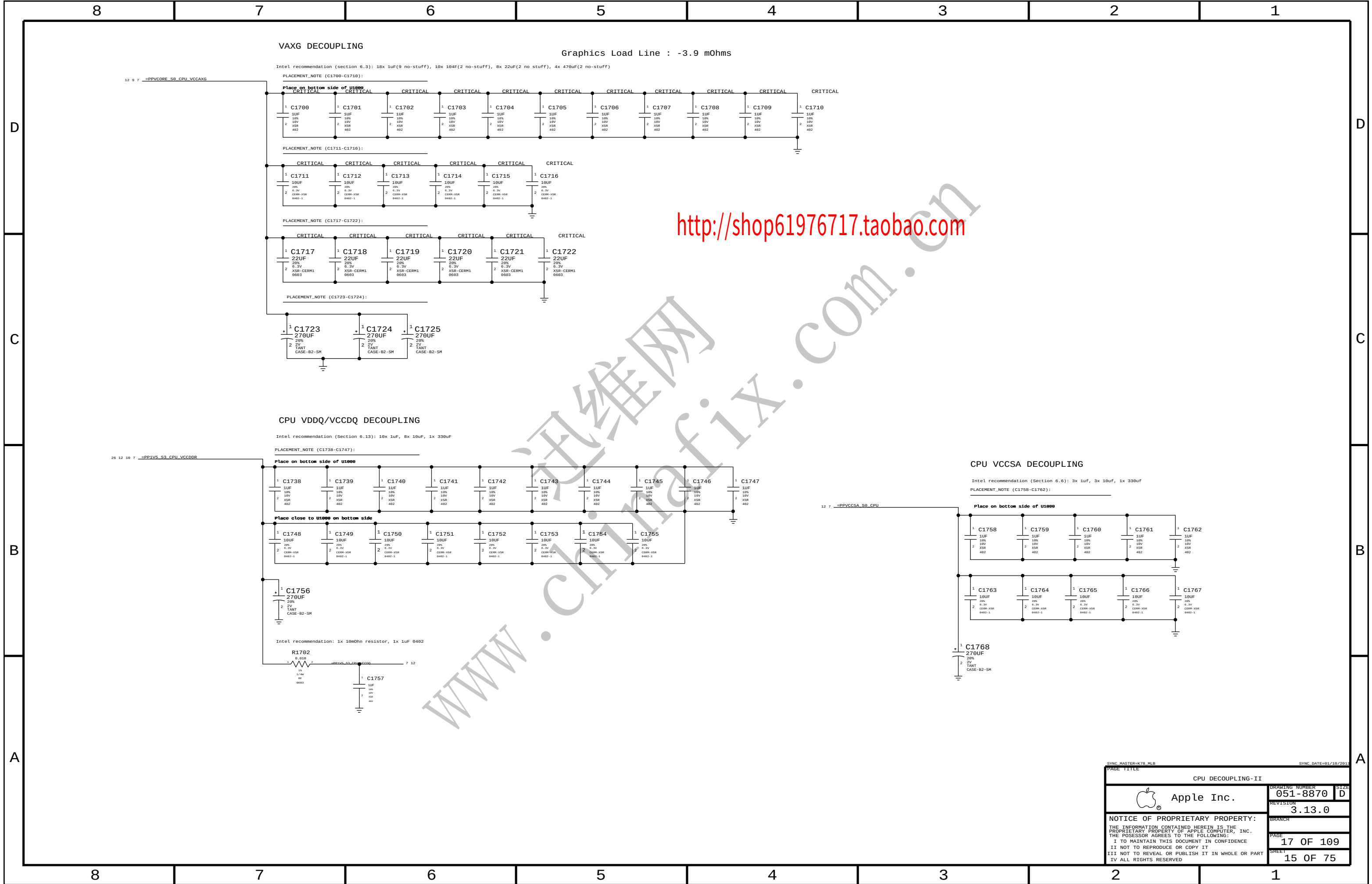
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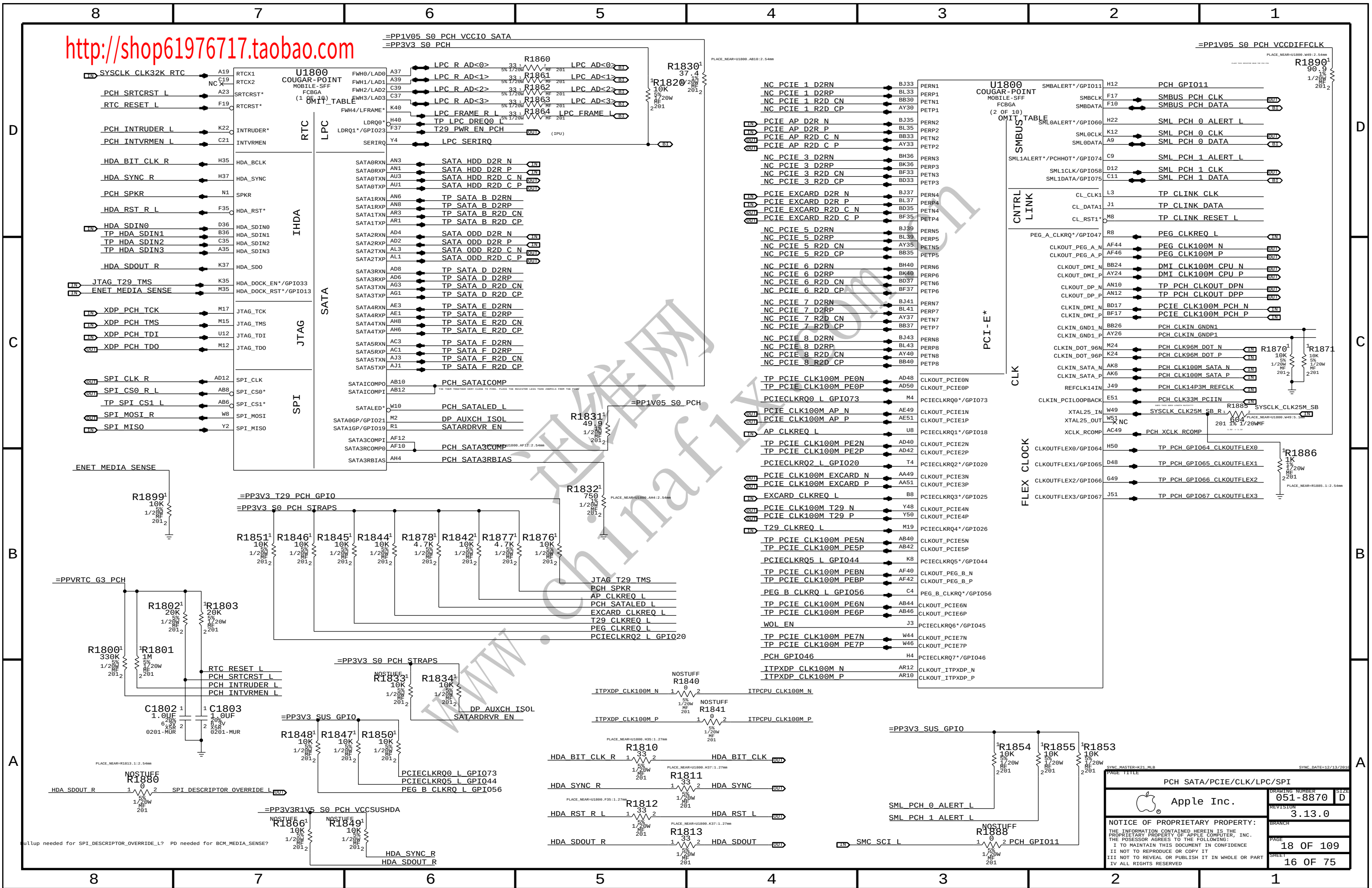






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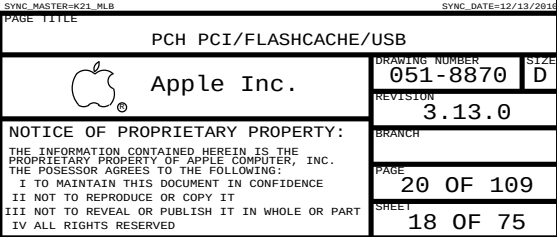
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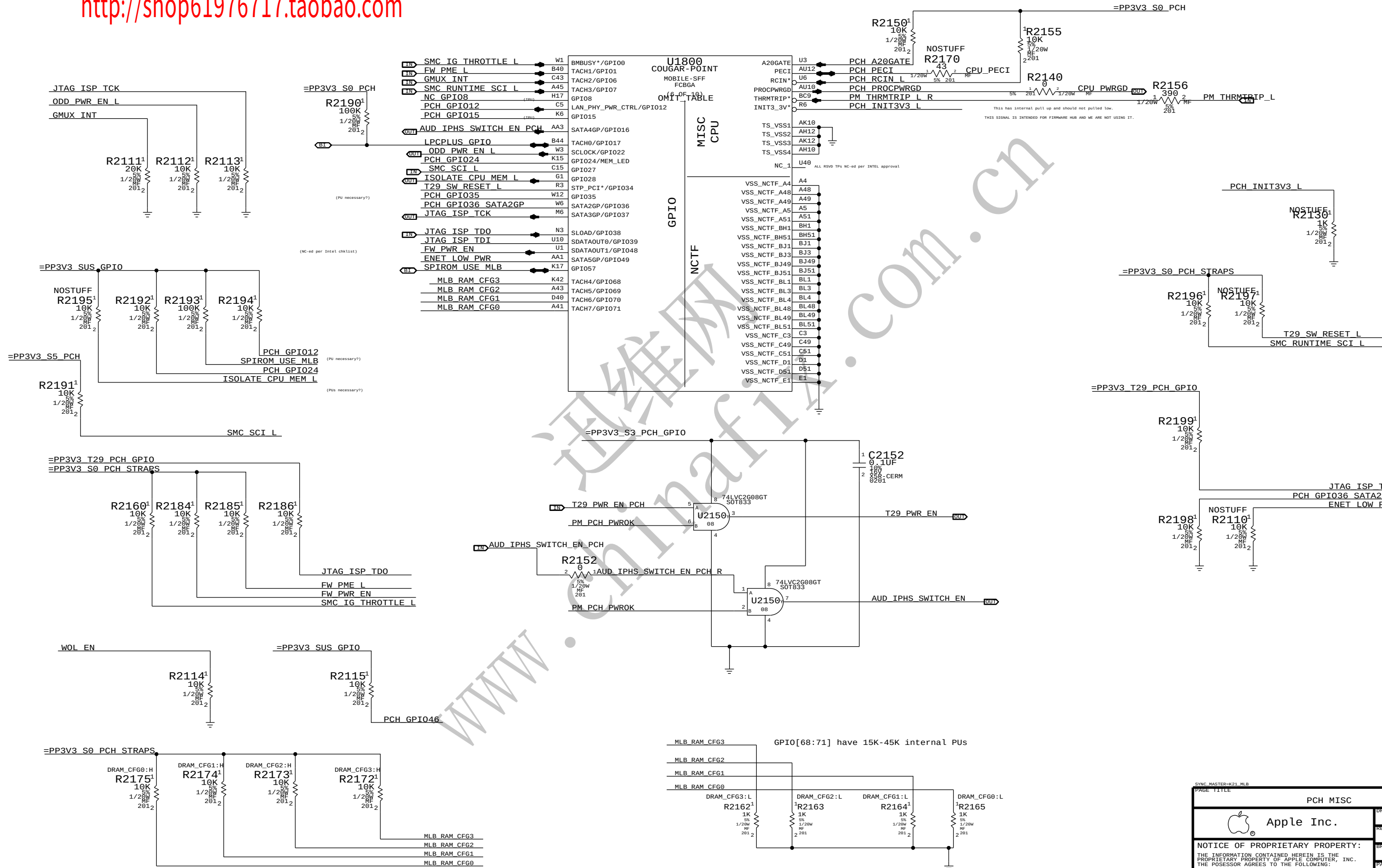
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SATA

VCC3_3_AC19
VCC3_3_R40
VCC3_3_AF6
VCCIO_AA13
VCCIO_AG13
VCCIO_AG15
VCCIO_AF15
VCCAPLLSATA
VCCVRM_AE19
VCCVRM_AF17
VCCIO_AB15
VCCIO_AC13
VCCIO_AC15
VCCASW_U19
VCCASW_R19
VCCASW_V19
VCCSUSDA

HDA

VCC3_3_AC19
VCC3_3_R40
VCC3_3_AF6
VCCIO_AA13
VCCIO_AG13
VCCIO_AG15
VCCIO_AF15
VCCAPLLSATA
VCCVRM_AE19
VCCVRM_AF17
VCCIO_AB15
VCCIO_AC13
VCCIO_AC15
VCCASW_U19
VCCASW_R19
VCCASW_V19
VCCSUSDA

Power Planes

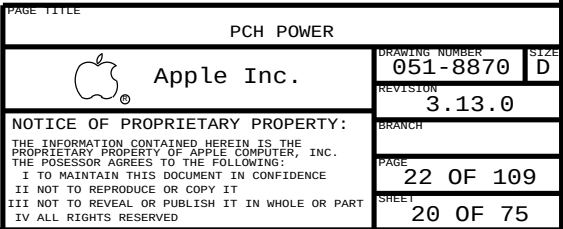
PP3V3 S0 PCH VCC3 3 SATA
PP1V05 S0 PCH VCCIO SATA
PP1V8R1V5 S0 PCH VCCVRM
PP1V05 S0 PCH VCCASW
PP3V3R1V5 S0 PCH VCCSUSDA

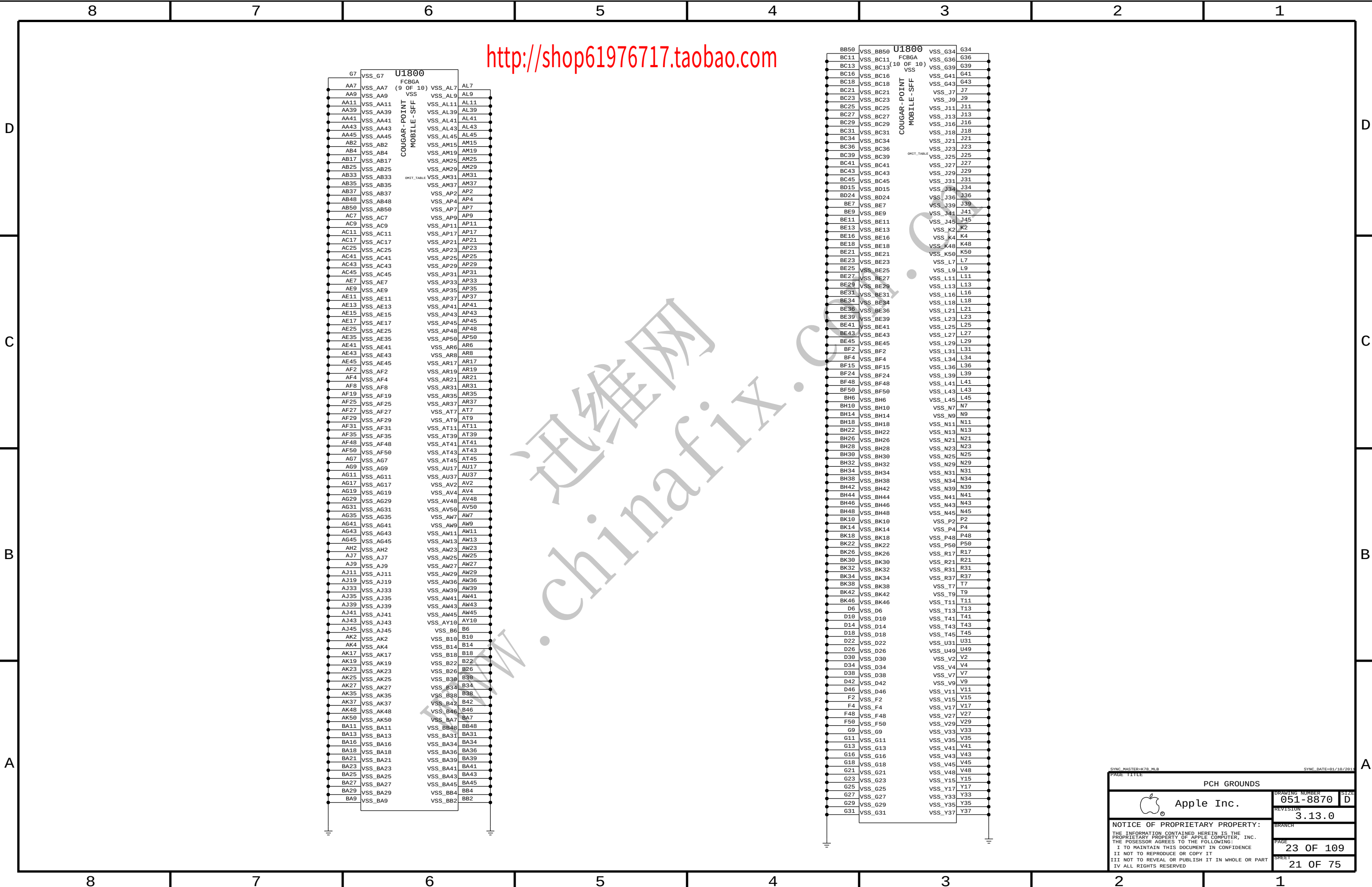
Components

R2260
R2265
C2260
C2261
C2265
C2266
D2260
D2261
D2265
D2266

Notes

VCCAPLLSATA pin left as NC per DG
VCCVRM_AE19
VCCVRM_AF17
VCCIO_AB15
VCCIO_AC13
VCCIO_AC15
VCCASW_U19
VCCASW_R19
VCCASW_V19
VCCSUSDA
10 mA Max, 1ms Totl





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SYNC MASTER=k78_MLB

SYNC DATE=01/10/2011

PAGE TITLE		PCH GROUNDS	
		DRAWING NUMBER	051-8870
		REVISION	3.13.0
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		SHEET	21 OF 75

PROCESSOR MICRO2-XDP CONNECTOR

NOTE: This is not the standard XDP pinout
Use with 920-0782 Adapter Flex to support chipset debug

D

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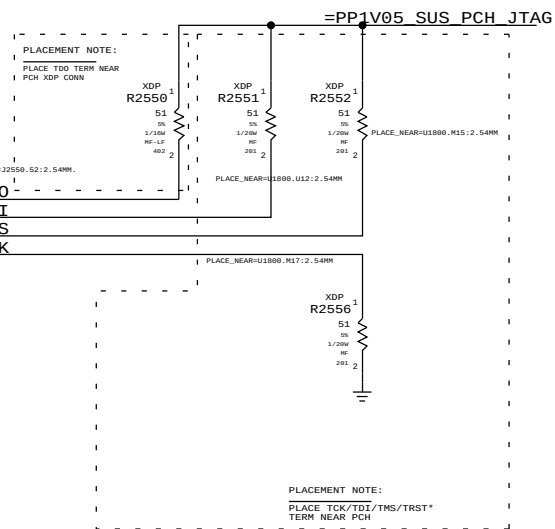
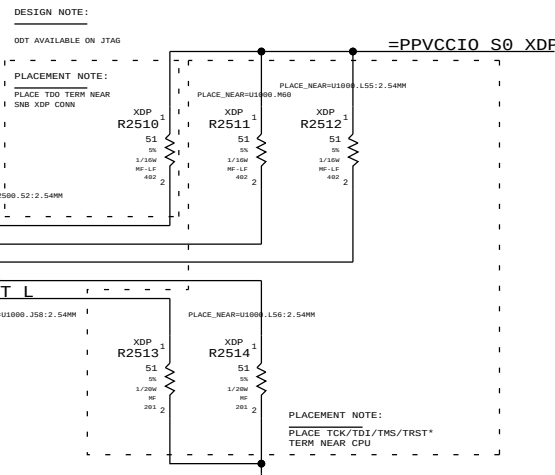
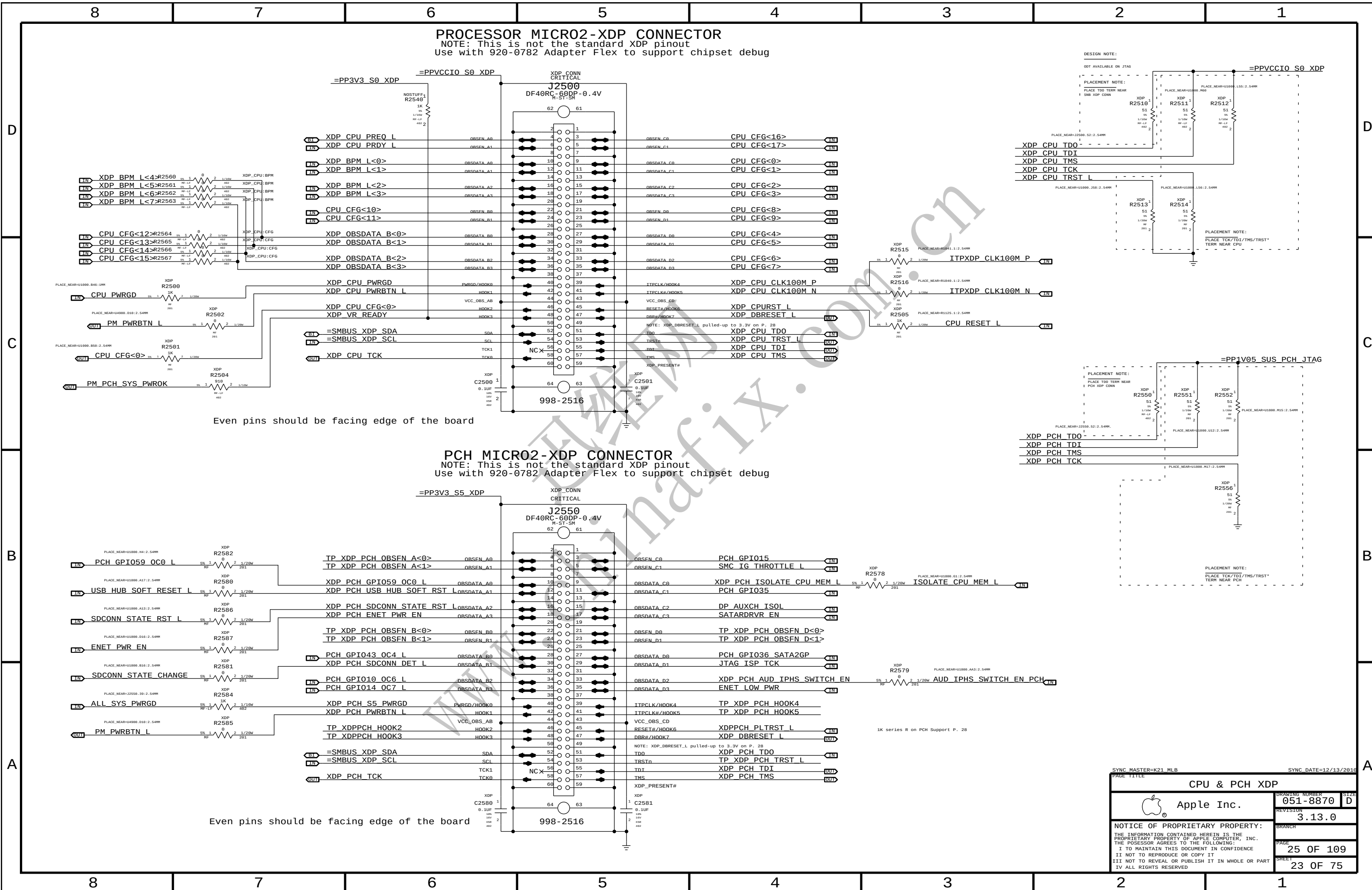
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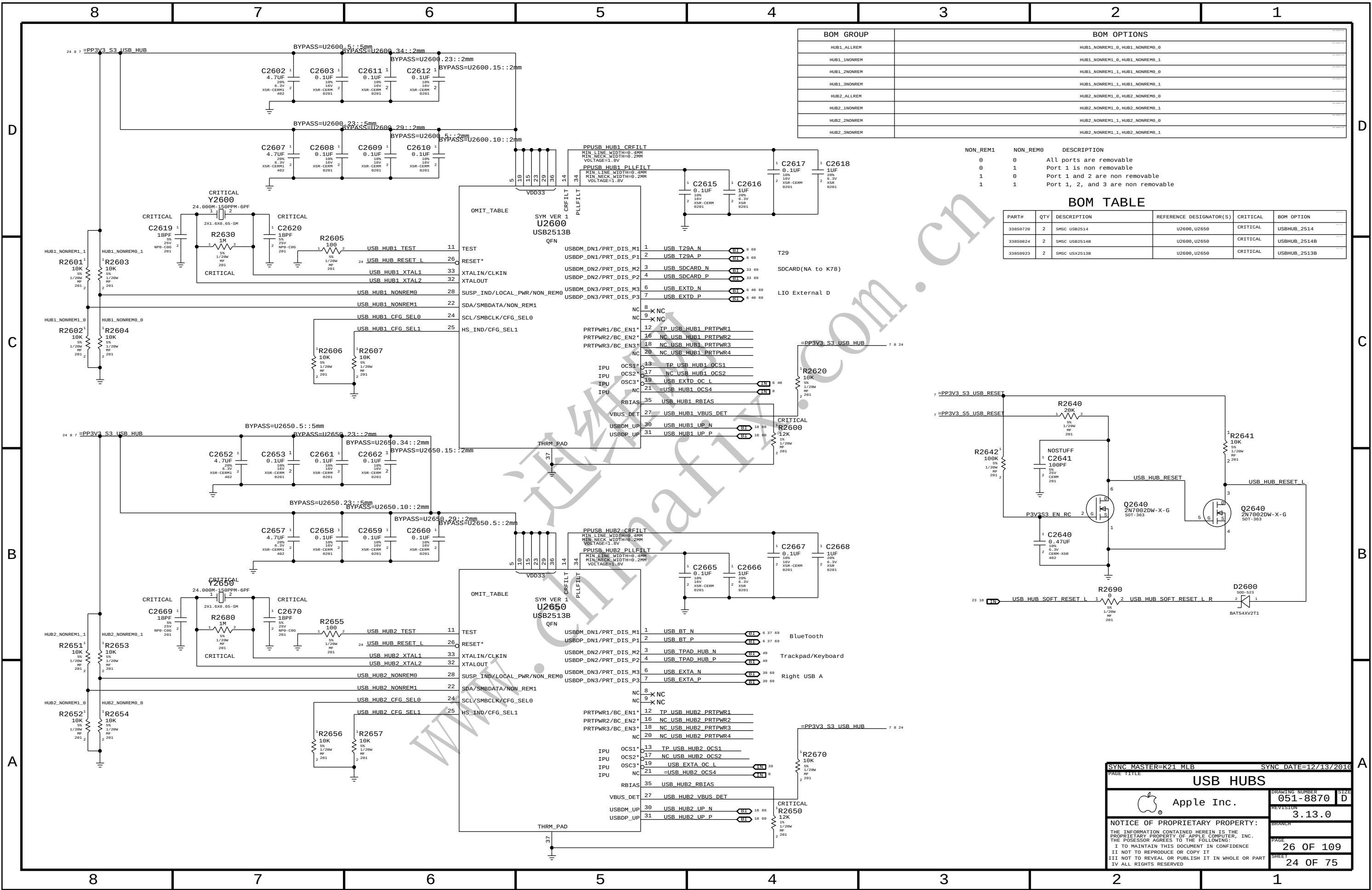
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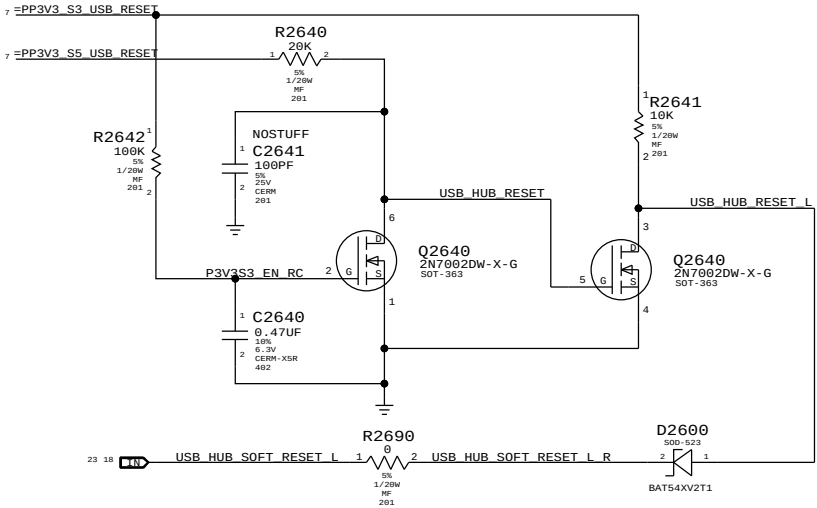
PAGE TITLE		SYNC MASTER=K21 MLB		SYNC DATE=12/13/2016	
CPU & PCH XDP		DRAWING NUMBER		SIZE	
Apple Inc.		051-8870		D	
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BOM GROUP		BOM OPTIONS	
HUB1_ALLREM		HUB1_NONREM1_0, HUB1_NONREM0_0	
HUB1_1NONREM		HUB1_NONREM1_0, HUB1_NONREM0_1	
HUB1_2NONREM		HUB1_NONREM1_1, HUB1_NONREM0_0	
HUB1_3NONREM		HUB1_NONREM1_1, HUB1_NONREM0_1	
HUB2_ALLREM		HUB2_NONREM1_0, HUB2_NONREM0_0	
HUB2_1NONREM		HUB2_NONREM1_0, HUB2_NONREM0_1	
HUB2_2NONREM		HUB2_NONREM1_1, HUB2_NONREM0_0	
HUB2_3NONREM		HUB2_NONREM1_1, HUB2_NONREM0_1	

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

BOM TABLE					
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33850720	2	SMSC USB2514	U2600, U2650	CRITICAL	USBHUB_2514
33850824	2	SMSC USB2514B	U2600, U2650	CRITICAL	USBHUB_2514B
33850923	2	SMSC USB2513B	U2600, U2650	CRITICAL	USBHUB_2513B



SYNC MASTER=K21 MLB

SYNC DATE=12/13/2016

USB HUBS

Apple Inc.

051-8870

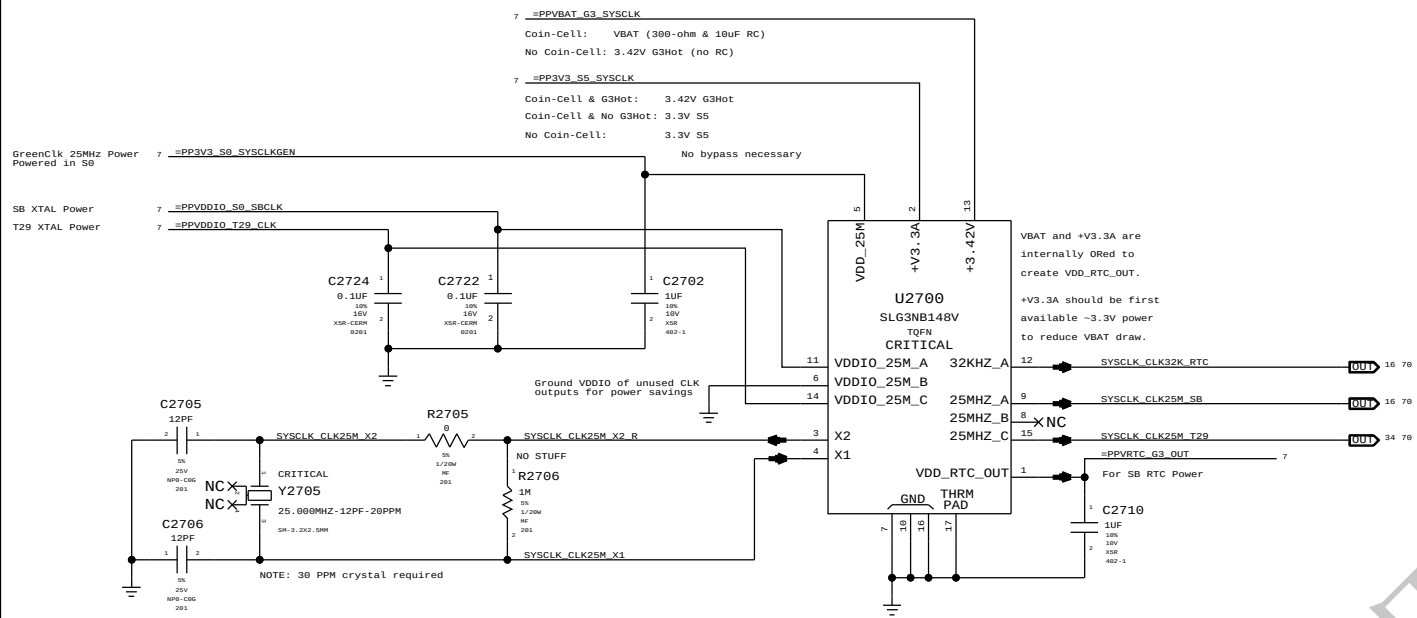
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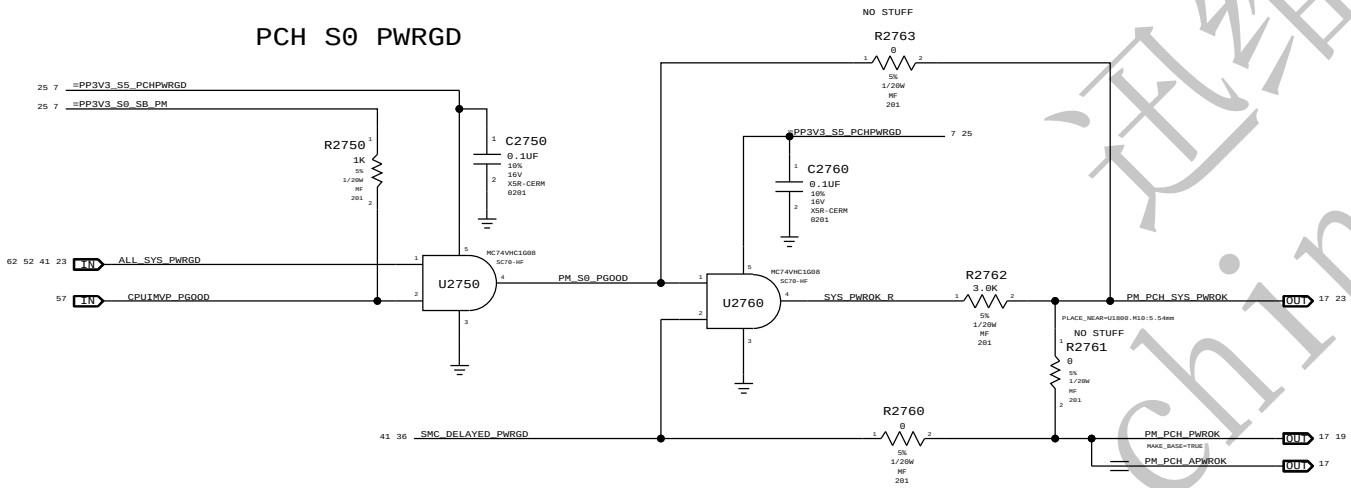
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System RTC Power Source & 32kHz / 25MHz Clock Generator

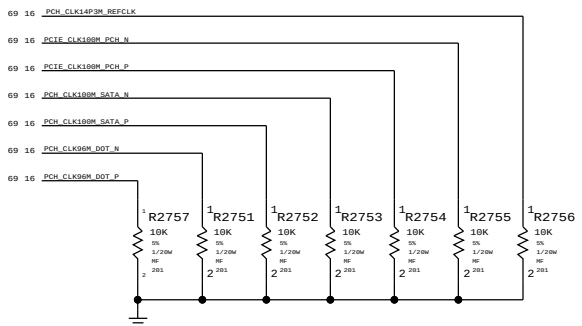


PCH S0 PWRGD

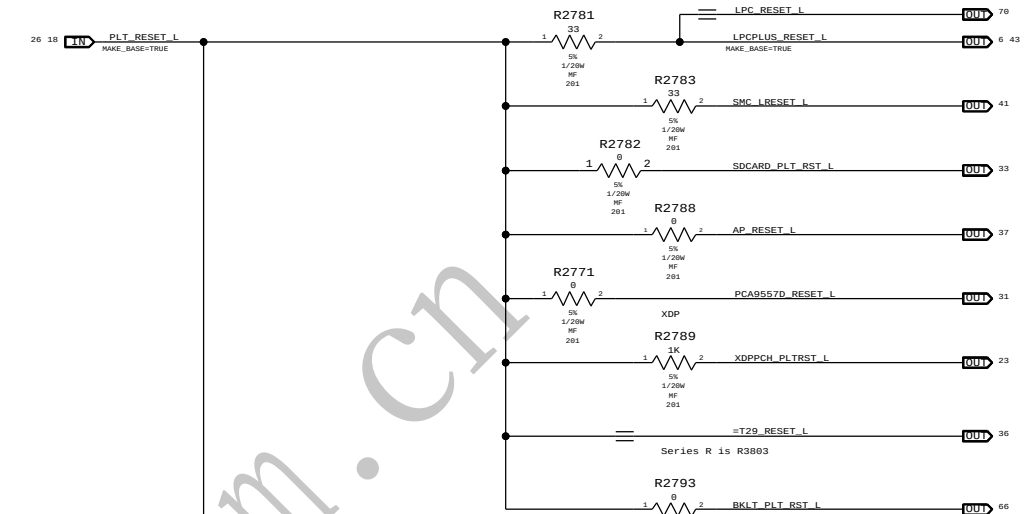


CLOCK (CK505)

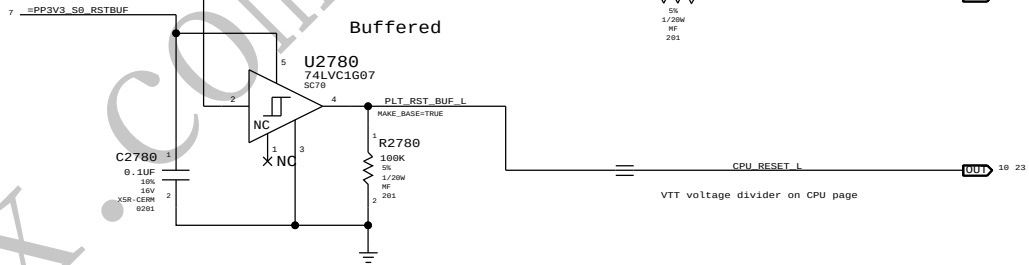
UNUSED clock terminations for FCIM MODE



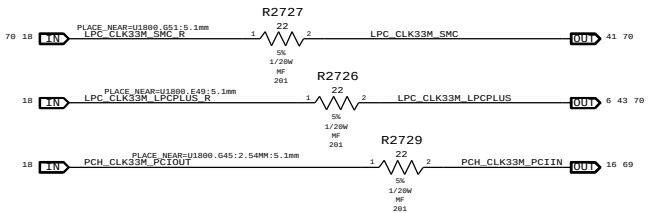
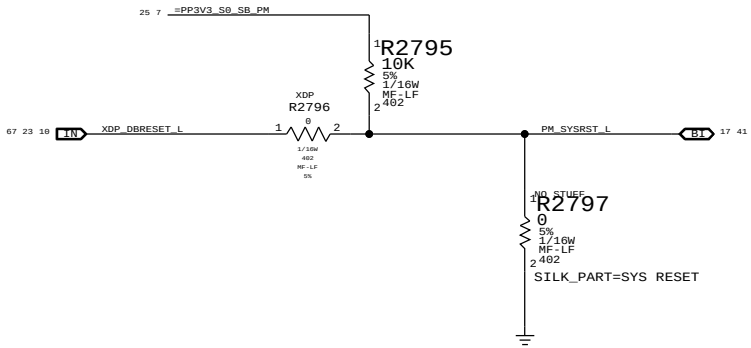
Platform Reset Connections
Unbuffered



Buffered



PCH Reset Button

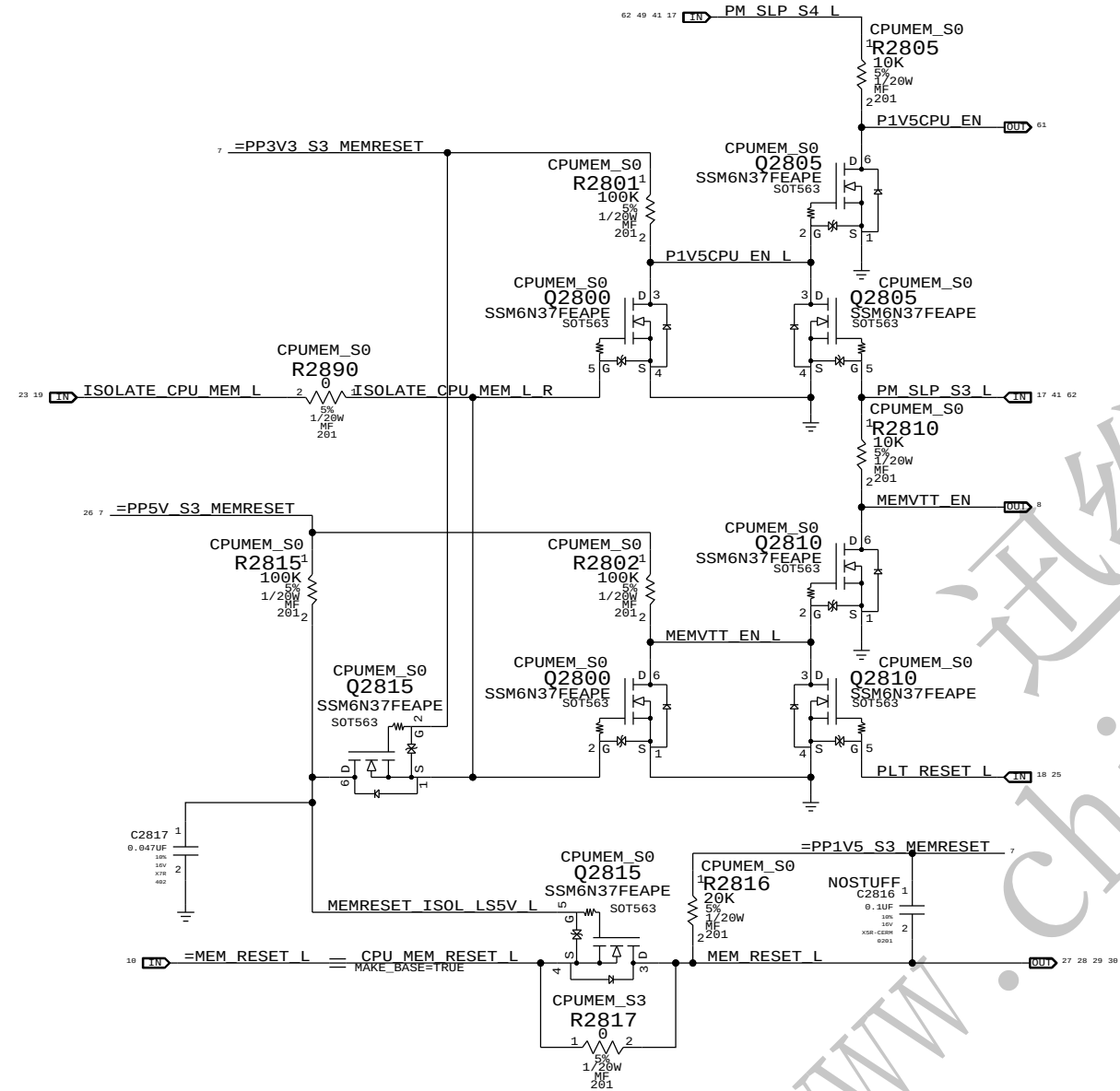


PAGE TITLE		PAGE TITLE	
Clock (CK505) and Chipset Support		DRAWING NUMBER	
Apple Inc.		051-8870	
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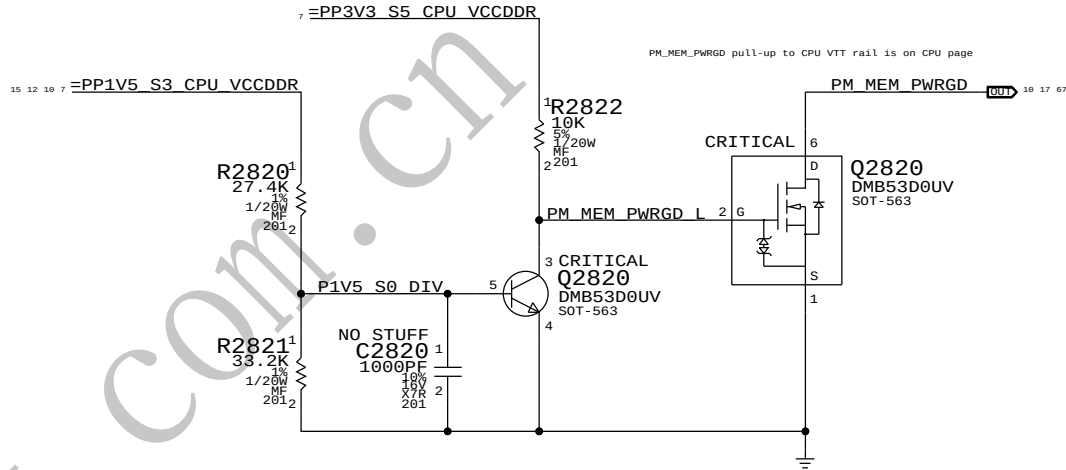
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

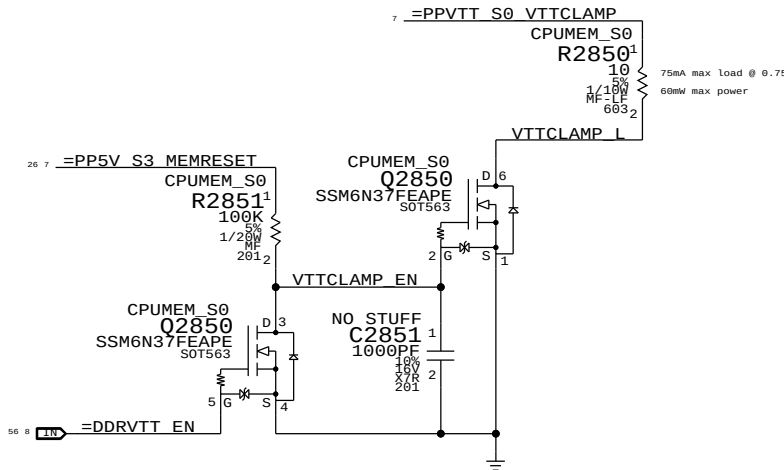


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

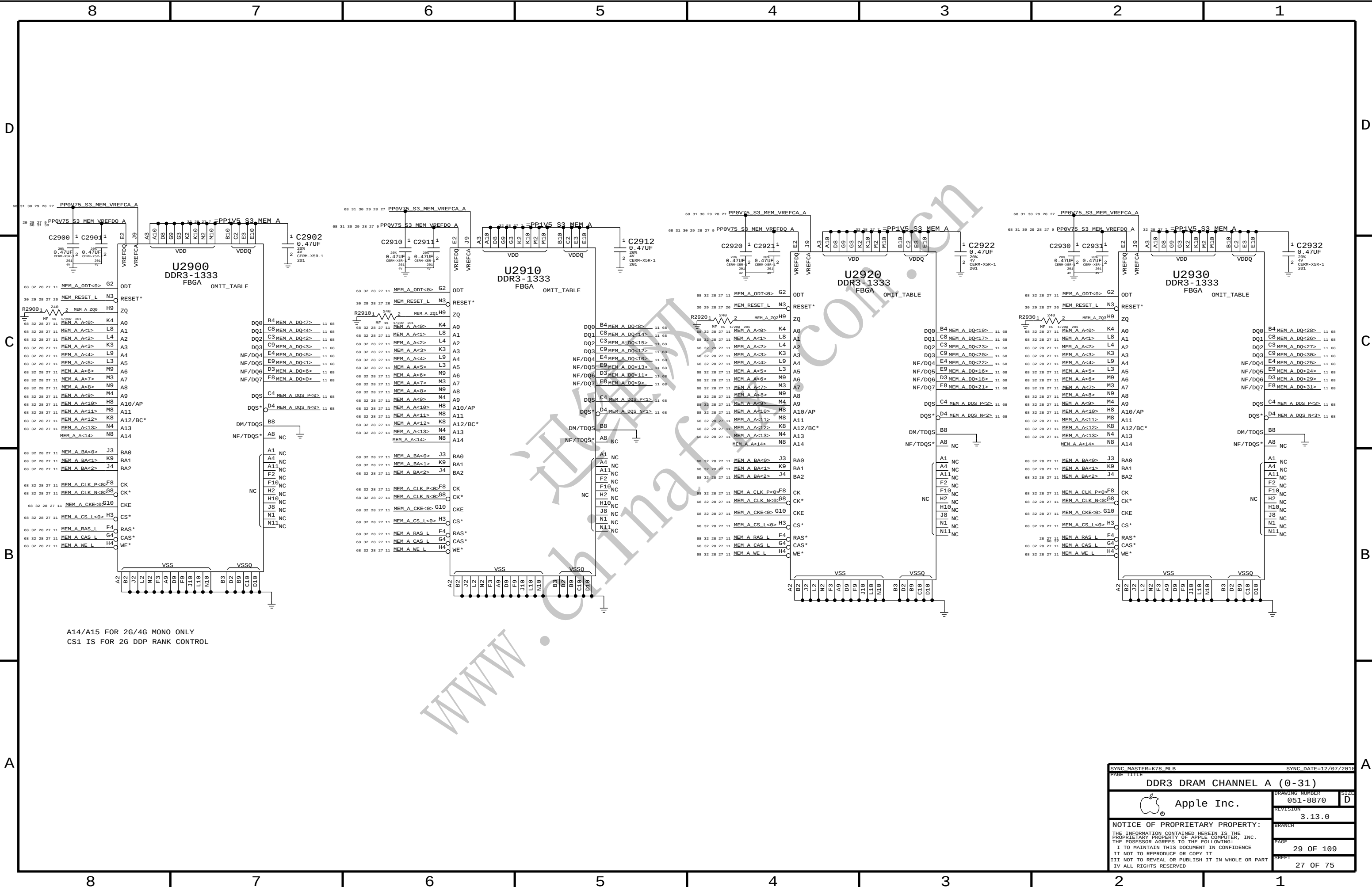


Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPUMEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPUMEM_RESET_L	1	1
1	0	1	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	1	1	1	1	1	CPUMEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

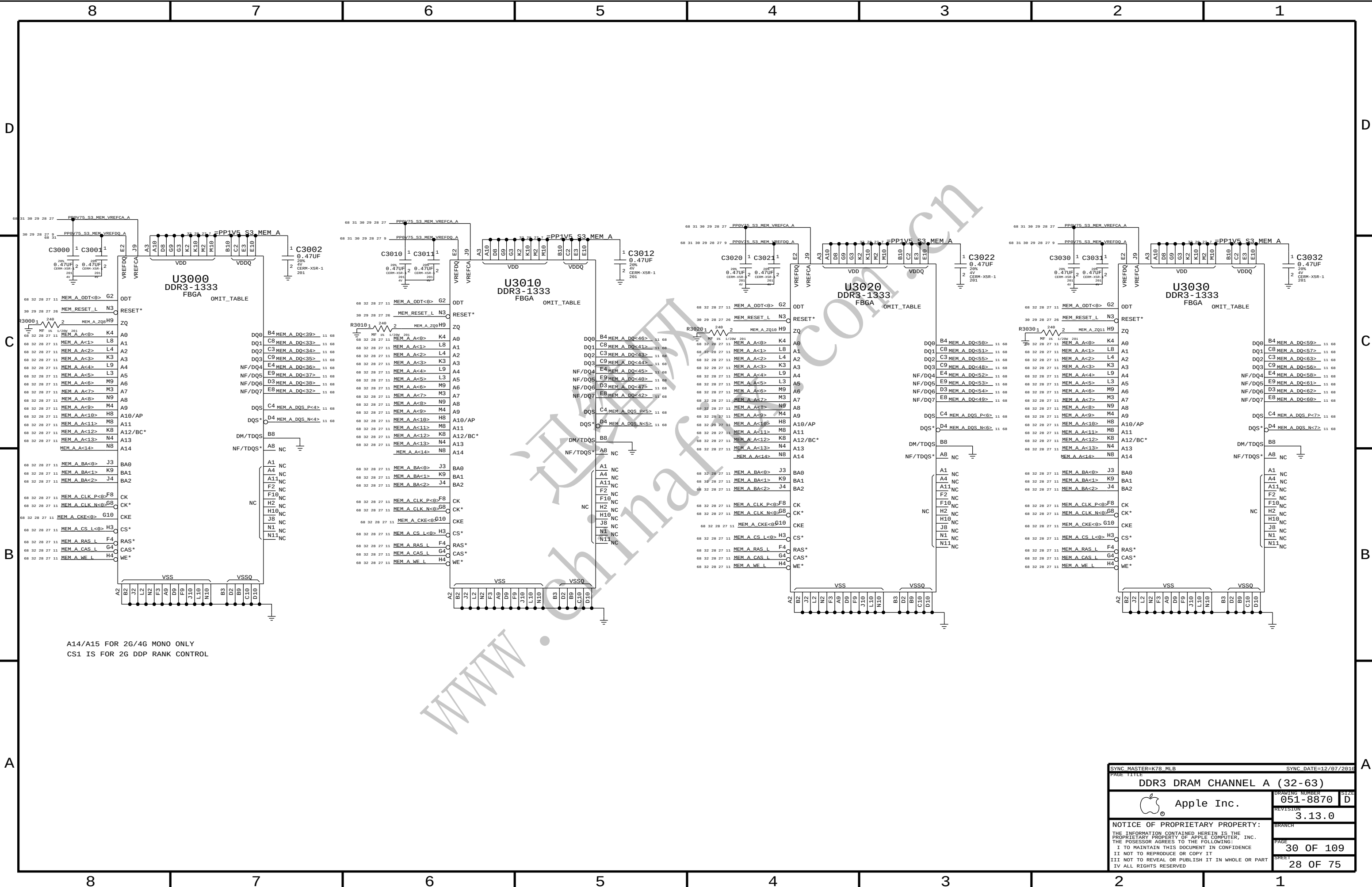
NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

CPU Memory S3 Support	
Apple Inc.	051-8870
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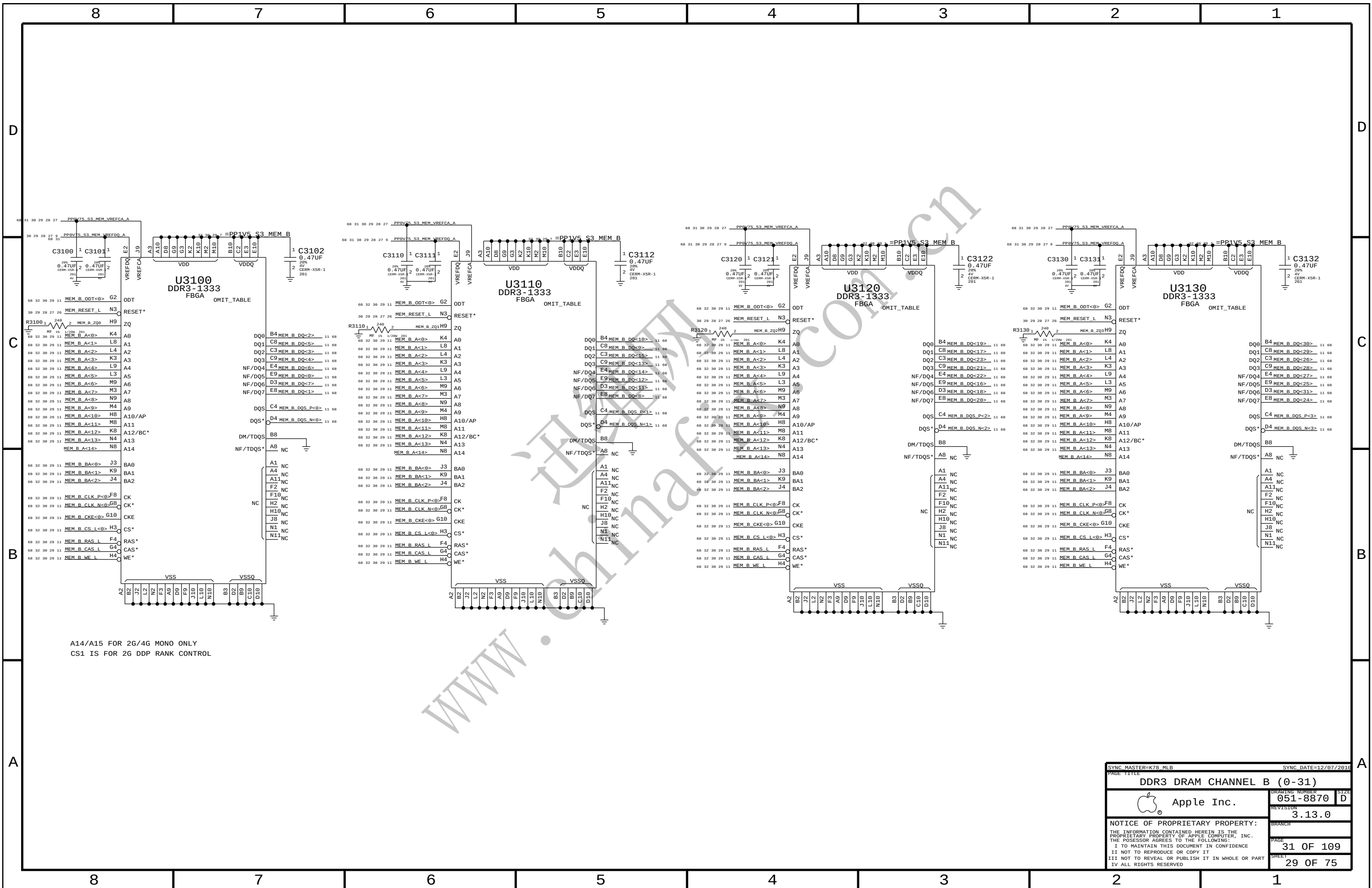
A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

SYNC MASTER=K78 MLB		SYNC DATE=12/07/2010	
PAGE TITLE			
DDR3 DRAM CHANNEL A (0-31)			
 Apple Inc.	DRAWING NUMBER	051-8870	SIZE D
	REVISION	3.13.0	
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A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

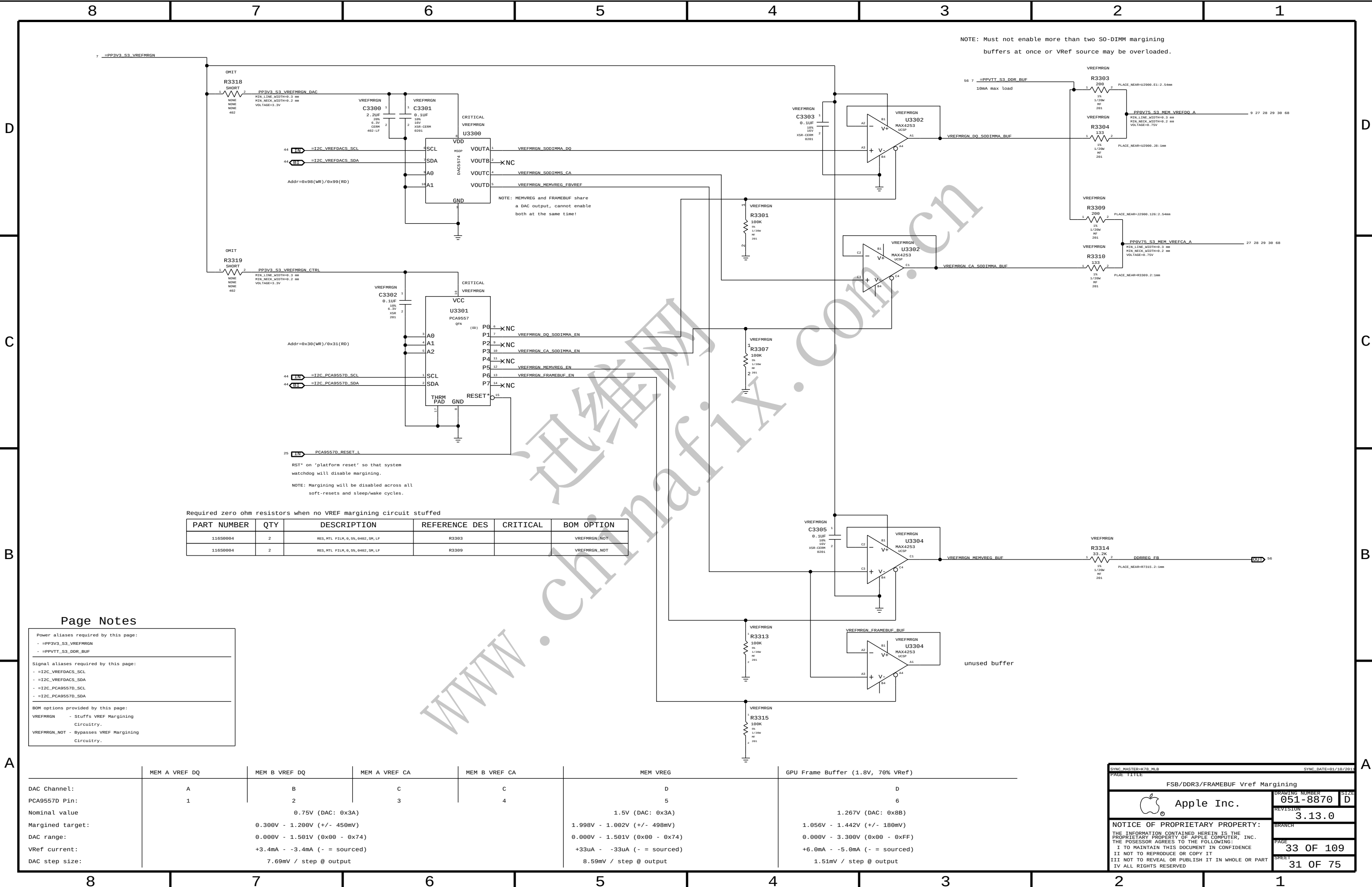
SYNC MASTER=K78 MLB		SYNC DATE=12/07/2010	
PAGE TITLE			
DDR3 DRAM CHANNEL A (32-63)			
 Apple Inc.		DRAWING NUMBER	051-8870
		REVISION	3.13.0
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A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

SYNC MASTER=K78 MLB		SYNC DATE=12/07/2010	
PAGE TITLE			
DDR3 DRAM CHANNEL B (0-31)			
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	REVISION	3.13.0	
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Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMGRN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

VREFMGRN - Stuffs VREF Margining Circuitry.

VREFMGRN_NOT - Bypasses VREF Margining Circuitry.

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% Vref)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.998V - 1.002V (+/- 498mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 1.501V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:		+3.4mA - -3.4mA (- = sourced)			+33uA - -33uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

SYNC MASTER=K70_M3.B

SYNC DATE=01/10/2011

PAGE TITLE

FSB/DDR3/FRAMEBUF Vref Margining

Apple Inc.

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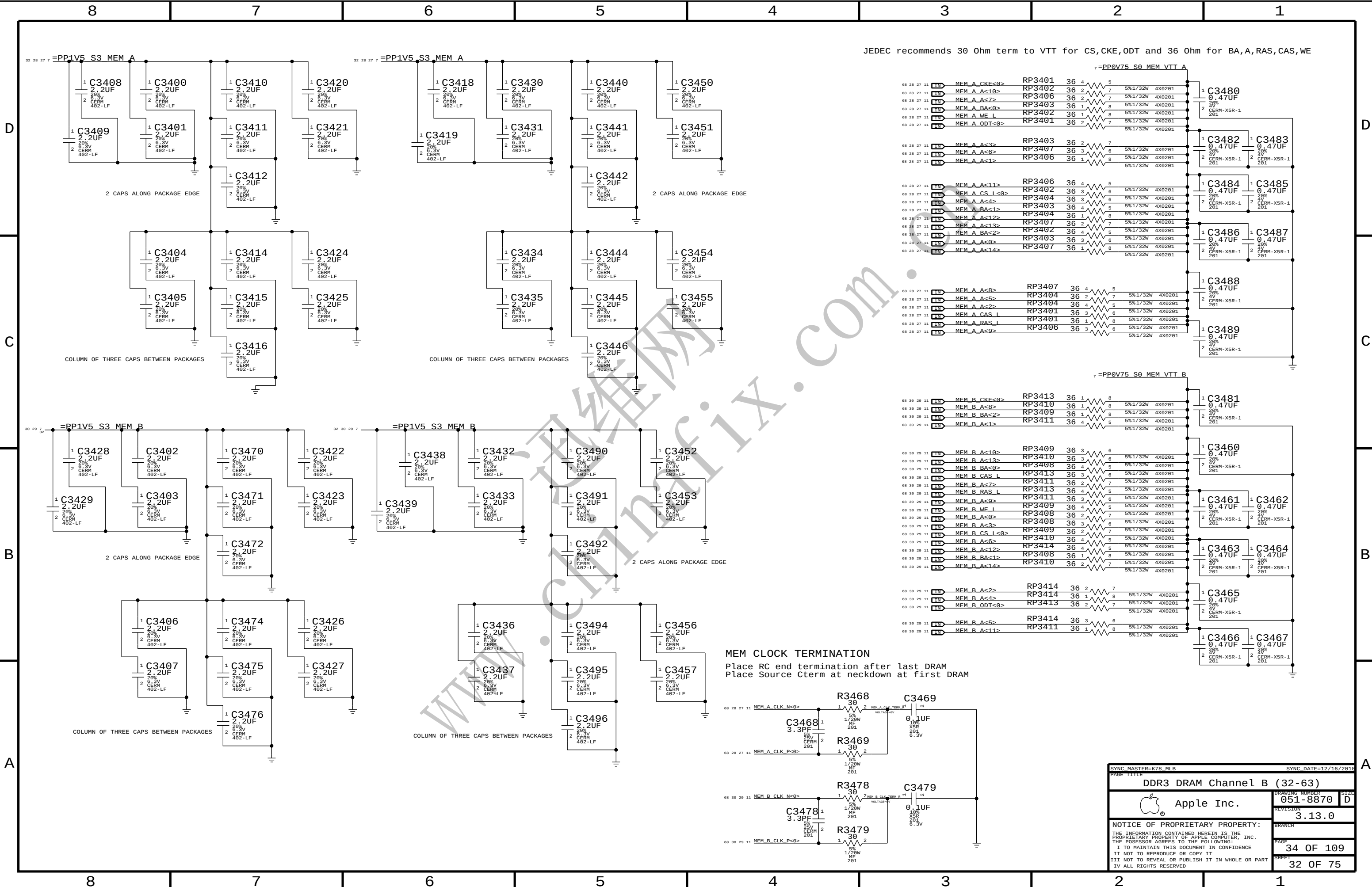
REVISION
3.13.0

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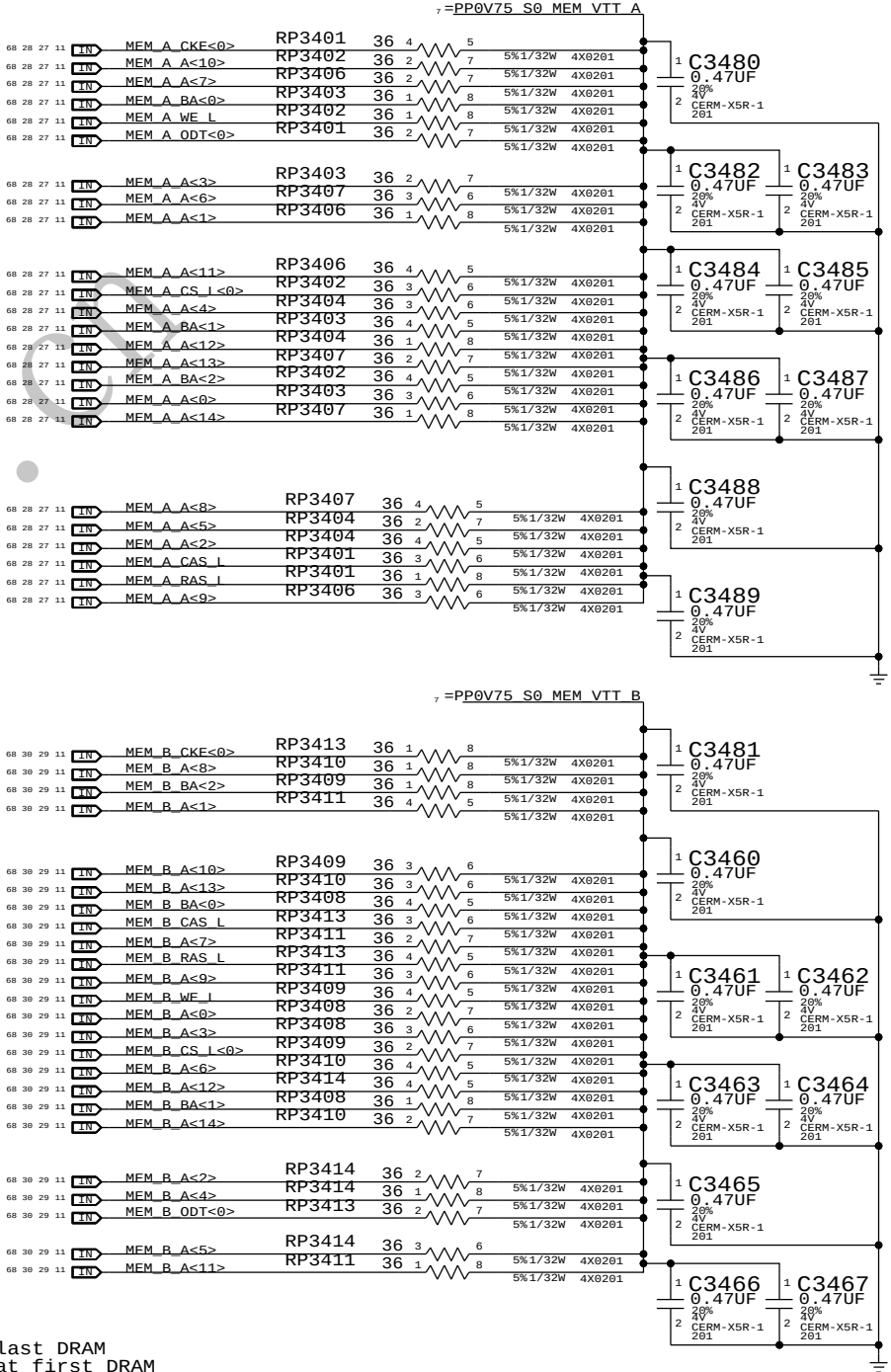
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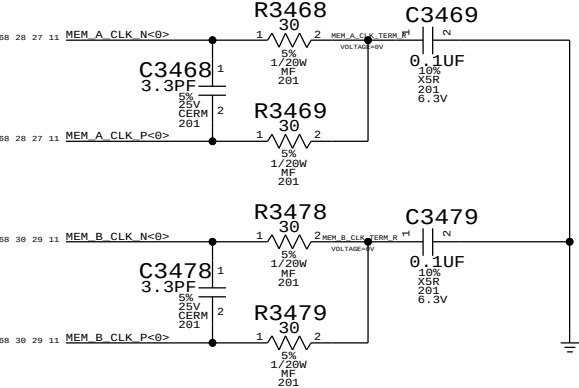
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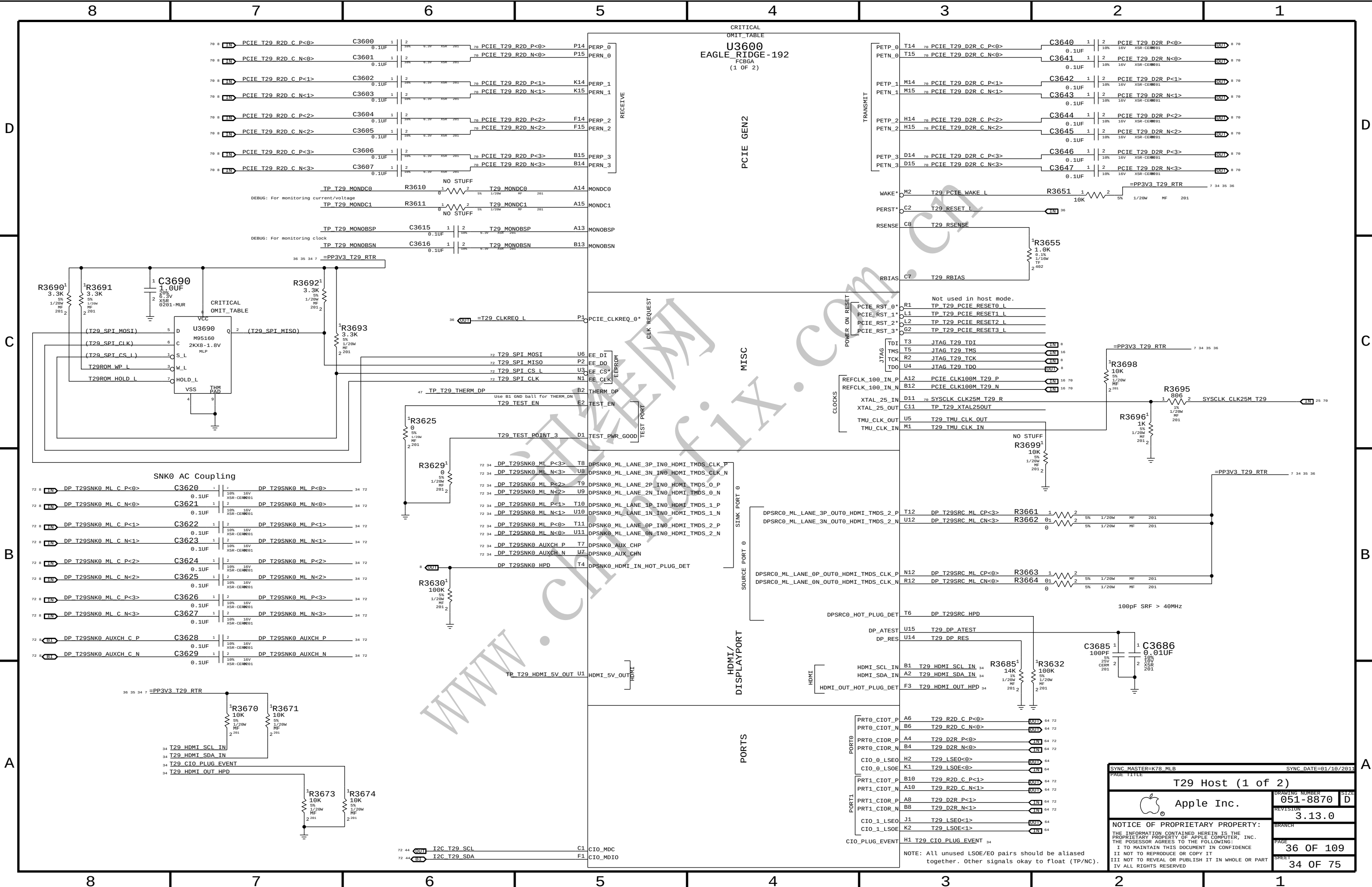
JEDEC recommends 30 Ohm term to VTT for CS,CKE,ODT and 36 Ohm for BA,A,RAS,CAS,WE



MEM CLOCK TERMINATION
Place RC end termination after last DRAM
Place Source Cterm at neckdown at first DRAM



SYNC MASTER=K78 MLB		SYNC DATE=12/16/2010	
PAGE TITLE			
DDR3 DRAM Channel B (32-63)			
	DRAWING NUMBER		SIZE
	051-8870		D
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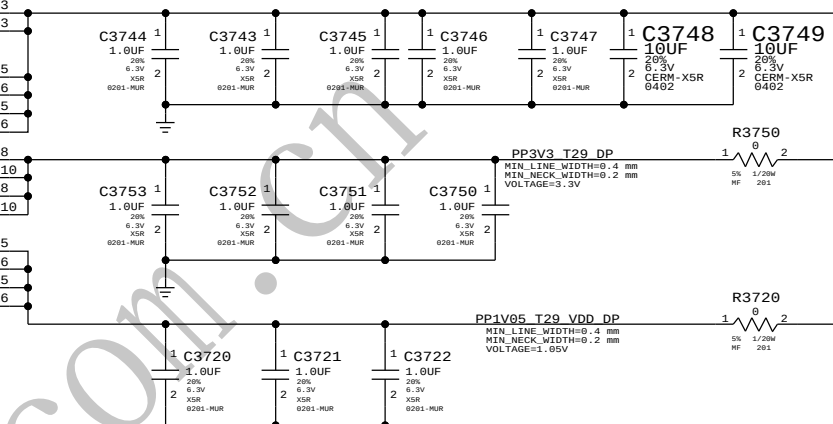
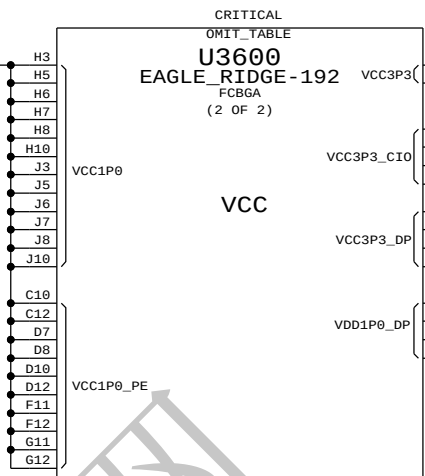
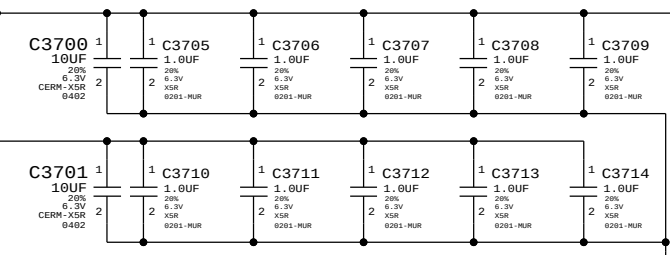
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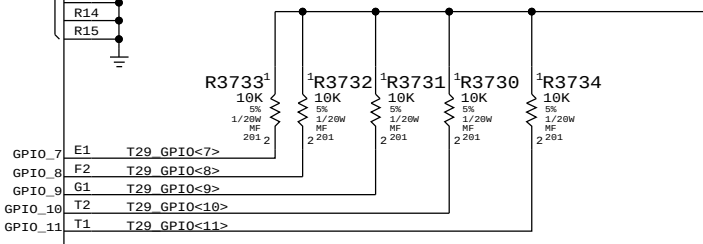
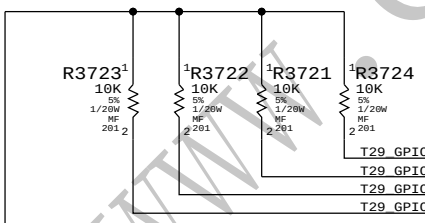
35 7 =PP1V05 T29_RTR
2100 mA (Single Port)
2250 mA (Dual Port)
EDP: 3000 mA



=PP3V3 T29_RTR
135 mA (Single-Port)
152 mA (Dual-Port)
EDP: 200 mA

0-ohms are placeholders for now, replace with proper values after characterization.

=PP1V05 T29_RTR 7 35
2100 mA (Single Port)
2250 mA (Dual Port)
EDP: 3000 mA



SYNC MASTER=K78 MLB		SYNC DATE=01/10/2011	
PAGE TITLE		T29 Host (2 of 2)	
 Apple Inc.		DRAWING NUMBER	051-8870
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Page Notes

Power aliases required by this page:

- PPVIN_SW_T29BST (8-13V Boost Input)
- PP18V_T29_REG (18V Boost Output)
- PP3V3_T29_P3V3T29FET (3.3V FET Input)
- PP3V3_T29_FET (3.3V FET Output)
- PP3V3_S0_T29PWRCTL
- PP1V05_T29_P1V05T29FET (1.05V FET Input)
- PP1V05_T29_FET (1.05V FET Output)

Signal aliases required by this page:

- T29_CLKREQ_L
- T29_RESET_L

BOM options provided by this page:

T29BST:Y - Stuffs 18V boost circuitry.

T29 18V Boost Regulator

SI8499DB:
Vds(max): -30V
Vgs(max): +/-12V
Vgs(th): -1.4V
Rds(on): 46mOhm @ 4.5V Vgs
Id(max): 3.7A @ 70C

CRITICAL
T29BST:Y
Q3880
SI8499DB
BGA

CRITICAL
T29BST:Y
L3895
6.8UH-4.0A
PIMB062D-SM

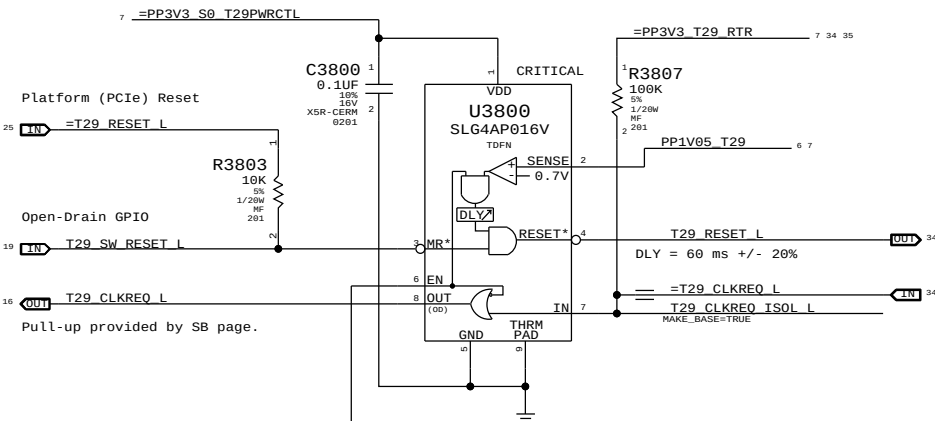
CRITICAL
T29BST:Y
D3895
POWER01-123
DFLS230L

Vout = 15.1V
Max Current = 1.0A
Freq = 300KHz

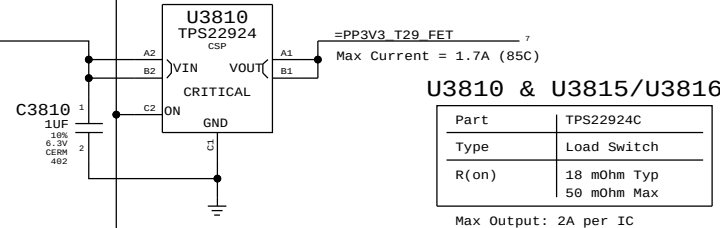
$$V_{out} = 1.6V * (1 + R_a / R_b)$$

$$UVLO(falling) = 1.22 * (R_1 + R_2) / R_2$$
$$UVLO(rising) = UVLO(falling) + (2\mu A * R_1)$$
$$UVLO = 4.55V (falling), 4.95V (rising)$$

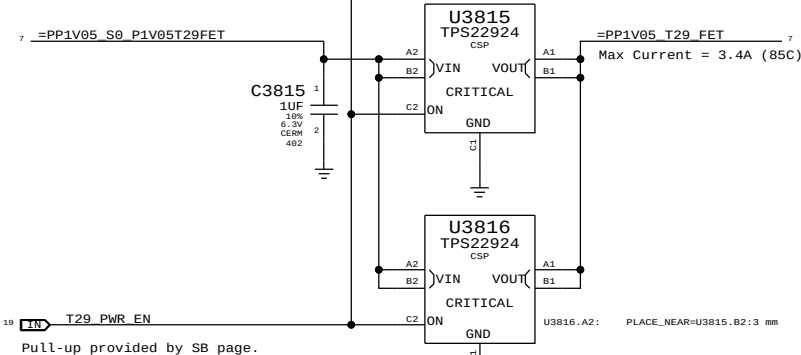
Supervisor & CLKREQ# Isolation



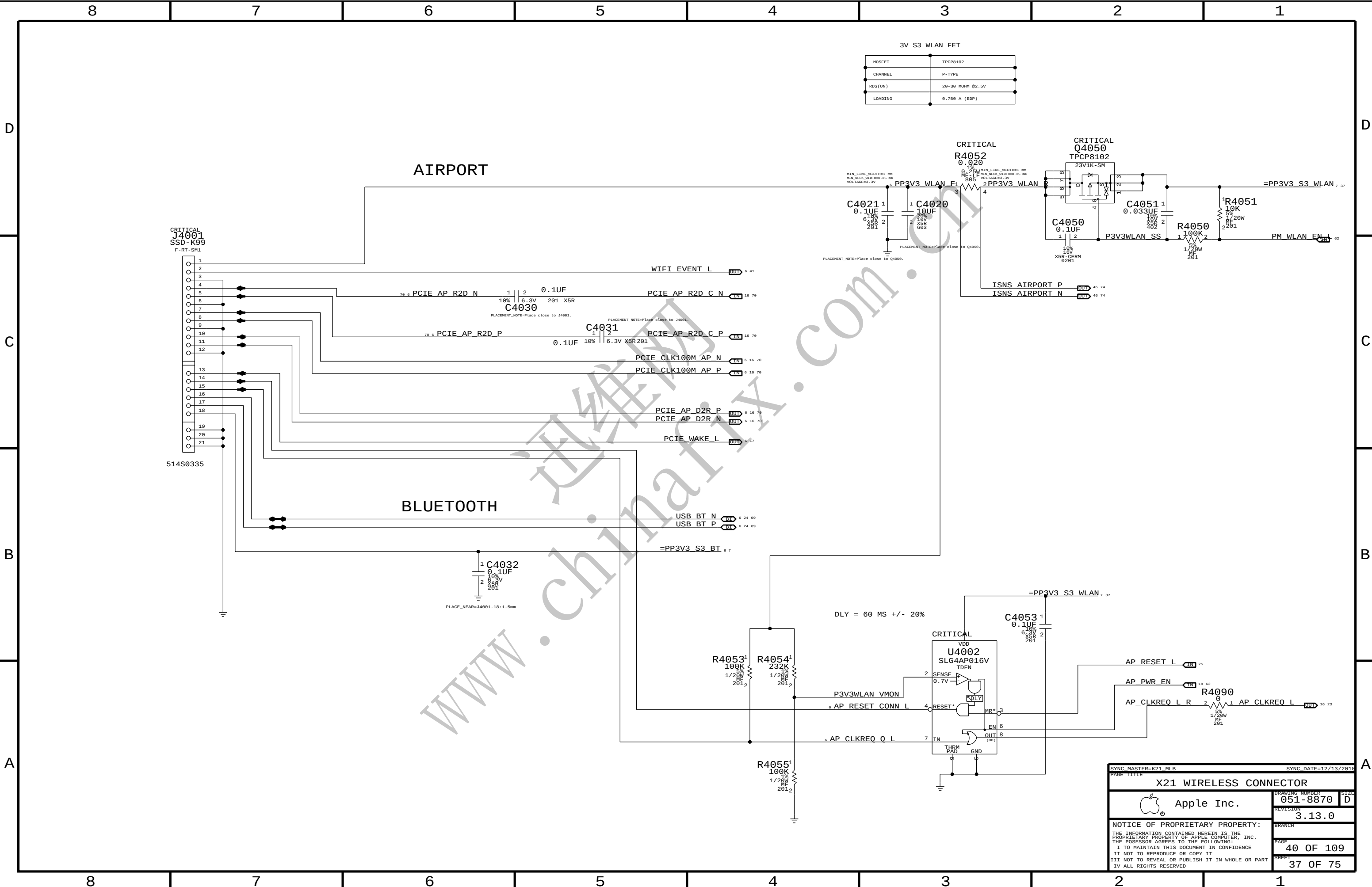
3.3V T29 Switch



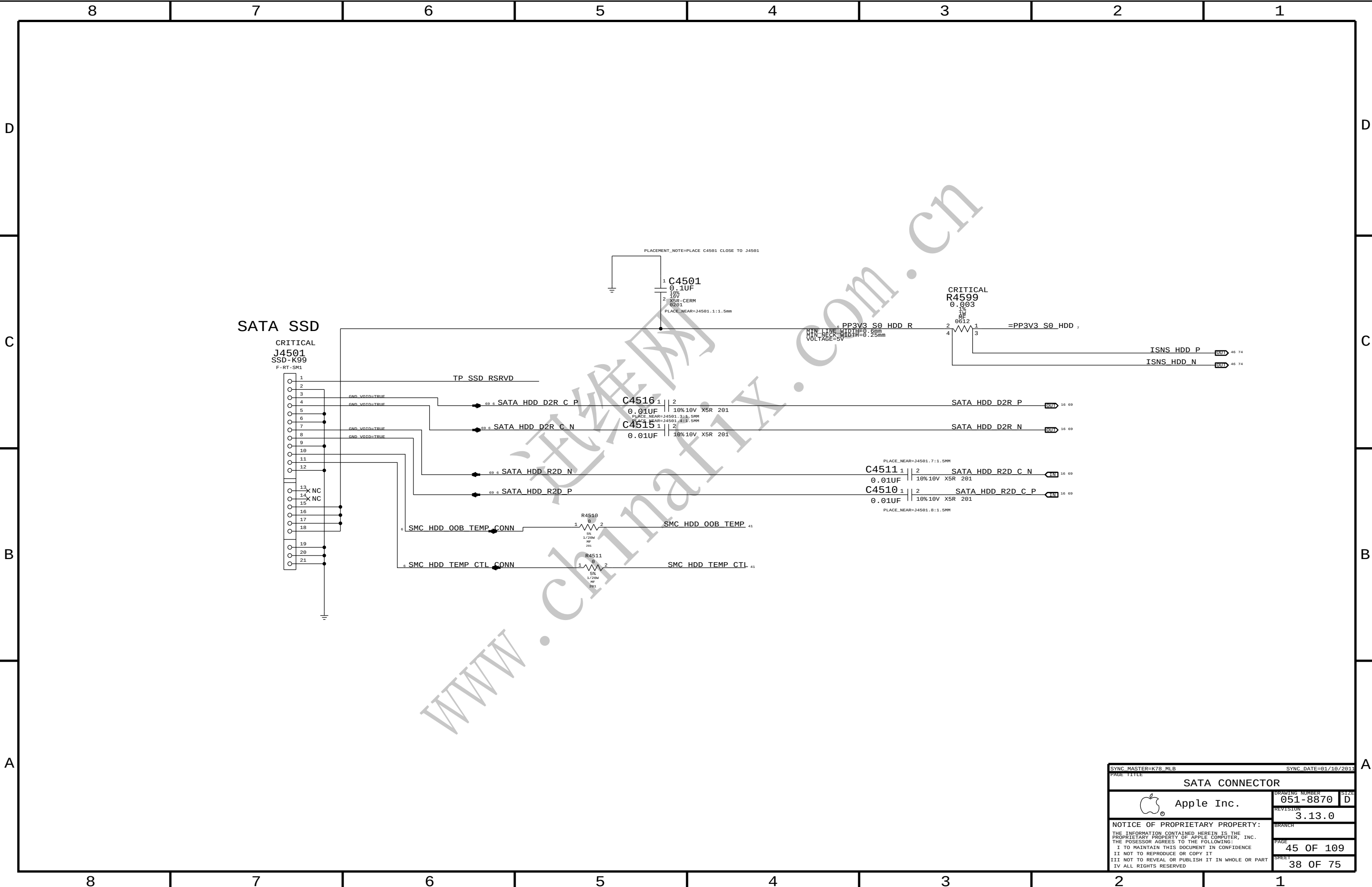
1.05V T29 Switch



SYNC MASTER=K78 MLB		SYNC DATE=01/18/2011	
PAGE TITLE			
T29 Power Support			
		DRAWING NUMBER	051-8870
Apple Inc.		SIZE	D
		REVISION	3.13.0
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SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE		DRAWING NUMBER	
X21 WIRELESS CONNECTOR		051-8870	
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		3.13.0	
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		40 OF 109	
		SHEET	
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SYNC MASTER=K78_MLB

SYNC DATE=01/10/2011

PAGE TITLE

SATA CONNECTOR

 Apple Inc.

DRAWING NUMBER
051-8870

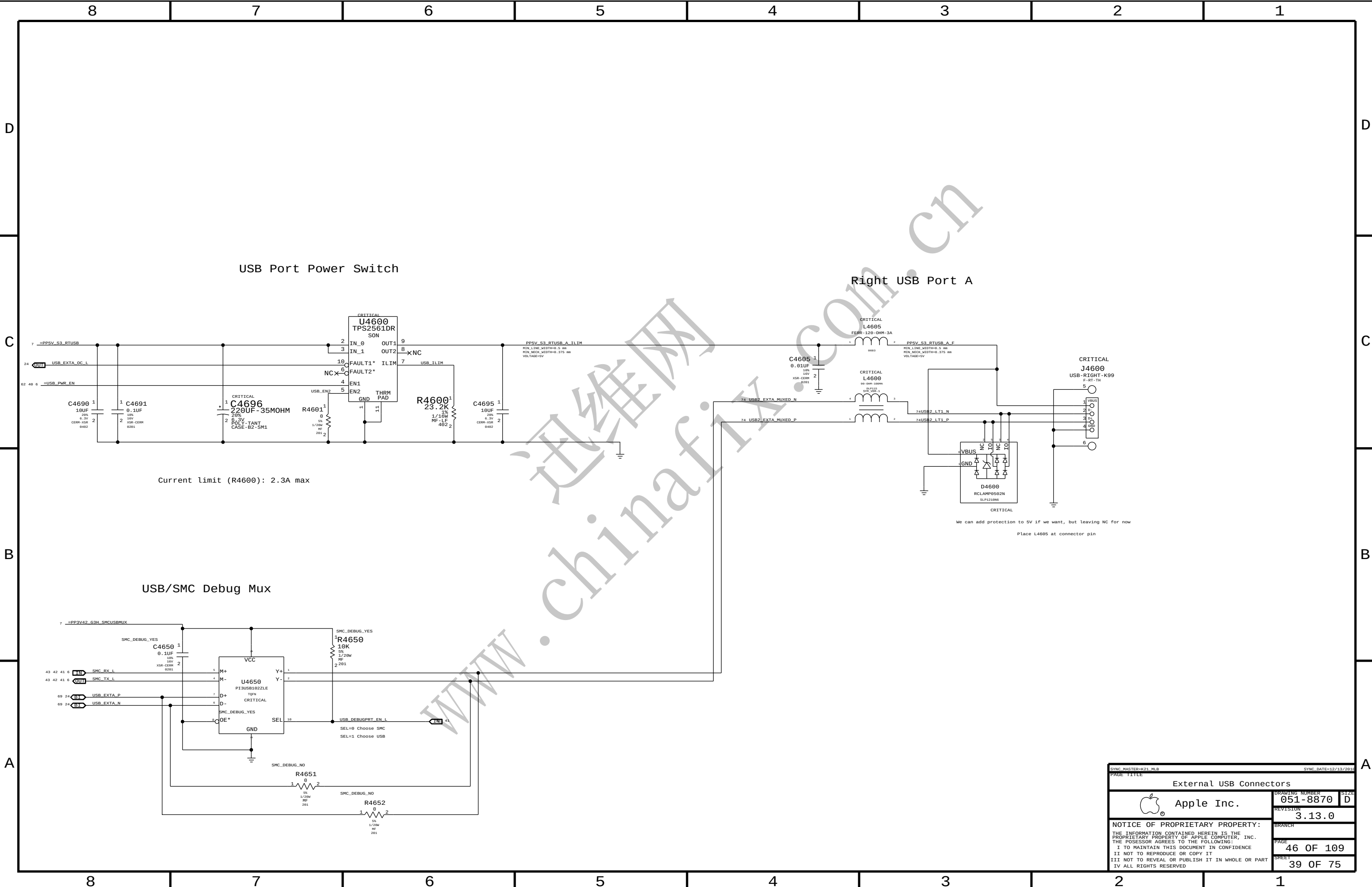
REVISION
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USB Port Power Switch

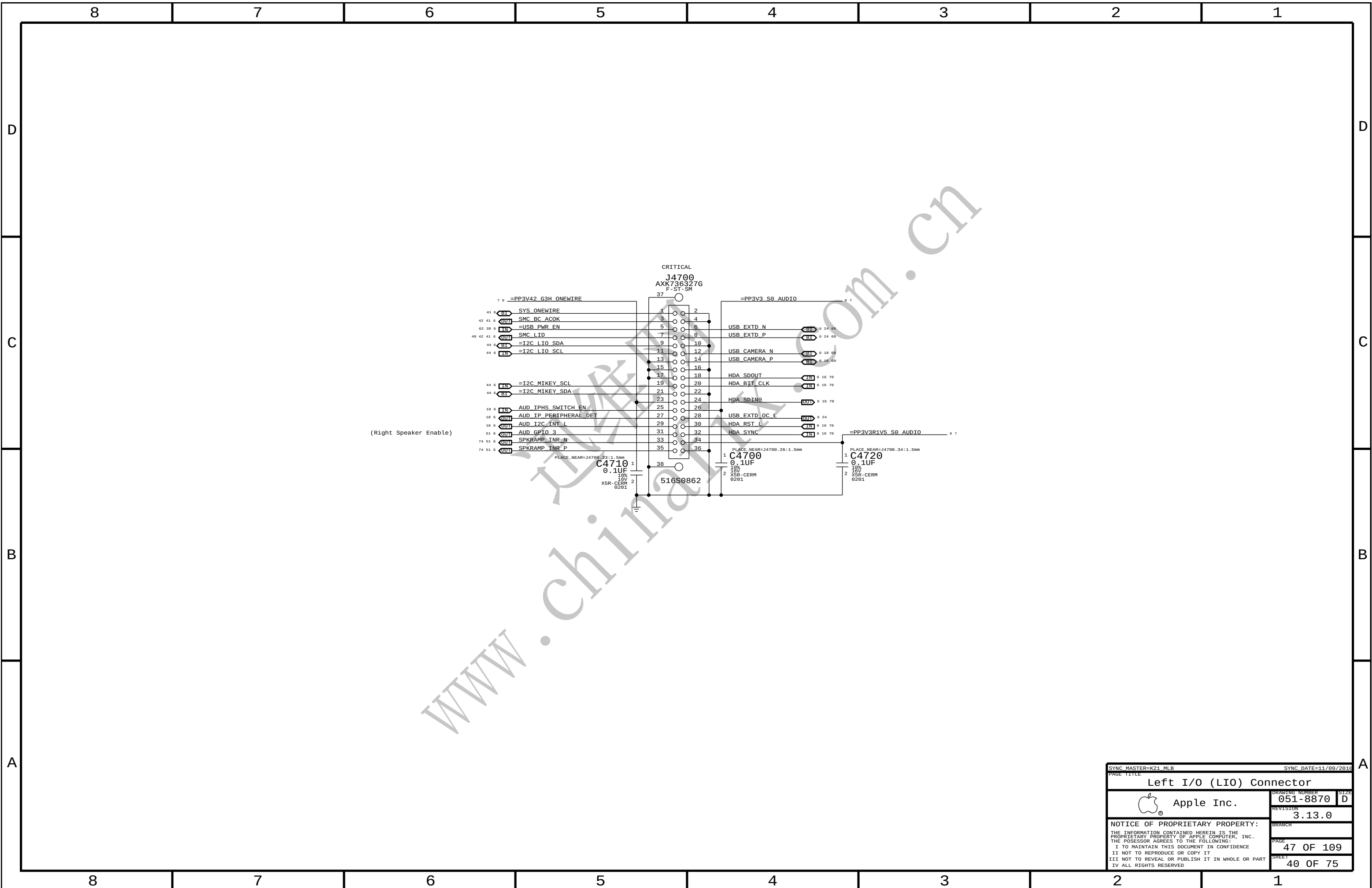
Right USB Port A

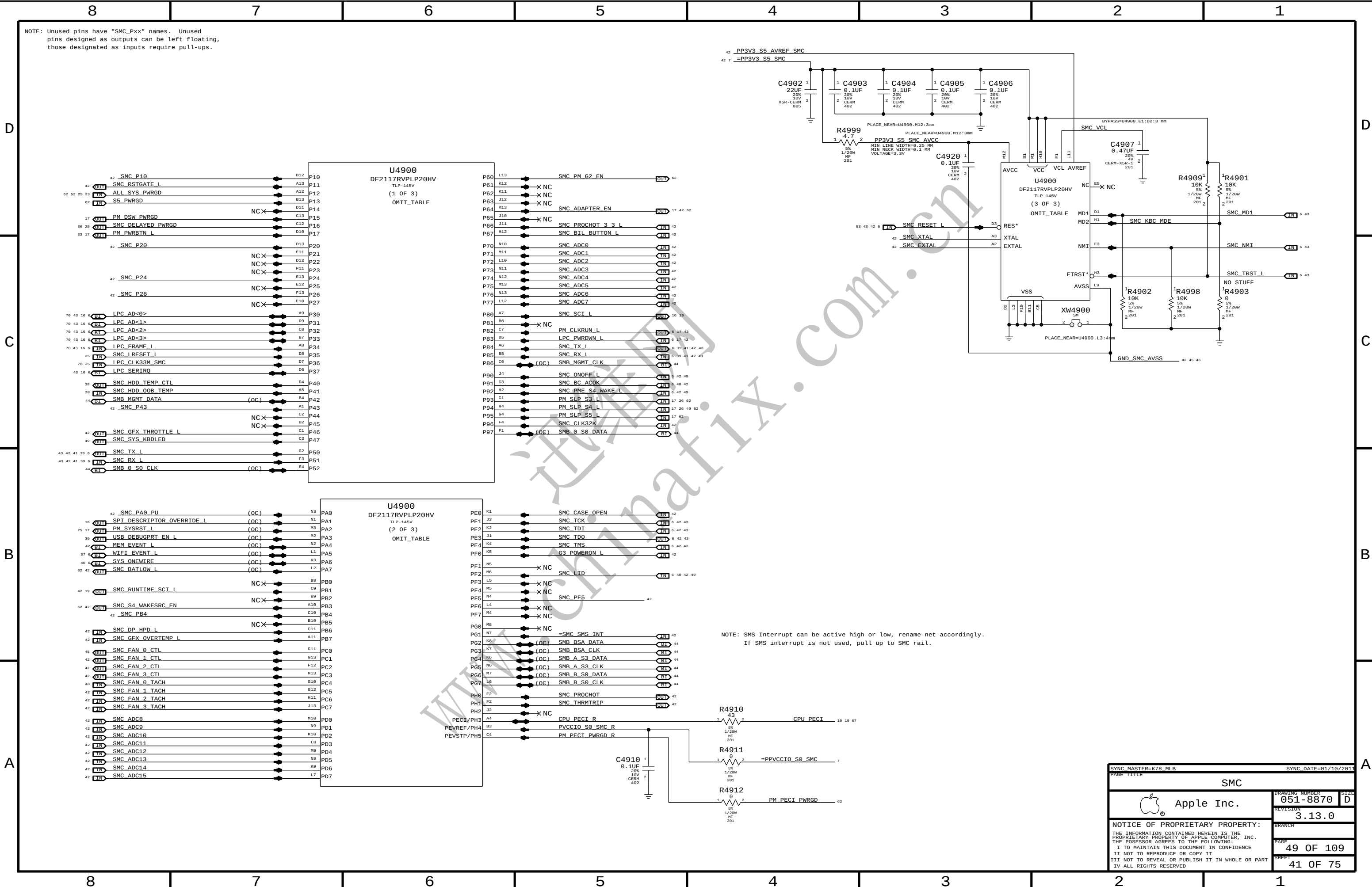
Current limit (R4600): 2.3A max

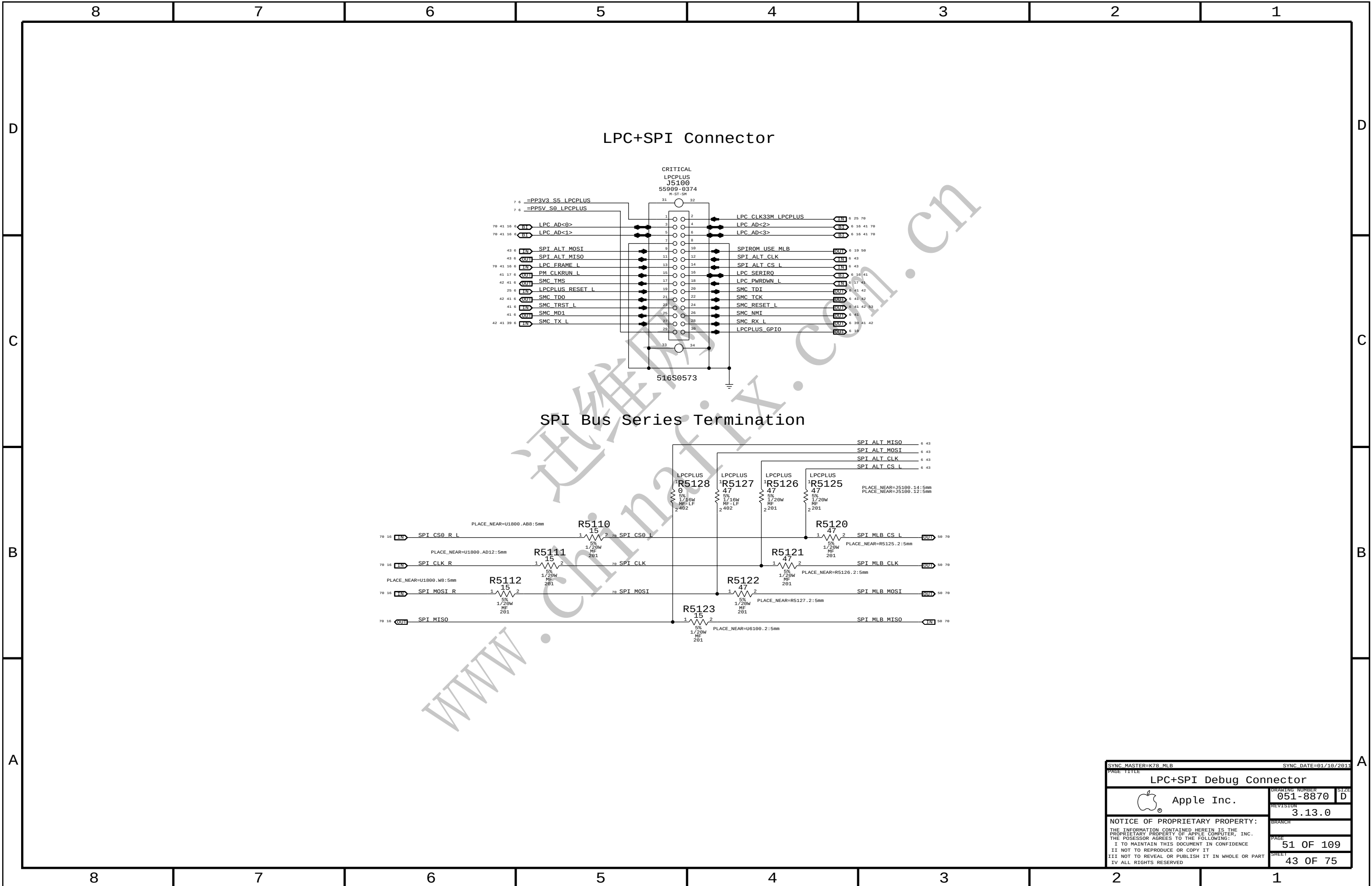
USB/SMC Debug Mux

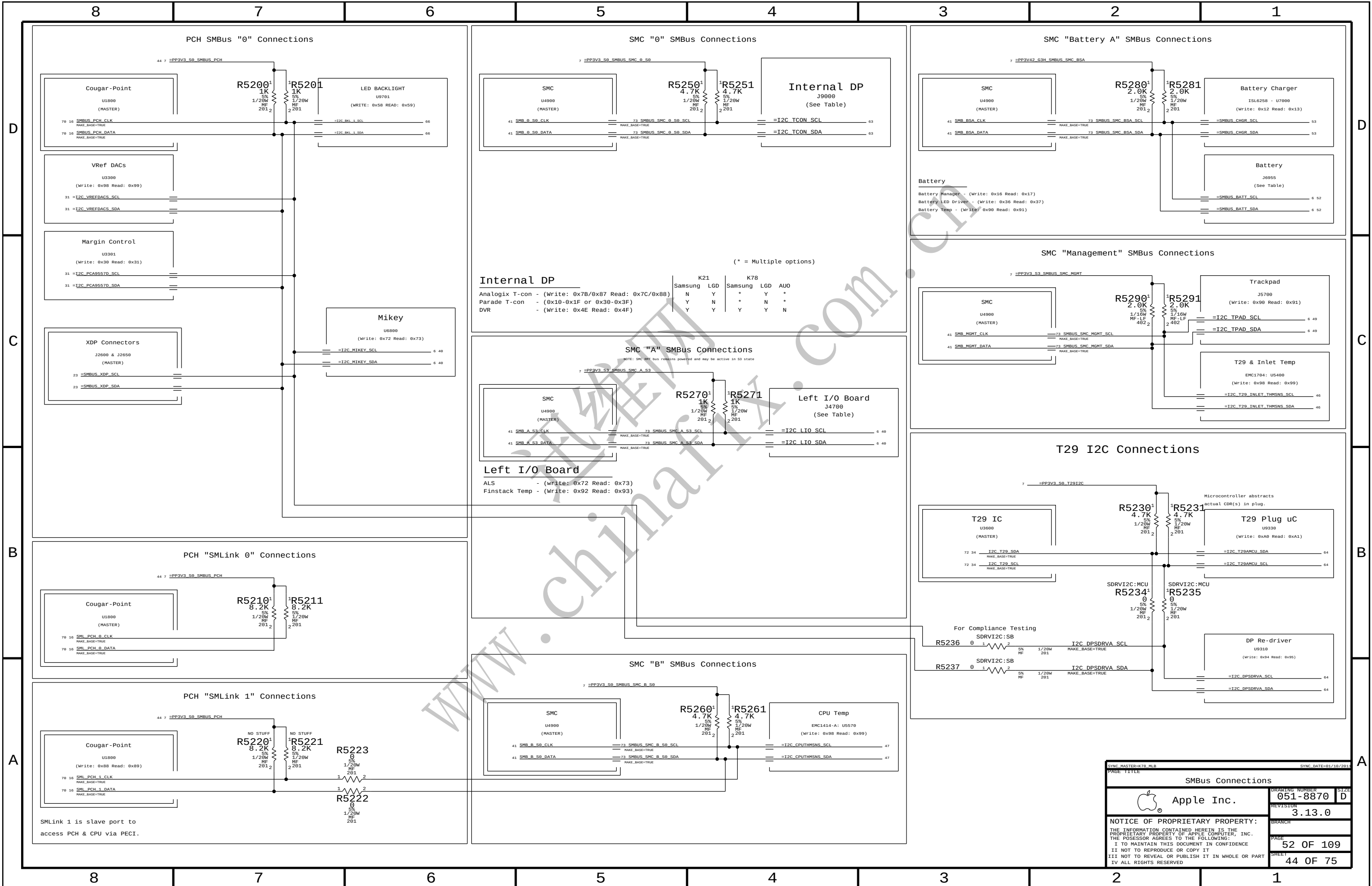
We can add protection to 5V if we want, but leaving NC for now
Place L4605 at connector pin

SYNC MASTER=K21_MLB		SYNC DATE=12/13/2011	
PAGE TITLE			
External USB Connectors			
	Apple Inc.	DRAWING NUMBER	051-8870
		REVISION	3.13.0
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		PAGE	46 OF 109
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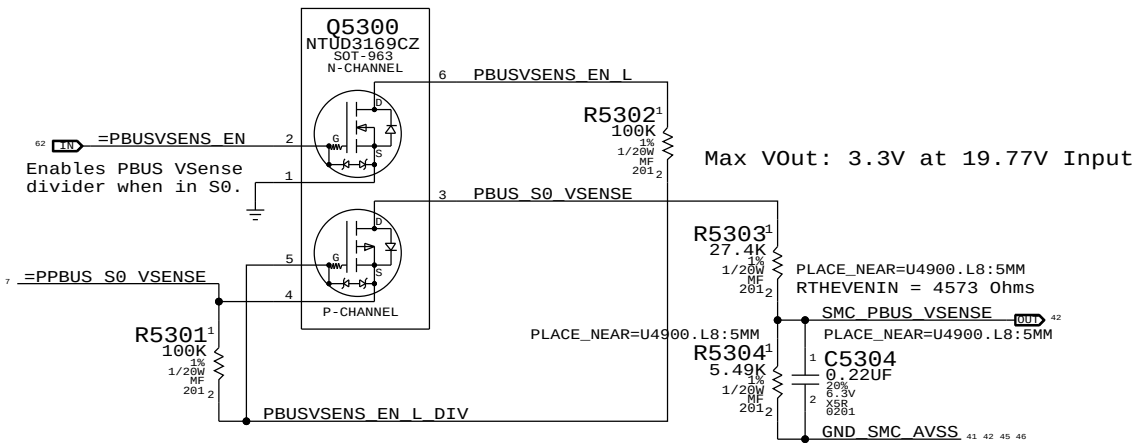




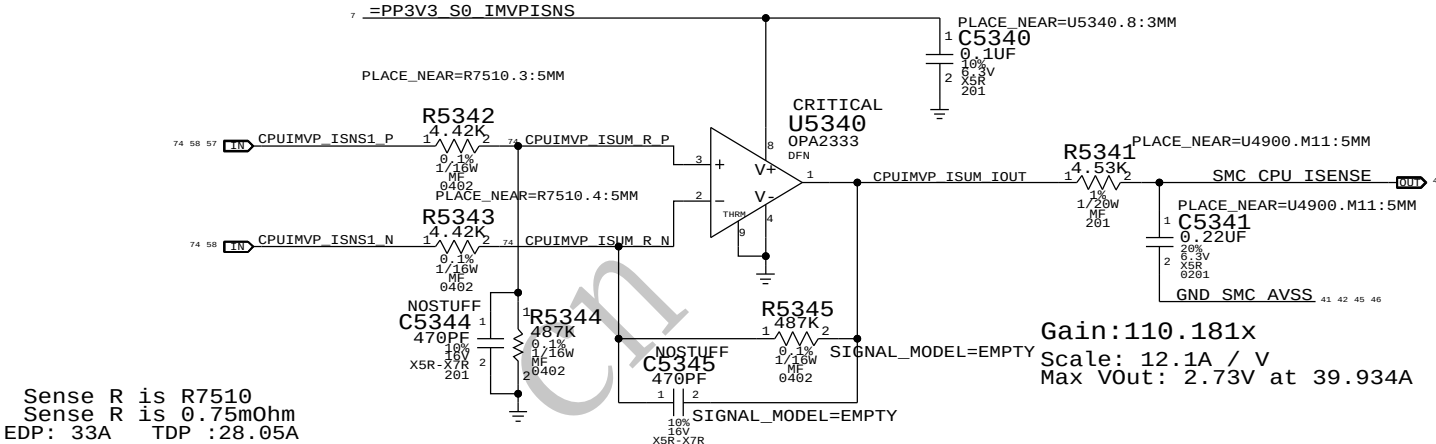




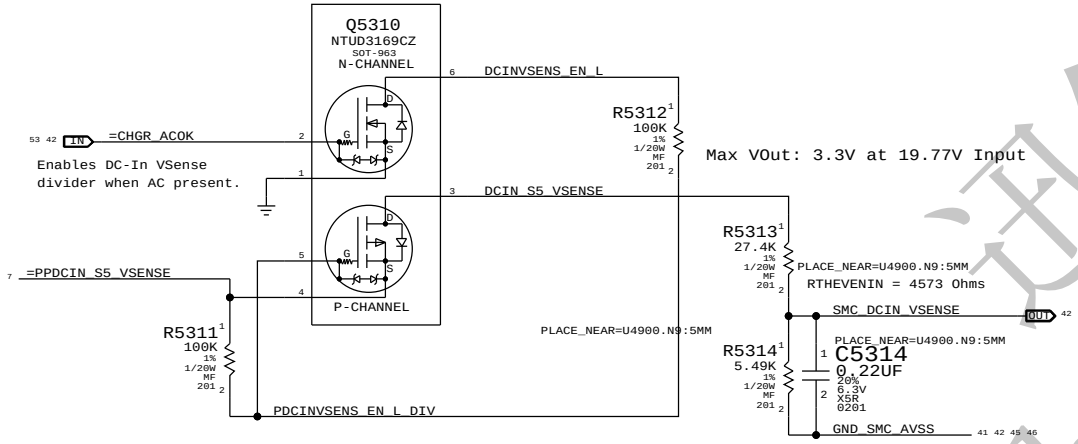
PBUS Voltage Sense Enable & Filter



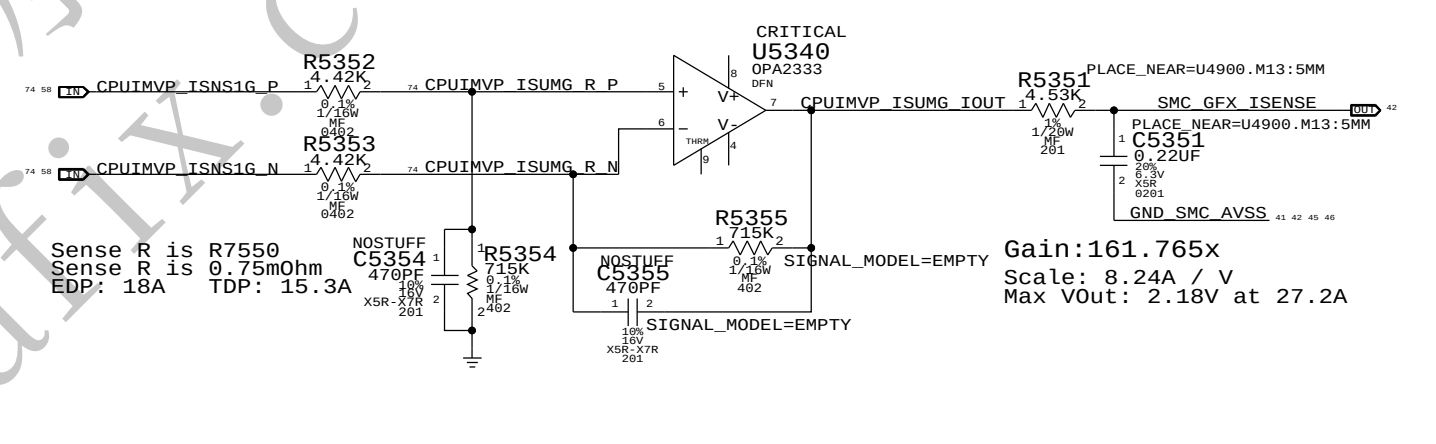
CPU VCore Load Side Current Sense / Filter



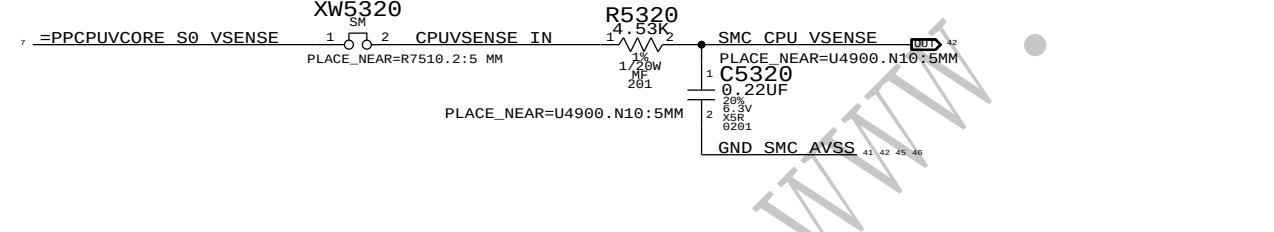
DC-In Voltage Sense Enable & Filter



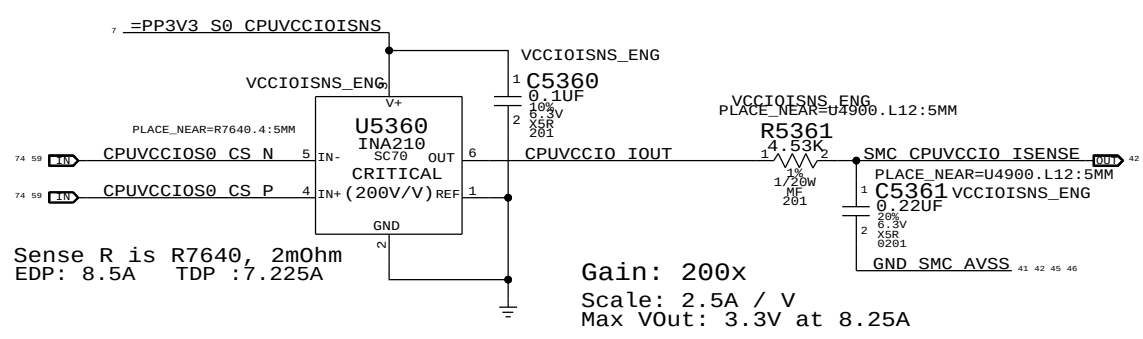
GFX/IG VCore Load Side Current Sense / Filter



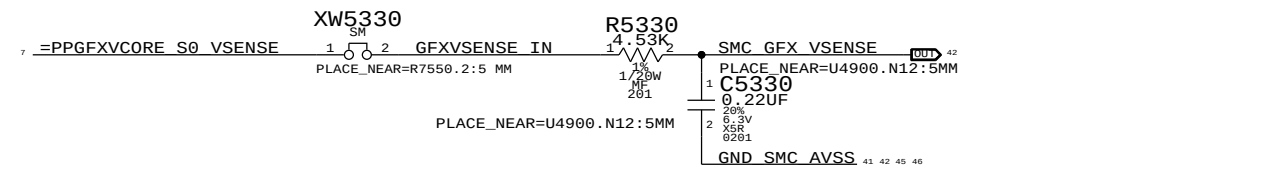
CPU Vcore Voltage Sense / Filter



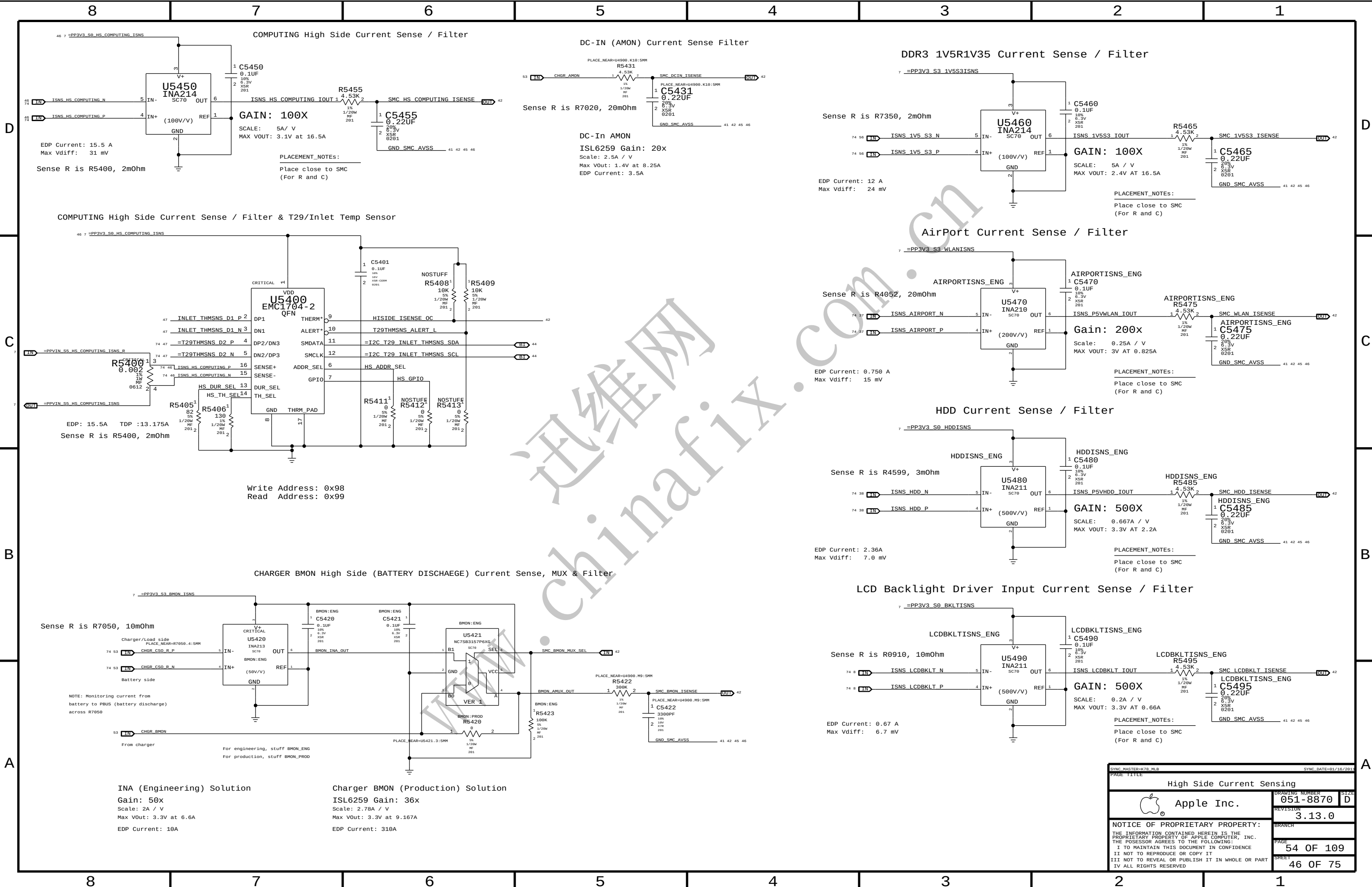
CPU 1.05V VCCIO Current Sense / Filter



GFX/IG Vcore Voltage Sense / Filter



SYNC MASTER=K7B MLB		SYNC DATE=01/10/2011	
PAGE TITLE		Voltage & Load Side Current Sensing	
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		REVISION	3.13.0
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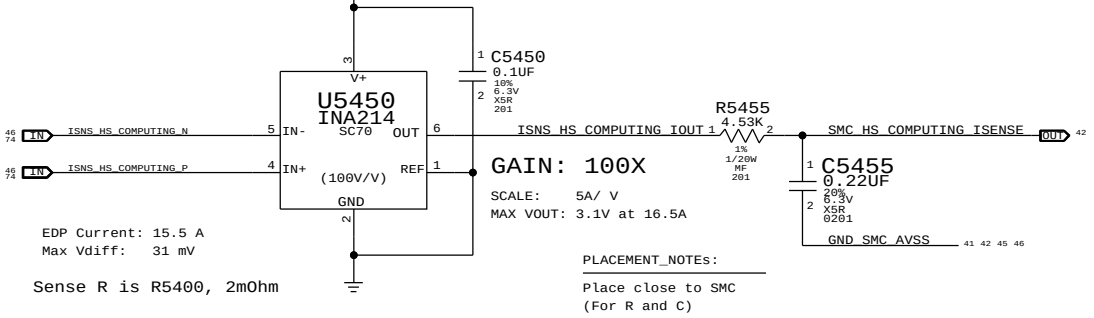
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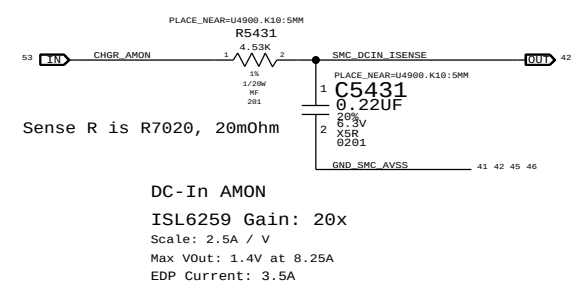
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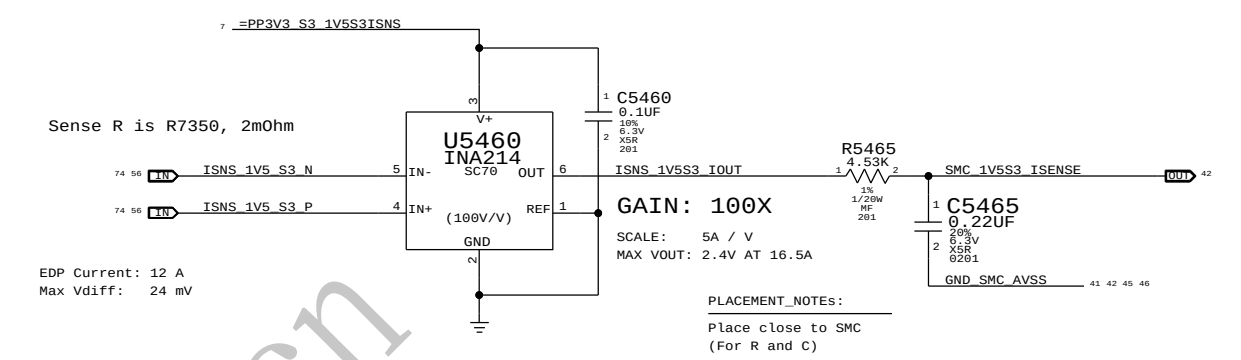
COMPUTING High Side Current Sense / Filter



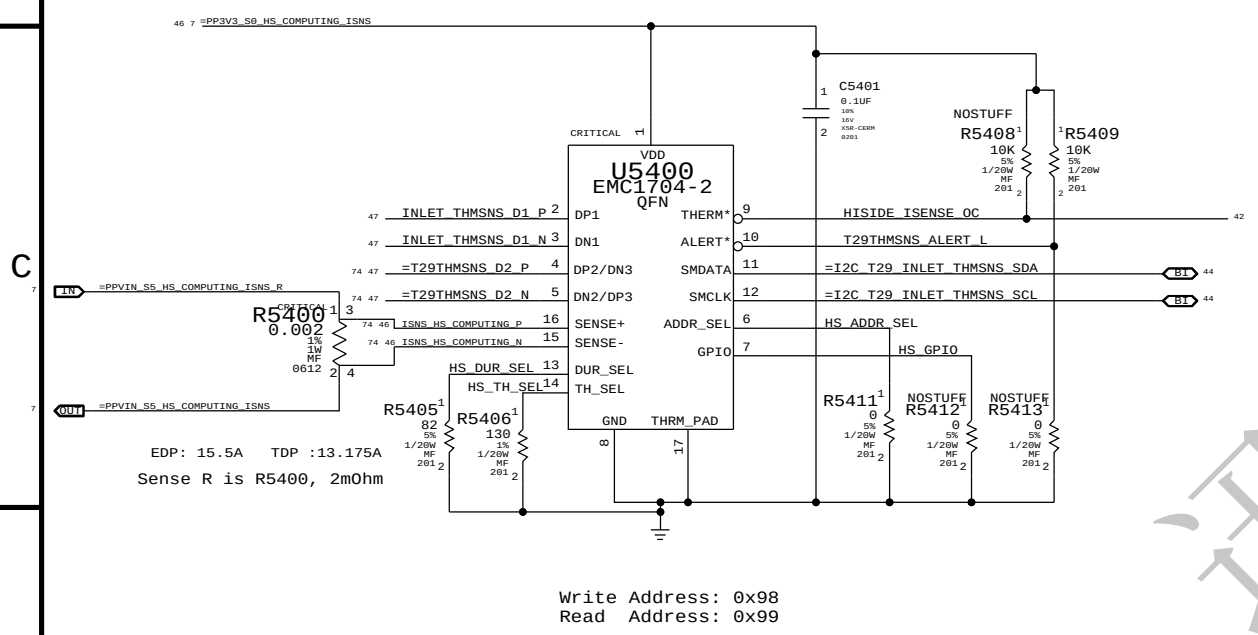
DC-IN (AMON) Current Sense Filter



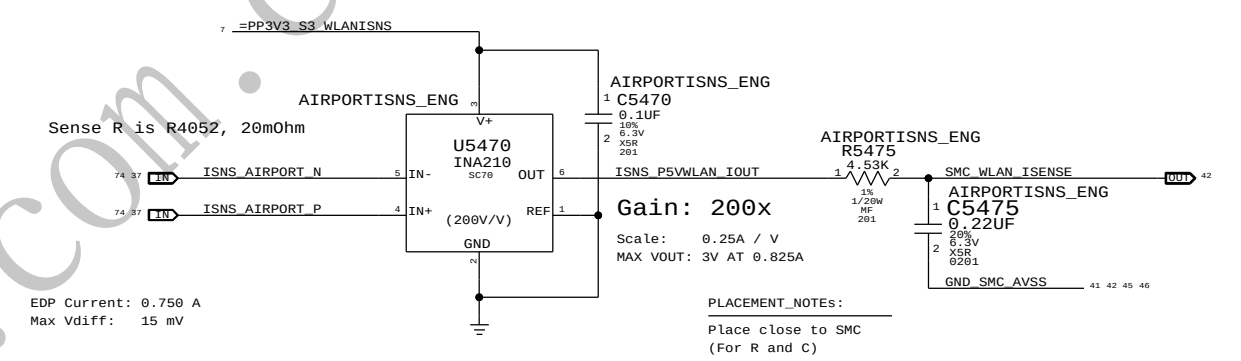
DDR3 1V5R1V35 Current Sense / Filter



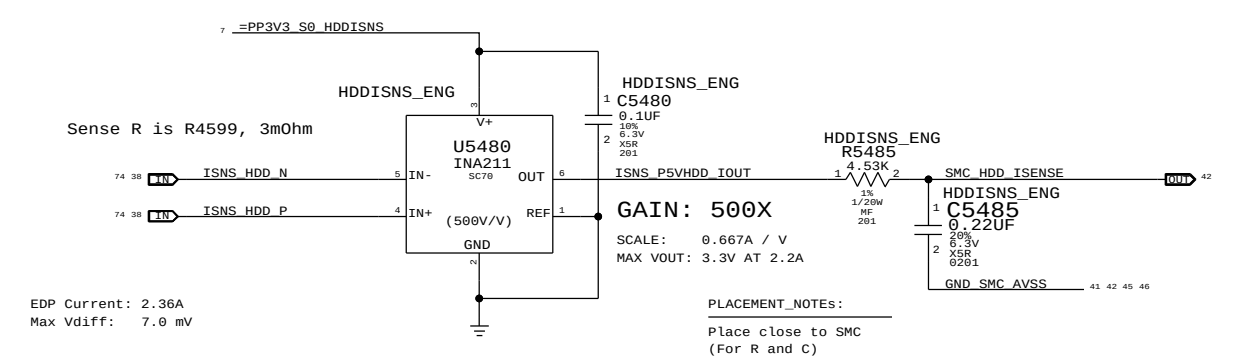
COMPUTING High Side Current Sense / Filter & T29/Inlet Temp Sensor



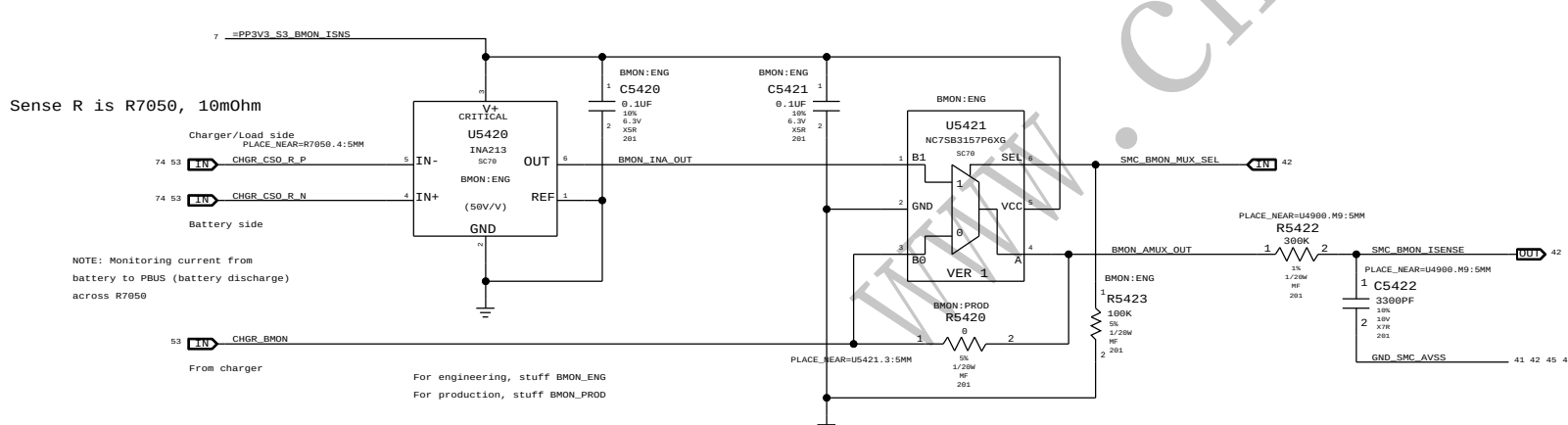
AirPort Current Sense / Filter



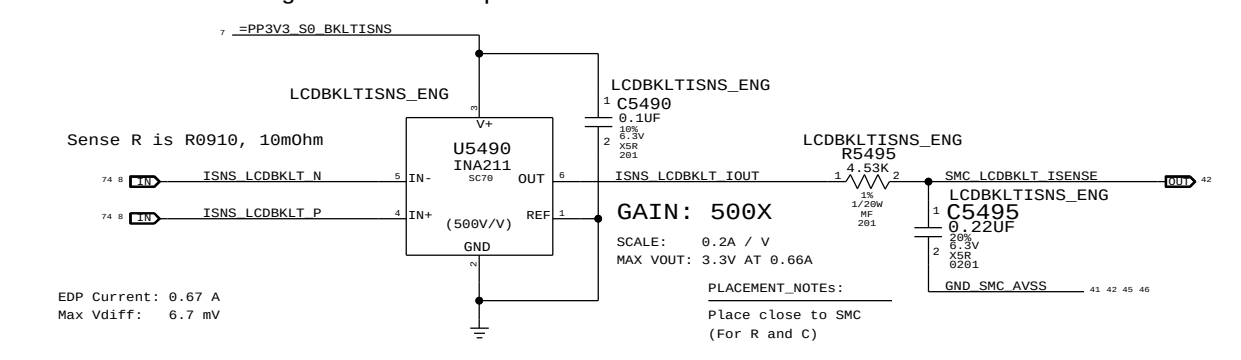
HDD Current Sense / Filter



CHARGER BMON High Side (BATTERY DISCHARGE) Current Sense, MUX & Filter



LCD Backlight Driver Input Current Sense / Filter

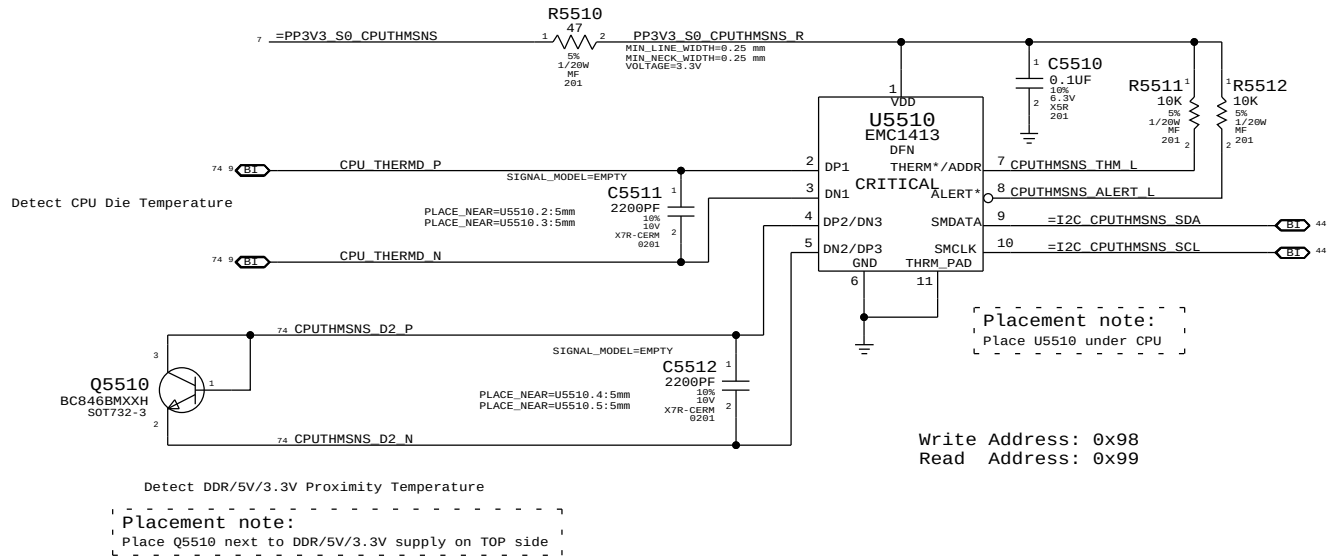


INA (Engineering) Solution
Gain: 50x
Scale: 2A / V
Max Vout: 3.3V at 6.6A
EDP Current: 10A

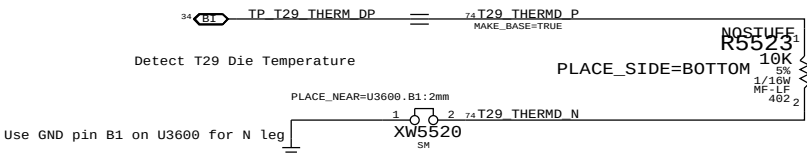
Charger BMON (Production) Solution
ISL6259 Gain: 36x
Scale: 2.78A / V
Max Vout: 3.3V at 9.167A
EDP Current: 310A

SYNC MASTER=K70_MLB		SYNC DATE=01/10/2011	
PAGE TITLE			
High Side Current Sensing			
	DRAWING NUMBER		SIZE
	051-8870		D
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CPU Proximity Sensor



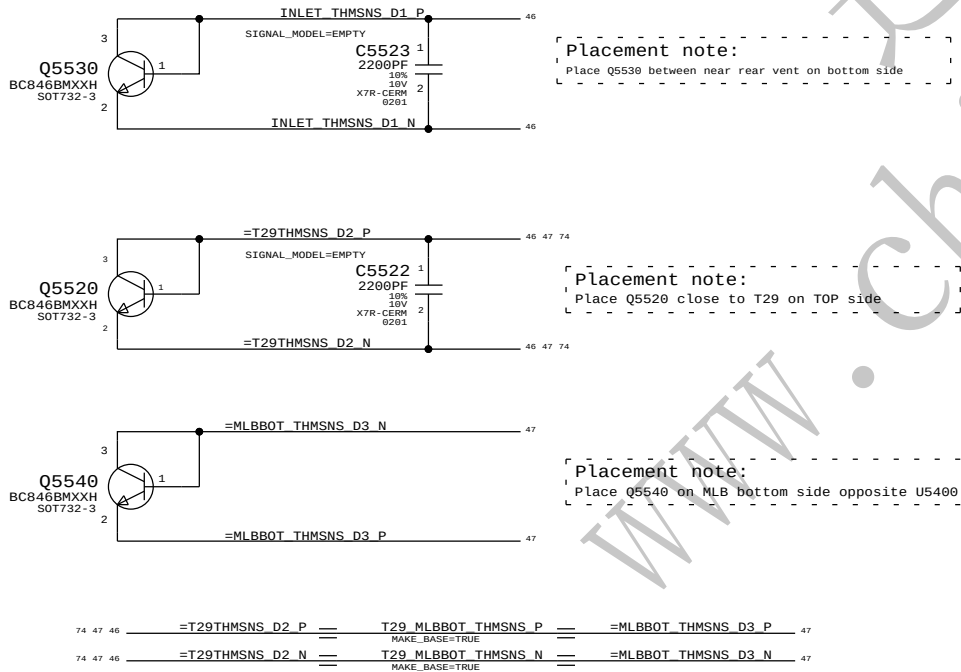
T29 Die



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES, MF, 1/20W, 100K OHM, 5, 0201, SMD	C5361		VCCIOISNS_PROD
117S0008	1	RES, MF, 1/20W, 100K OHM, 5, 0201, SMD	C5475		AIRPORTISNS_PROD
117S0008	1	RES, MF, 1/20W, 100K OHM, 5, 0201, SMD	C5485		HDDISNS_PROD
117S0008	1	RES, MF, 1/20W, 100K OHM, 5, 0201, SMD	C5495		LCDBKLTISNS_PROD

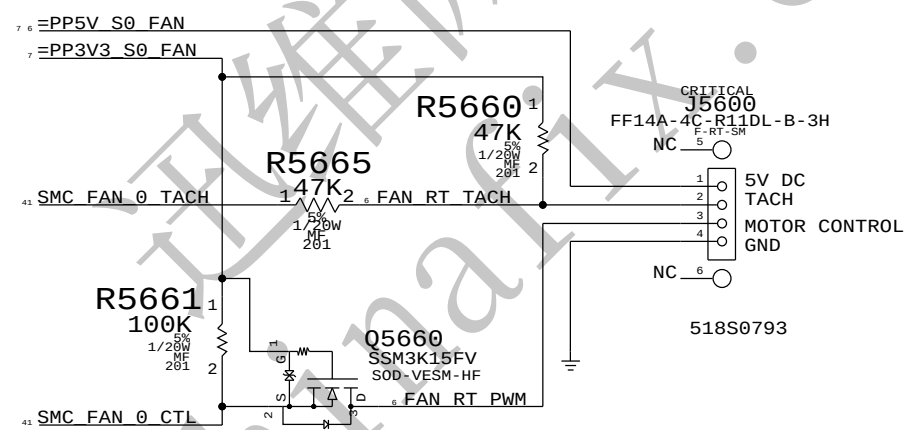
Replacing caps with 100K PD on ISENSE SMC inputs

T29,MLB Bottom & Inlet Proximity Sensors



SYNC MASTER=K78_MLB		SYNC DATE=01/16/2011	
PAGE TITLE		Thermal Sensors	
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FAN CONNECTOR



SYNC MASTER=K78 MLB		SYNC DATE=01/10/2011	
PAGE TITLE		Fan	
		DRAWING NUMBER	051-8870
		REVISION	3.13.0
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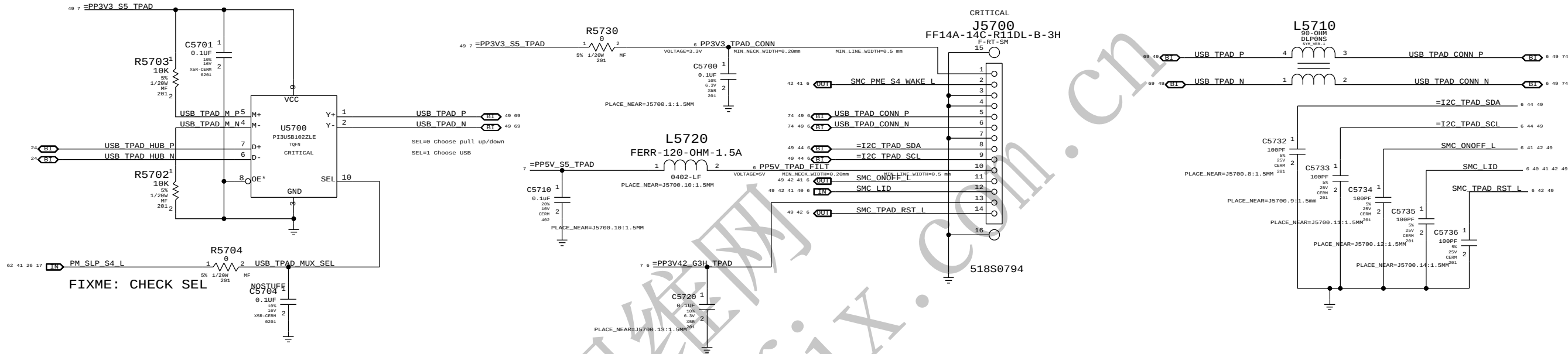
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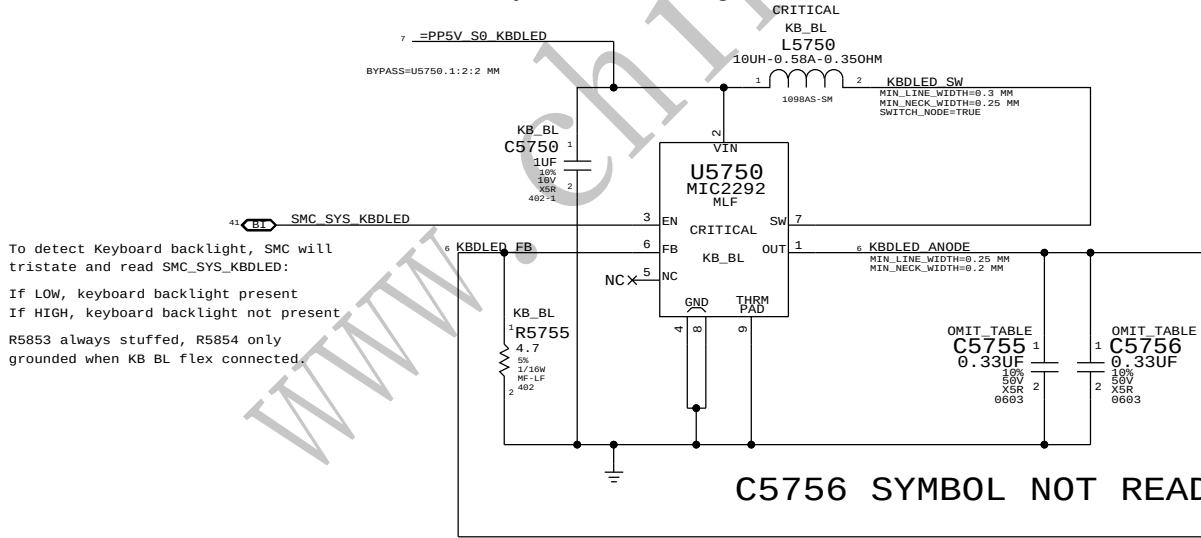
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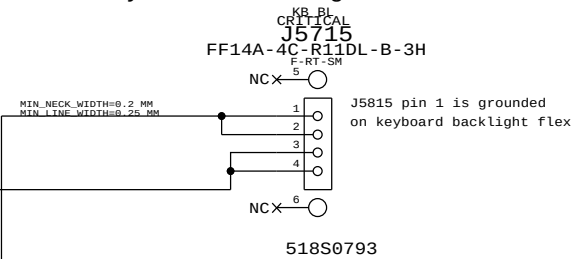
IPD Flex Connector



Keyboard Backlight Driver & Detection

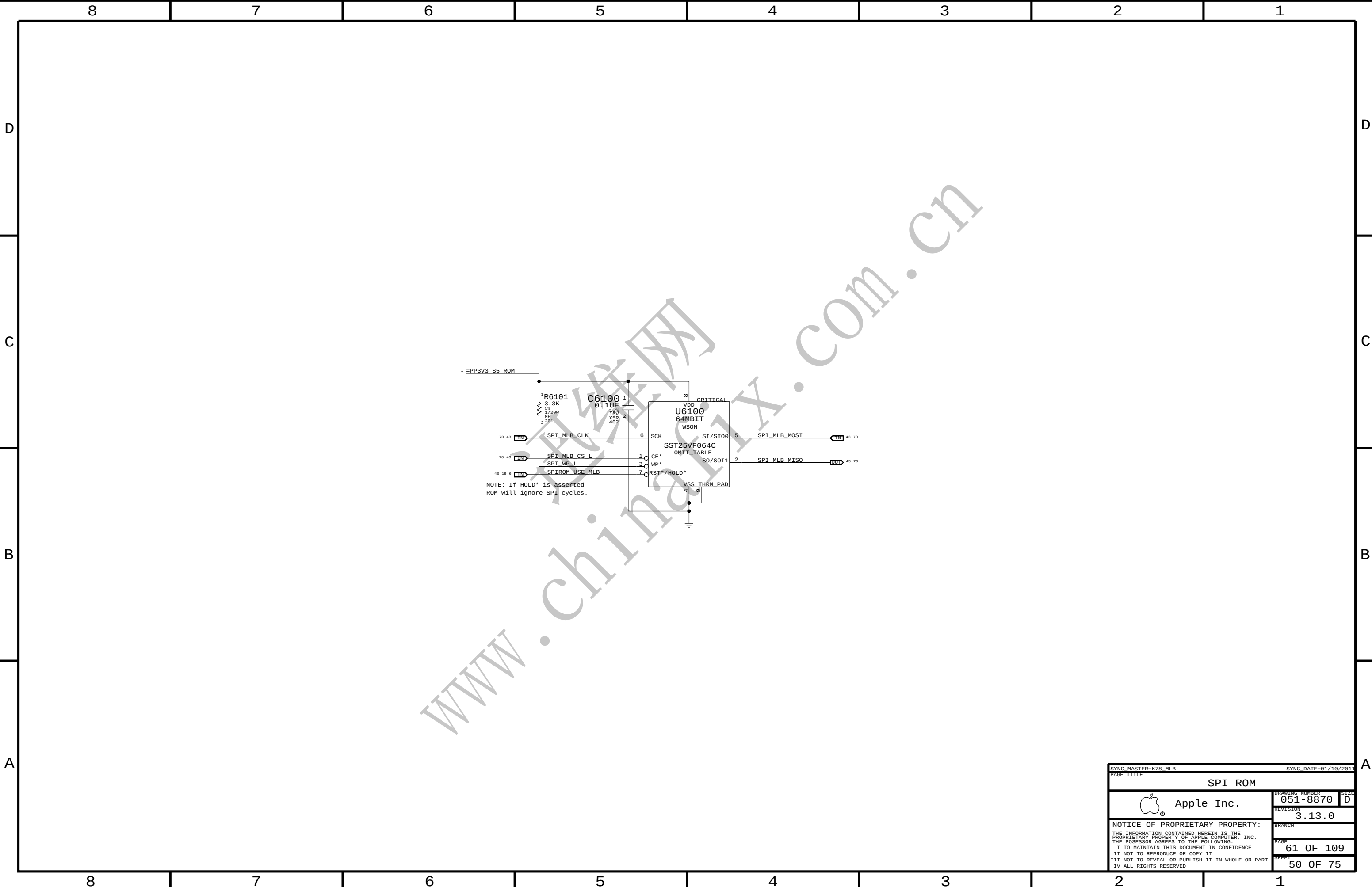


Keyboard Backlight Connector



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0704	2	CAP, CER, 0.22UF, 10%, 50V, XSR, 0603	C5755, C5756		KB_BL

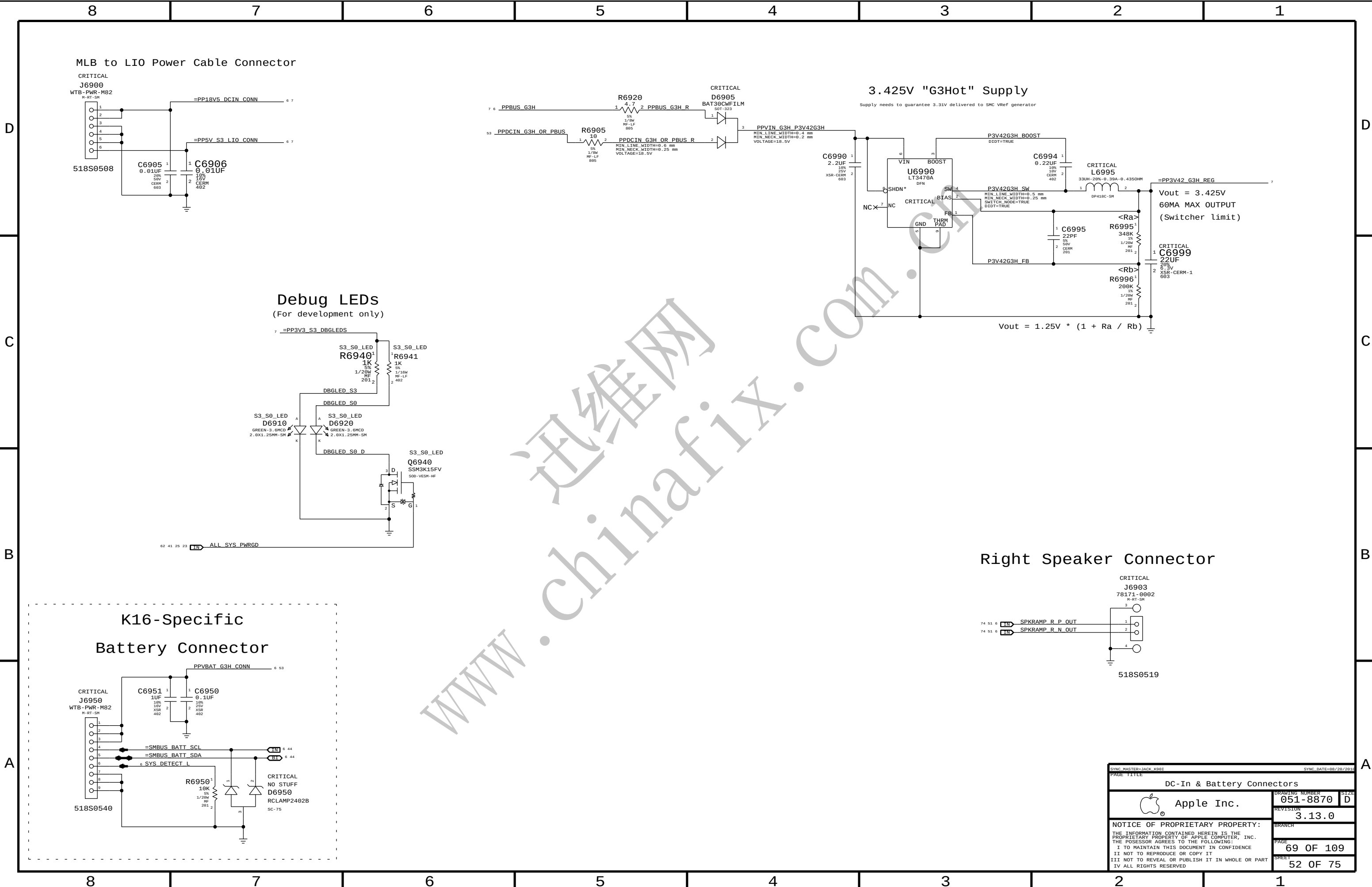
SYNC MASTER=K78 MLB		SYNC DATE=01/10/2011	
PAGE TITLE		IPD / KBD Backlight	
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		REVISION	3.13.0
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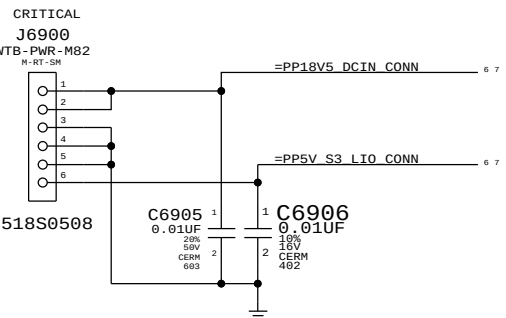
SYNC MASTER=K78_MLB		SYNC DATE=01/10/2011	
PAGE TITLE			
SPI ROM			
	Apple Inc.	DRAWING NUMBER	051-8870
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		REVISION	3.13.0
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APN: 353S2888

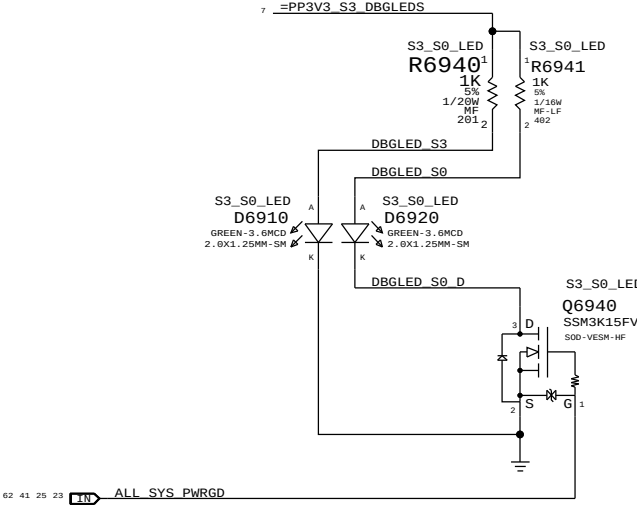
[illegible]



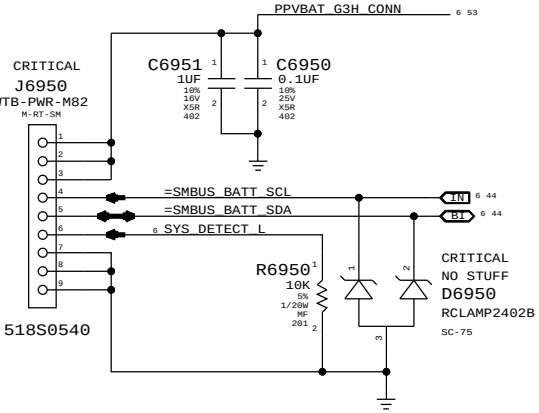
MLB to LIO Power Cable Connector



Debug LEDs
(For development only)

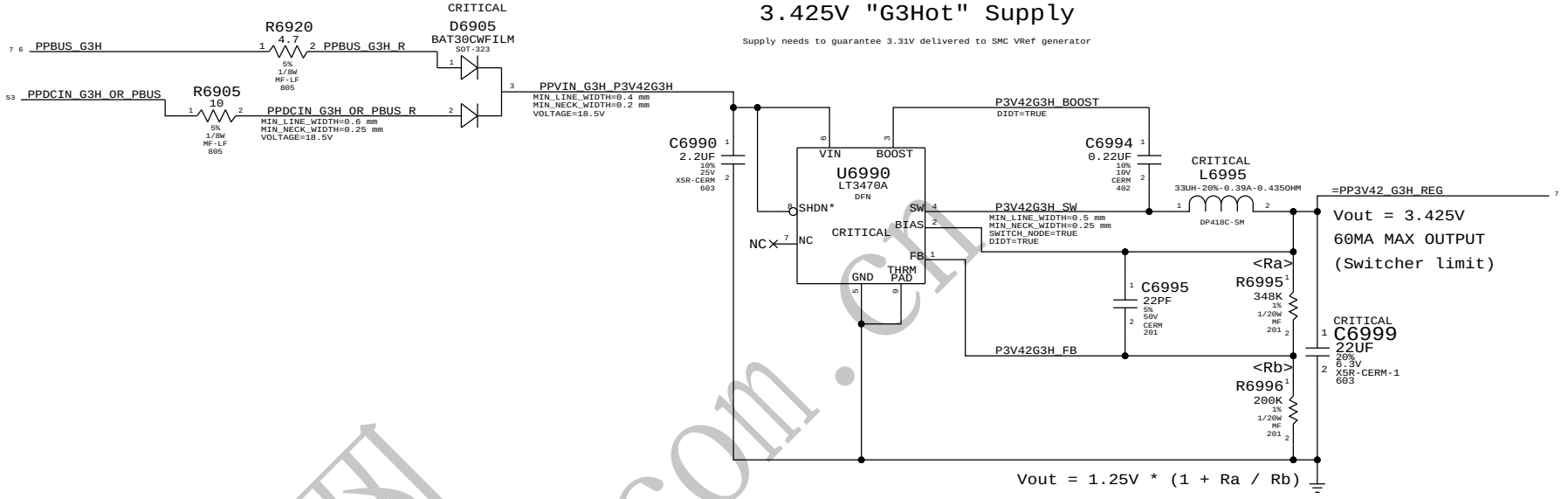


K16-Specific
Battery Connector

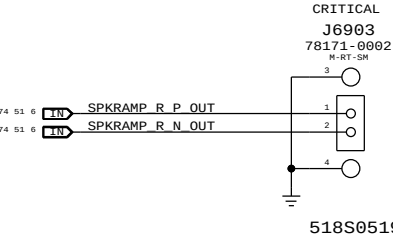


3.425V "G3Hot" Supply

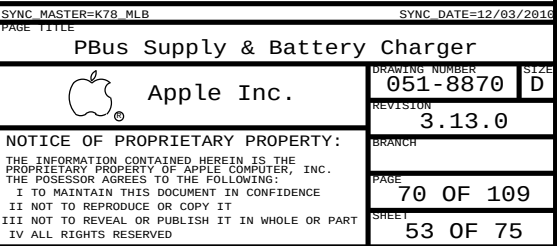
Supply needs to guarantee 3.31V delivered to SMC VRef generator

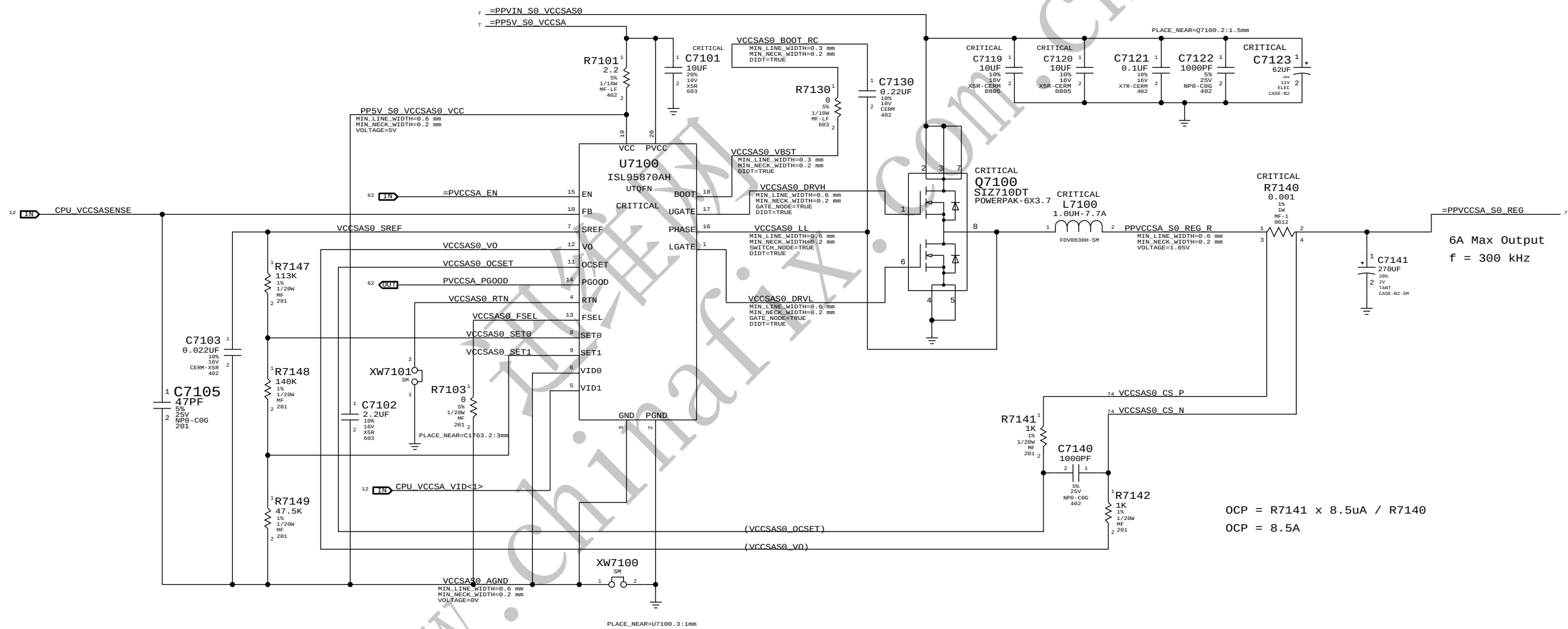


Right Speaker Connector



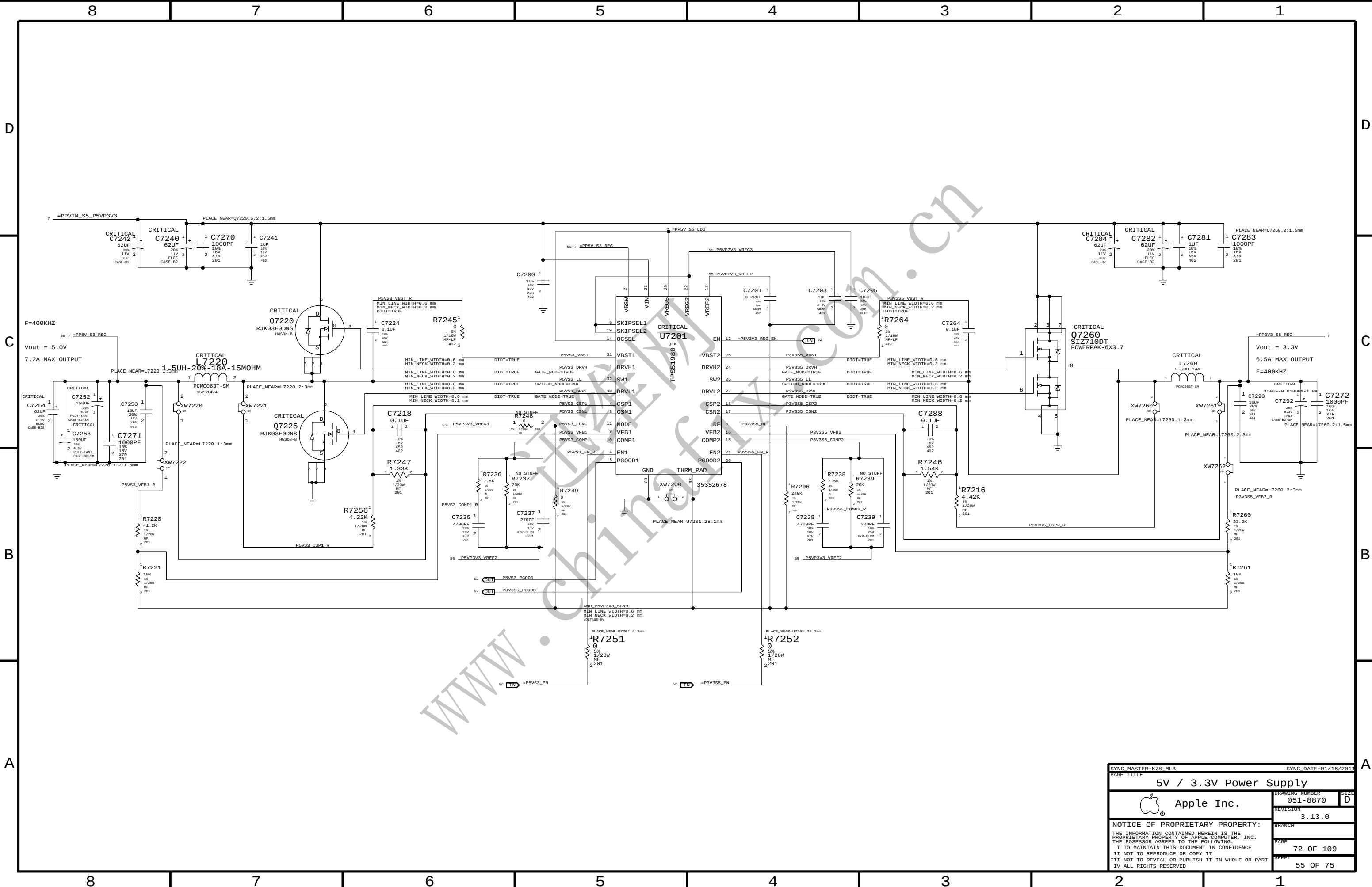
PAGE TITLE		PAGE NUMBER	
DC-In & Battery Connectors		051-8870	
Apple Inc.		3.13.0	
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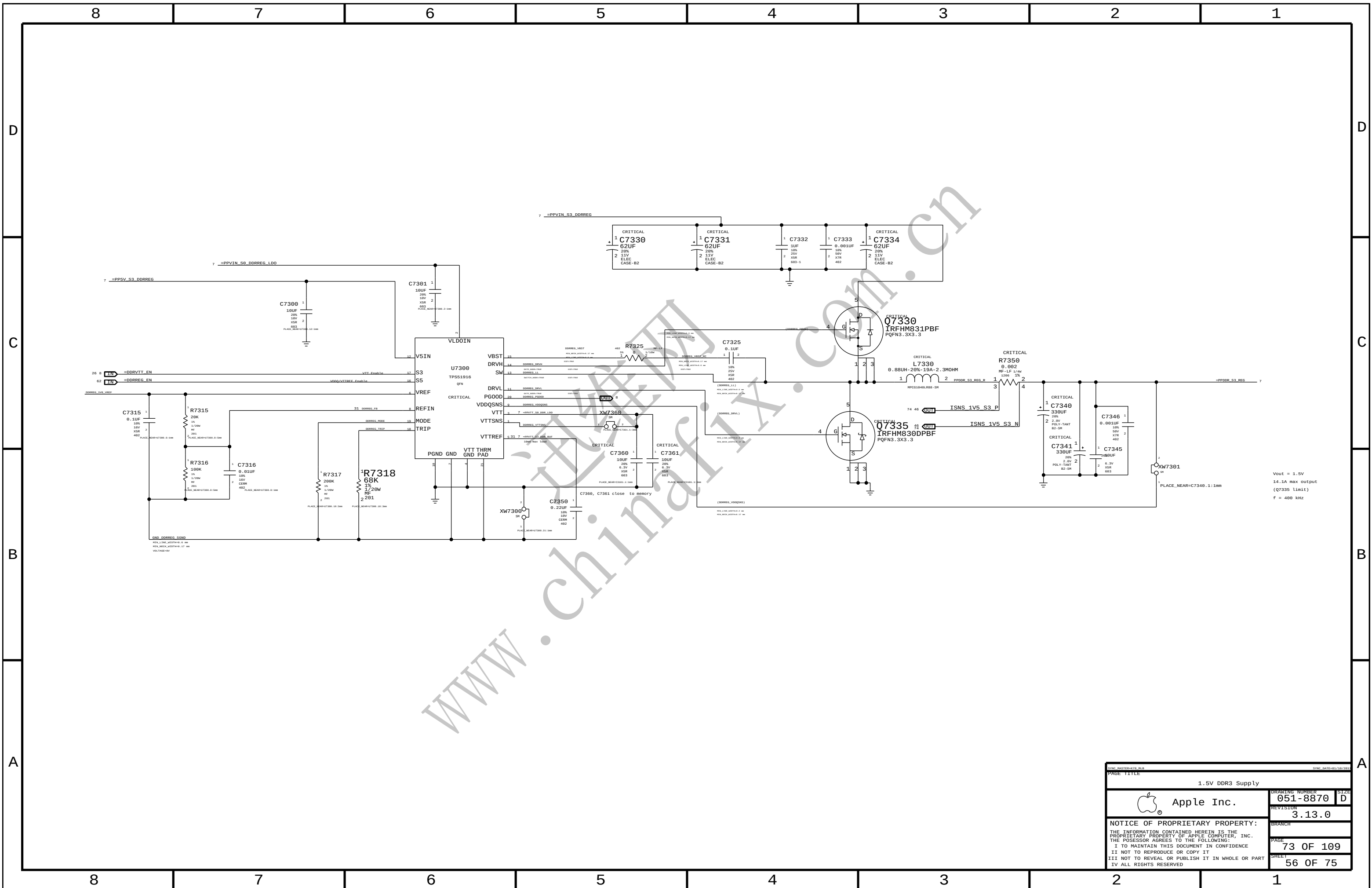




VID1	VID0	Voltage
0	0	0.9V
1	0	0.8V

$$OCP = R7141 \times 8.5\mu A / R7140$$
$$OCP = 8.5A$$

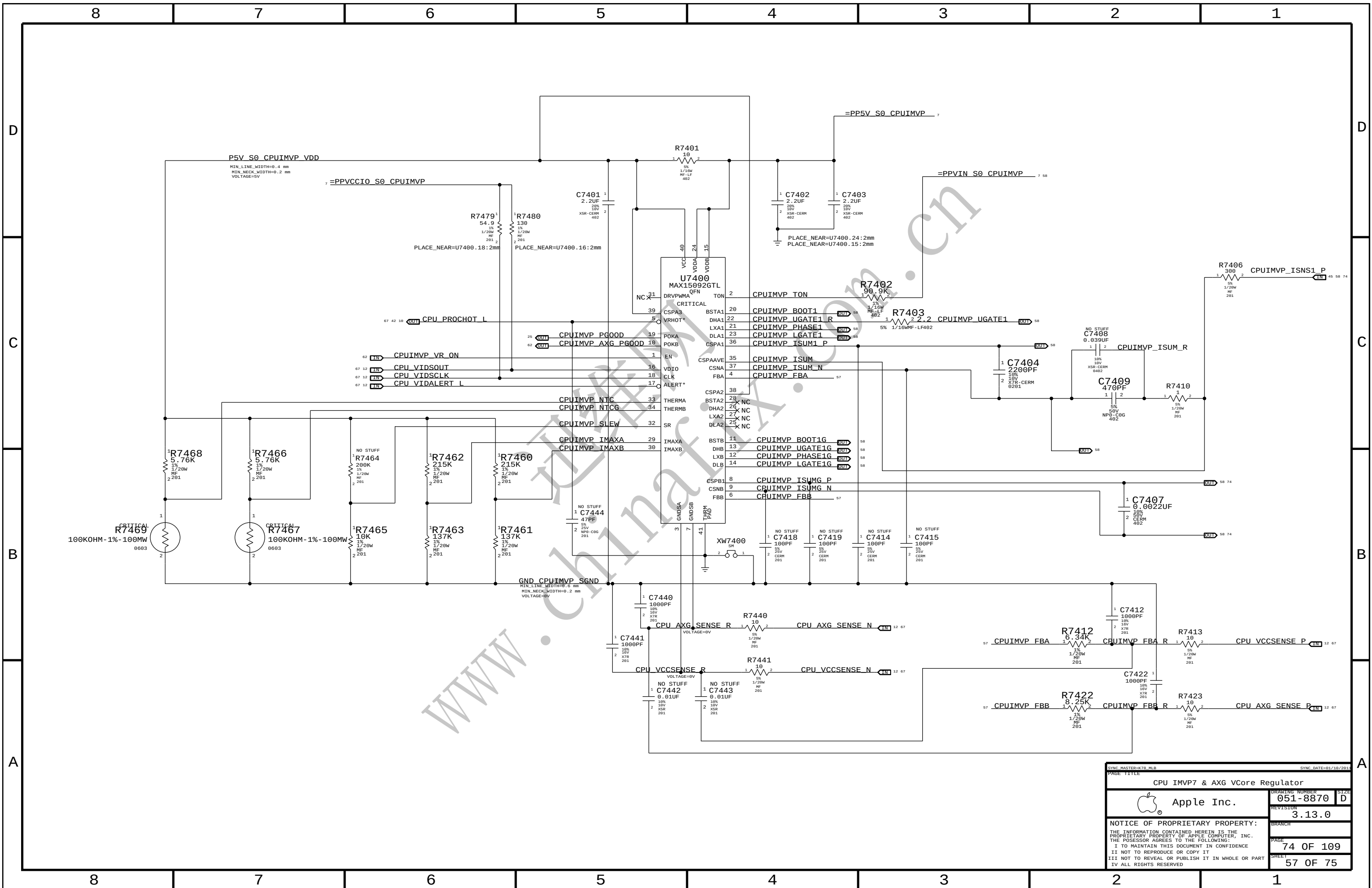




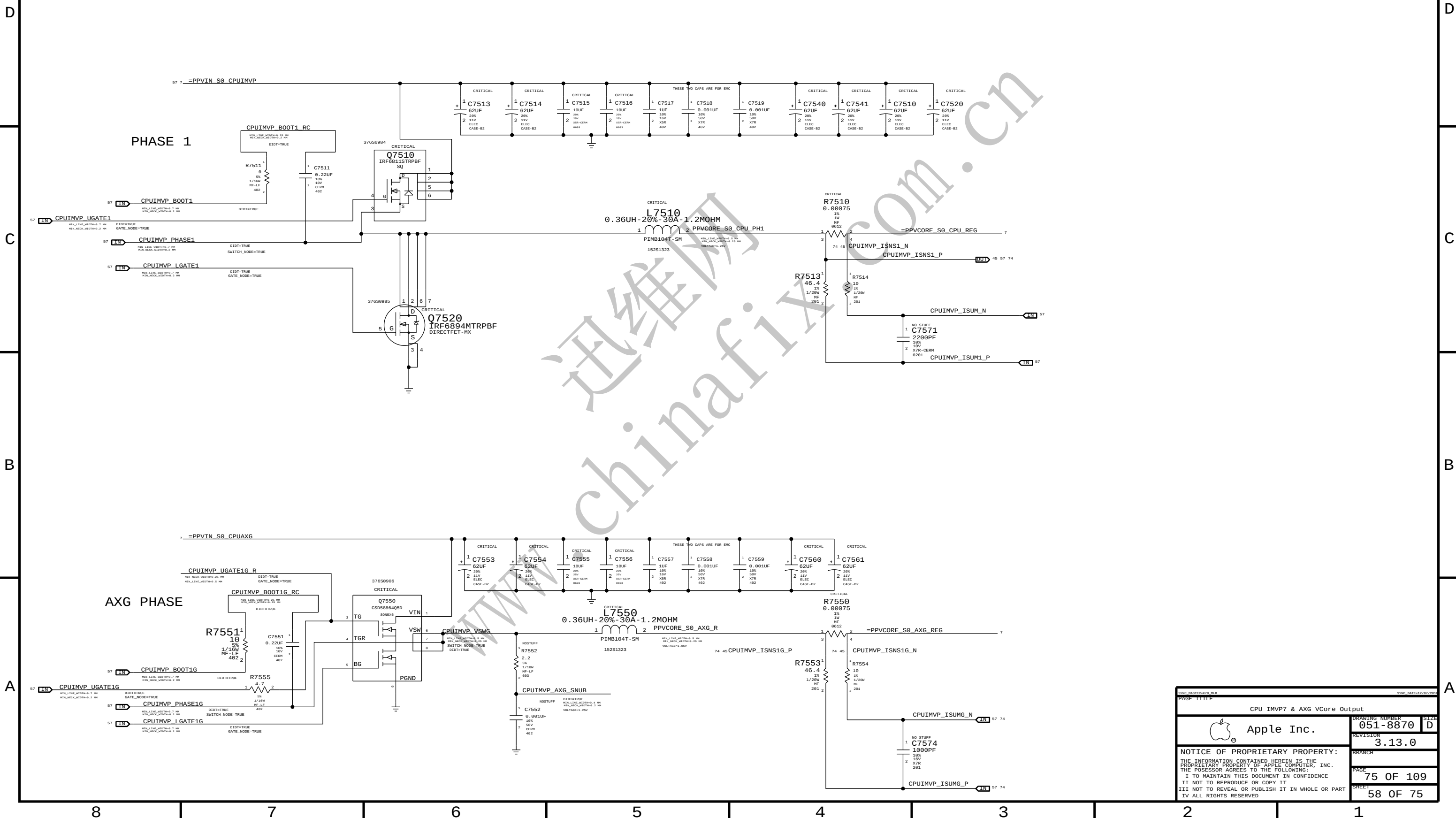
Vout = 1.5V
14.1A max output
(Q7335 limit)
f = 480 kHz

PAGE TITLE		1.5V DDR3 Supply	
DRAWING NUMBER		051-8870	SIZE D
REVISION		3.13.0	
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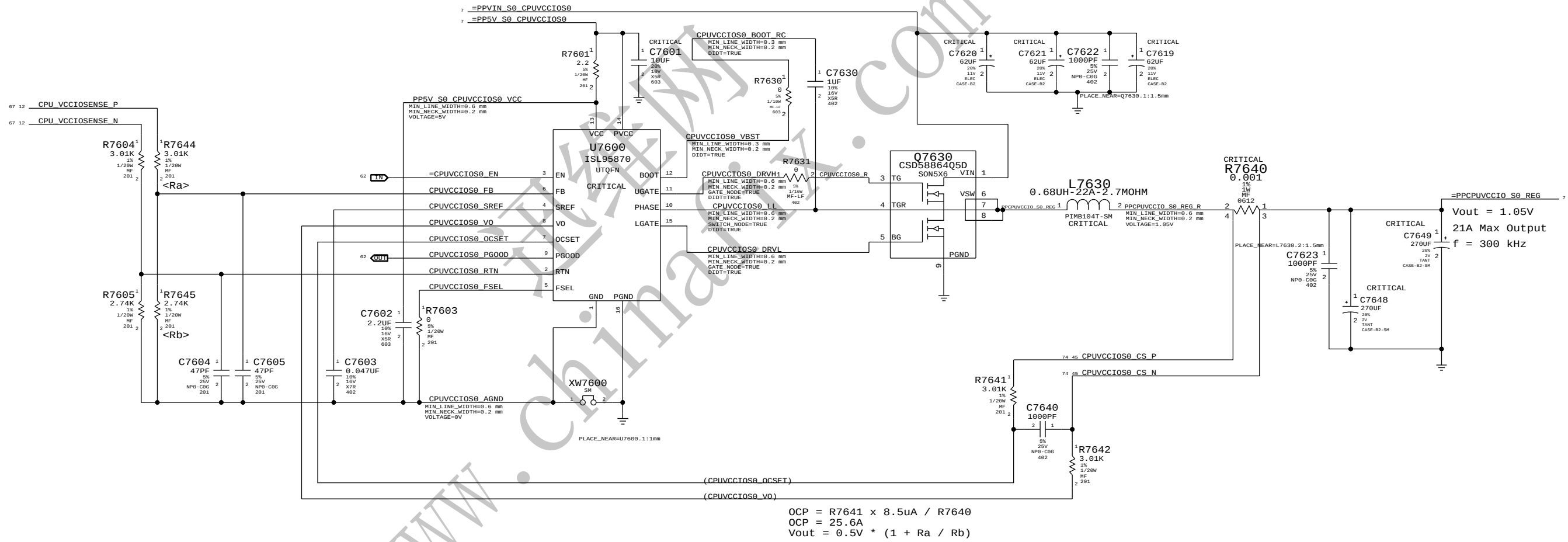
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CPU VCCIO (1.05V S0) Regulator



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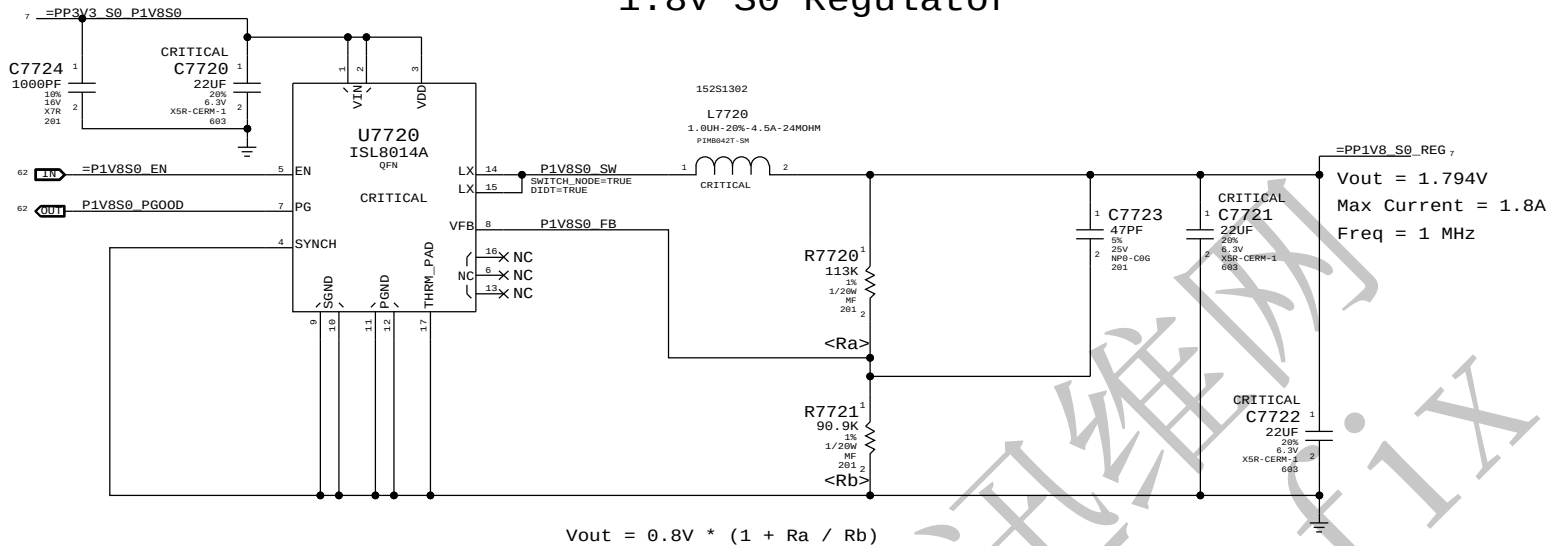
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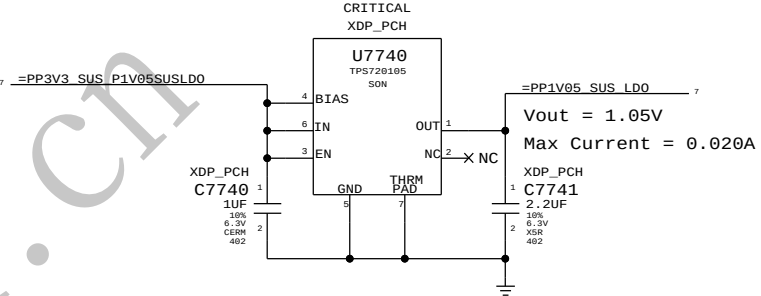
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1.8V S0 Regulator

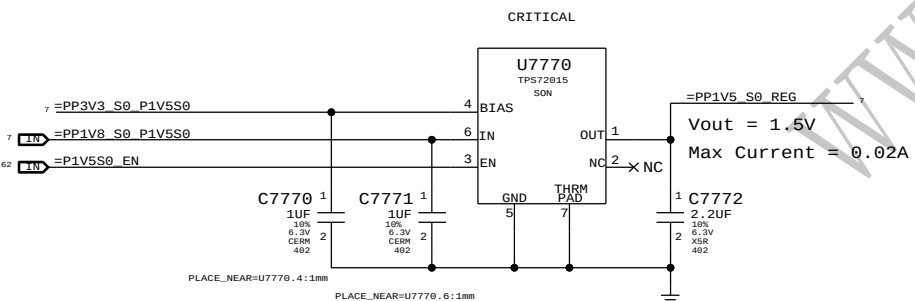


1.05V SUS LDO

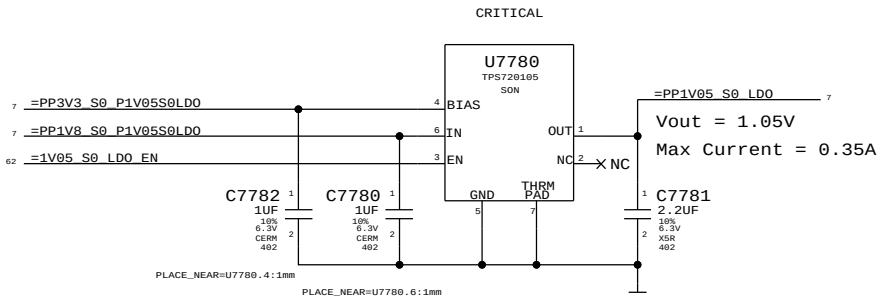
Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



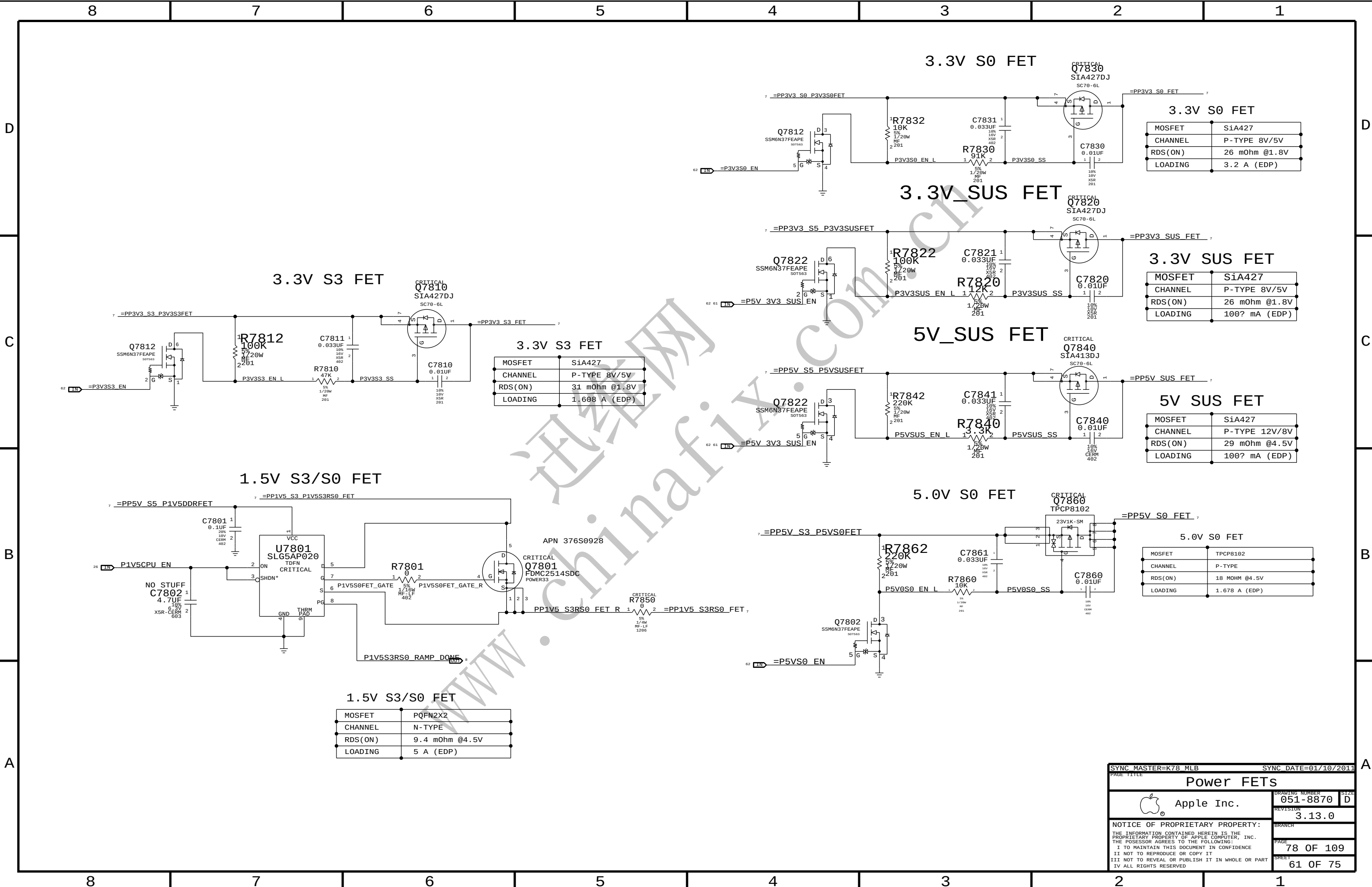
1.5V S0 LDO



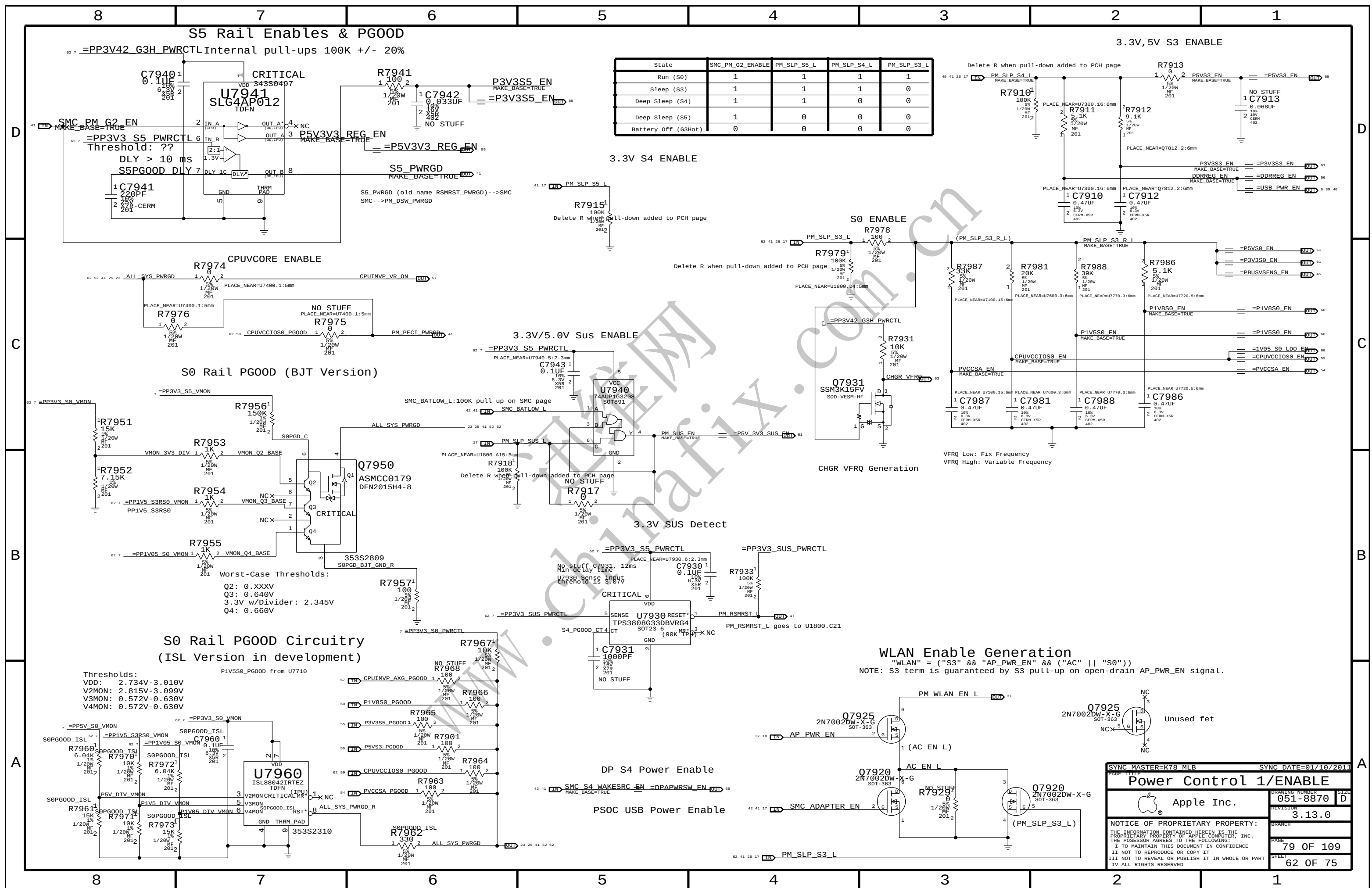
1.05V S0 LDO

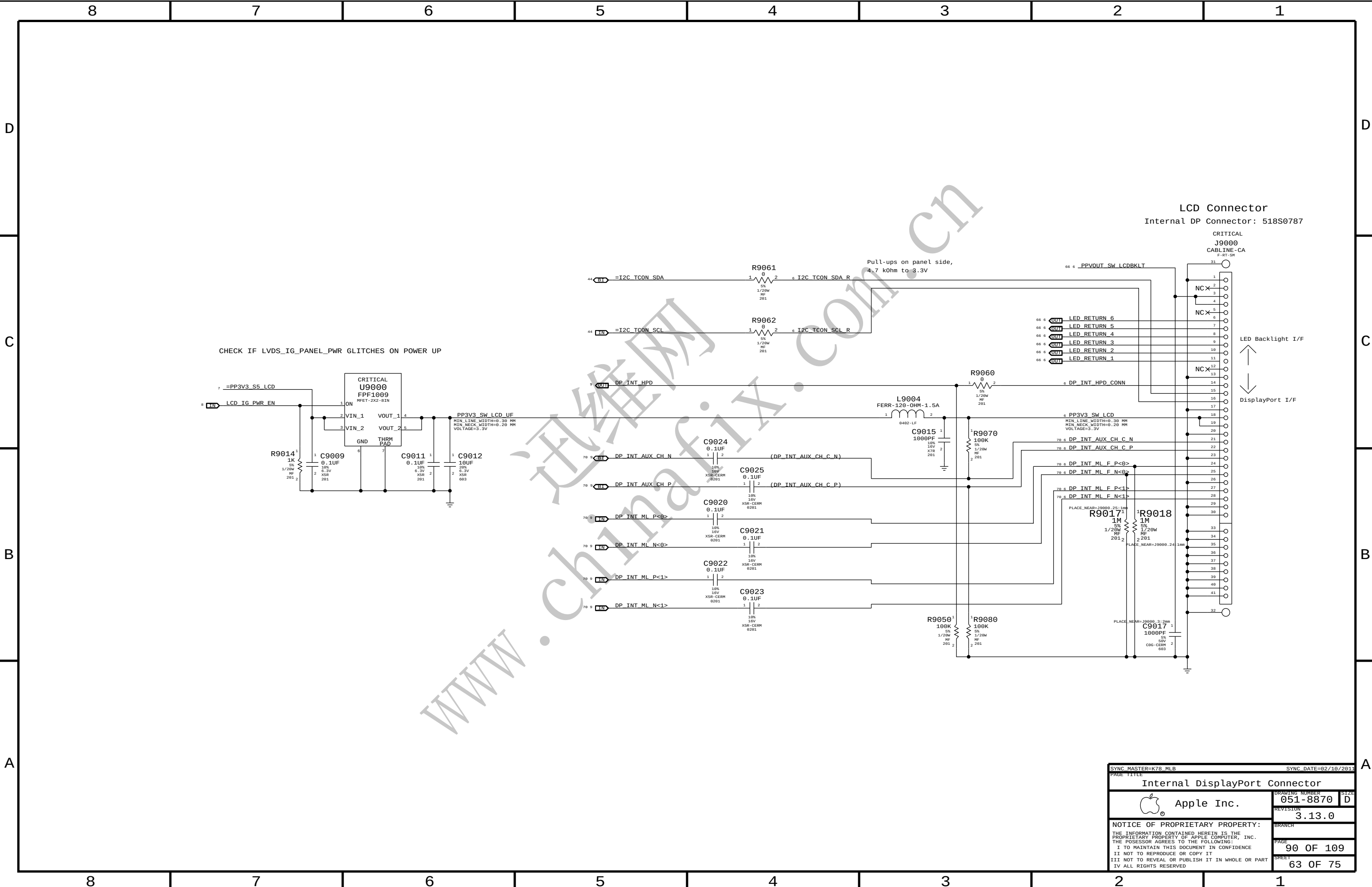


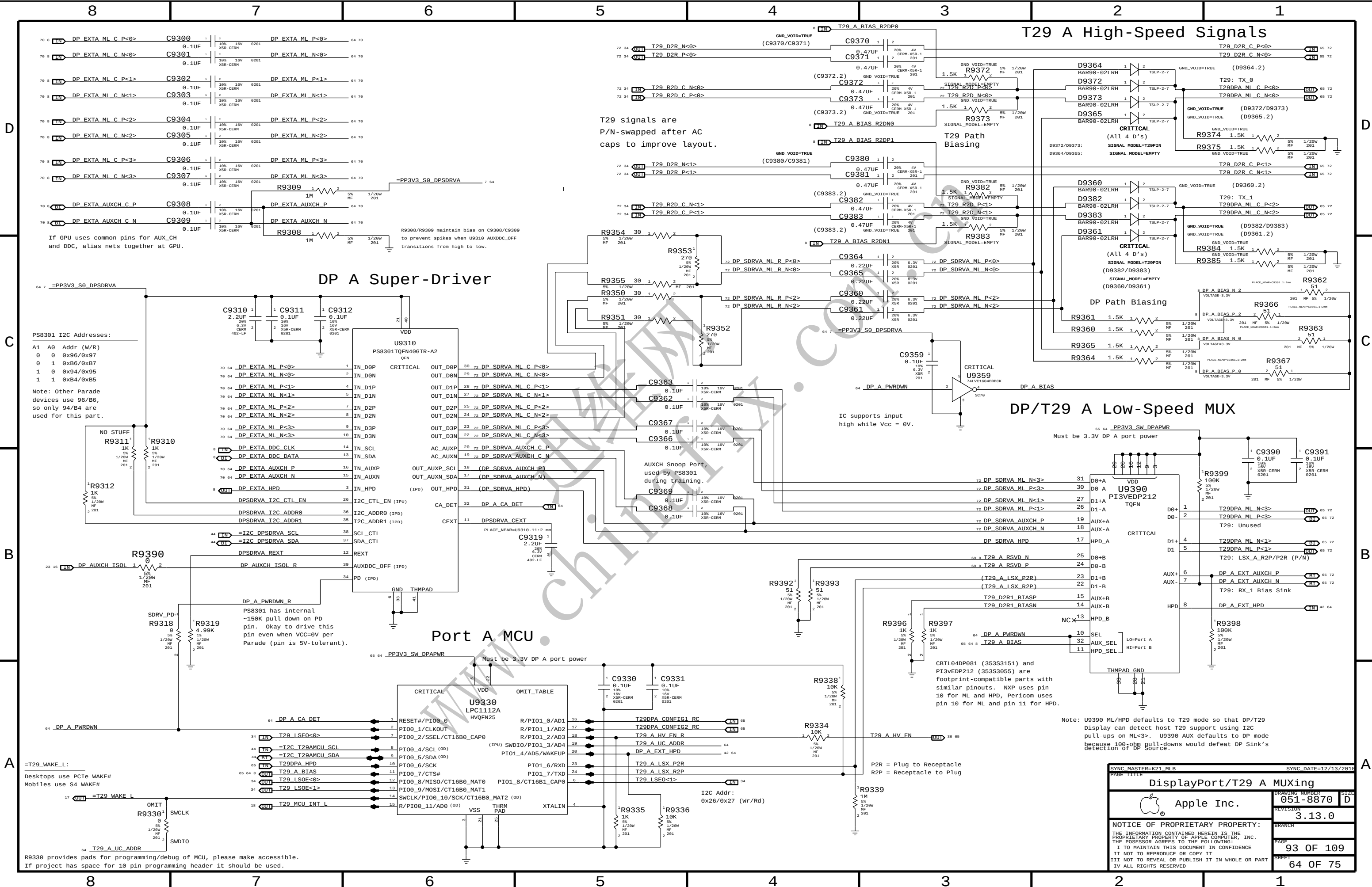
SYNC MASTER=K78 MLB		SYNC DATE=01/16/2011	
PAGE TITLE		PAGE	
Misc Power Supplies		DRAWING NUMBER	
Apple Inc.		051-8870	
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T29 signals are P/N-swapped after AC caps to improve layout.

T29 A High-Speed Signals

T29 Path Biasing

DP Path Biasing

DP/T29 A Low-Speed MUX

Port A MCU

PS8301 I2C Addresses:

- A1 A0 Addr (W/R)
 - 0 0 0x96/0x97
 - 0 1 0xB6/0xB7
 - 1 0 0x94/0x95
 - 1 1 0xB4/0xB5
- Note: Other Parade devices use 96/B6, so only 94/B4 are used for this part.

PS8301 has internal ~150K pull-down on PD pin. Okay to drive this pin even when VCC=0V per Parade (pin is 5V-tolerant).

IC supports input high while Vcc = 0V.

Note: U9390 ML/HPD defaults to T29 mode so that DP/T29 Display can detect host T29 support using I2C pull-ups on ML<3>. U9390 AUX defaults to DP mode because 100-ohm pull-downs would defeat DP Sink's detection of DP Source.

=T29_WAKE_L:
Desktops use PCIe WAKE#
Mobiles use S4 WAKE#

R9330 provides pads for programming/debug of MCU, please make accessible.
If project has space for 10-pin programming header it should be used.

SYNC MASTER=K21 MLB

SYNC DATE=12/13/2016

DisplayPort/T29 A MUXing

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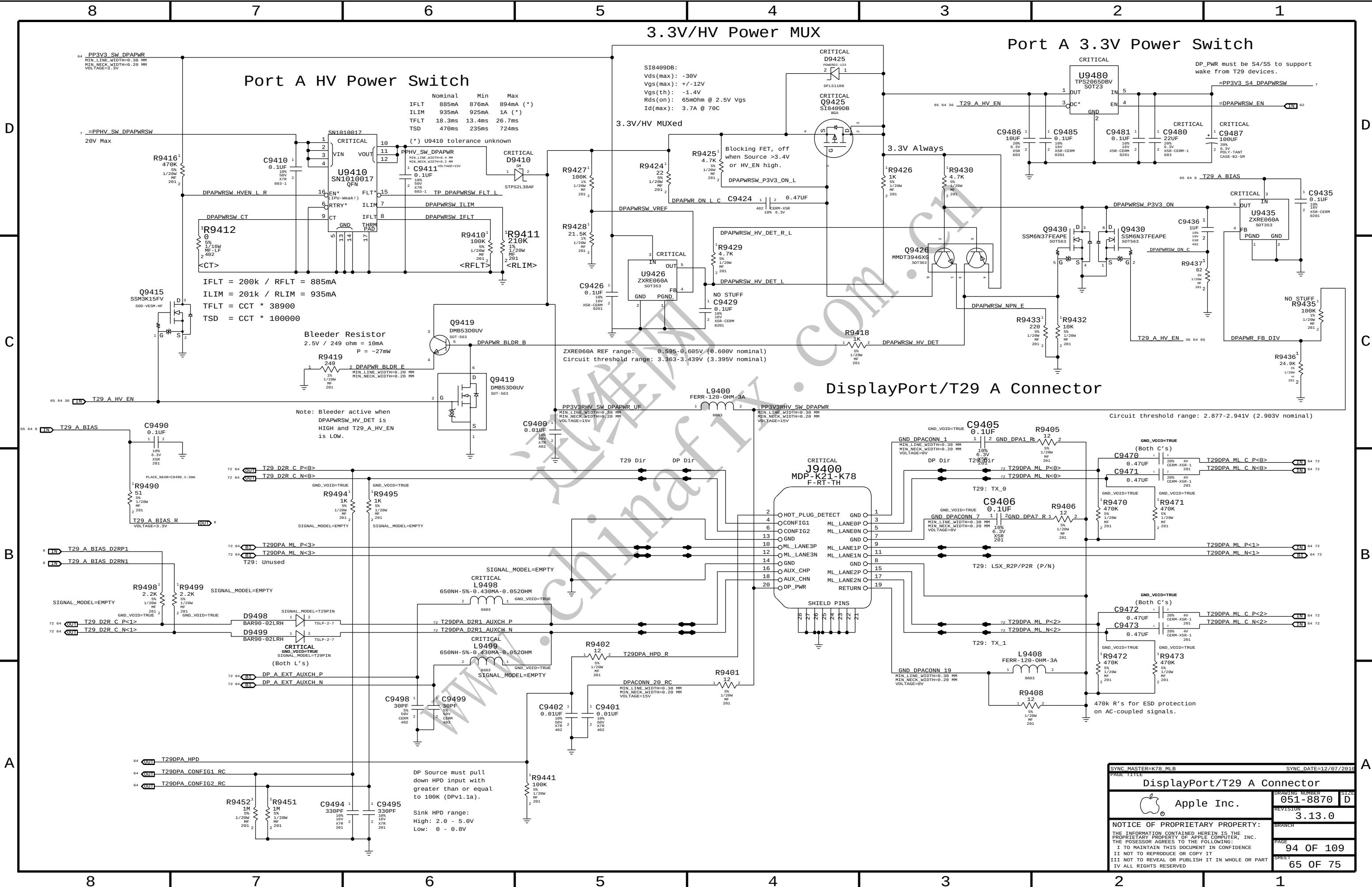
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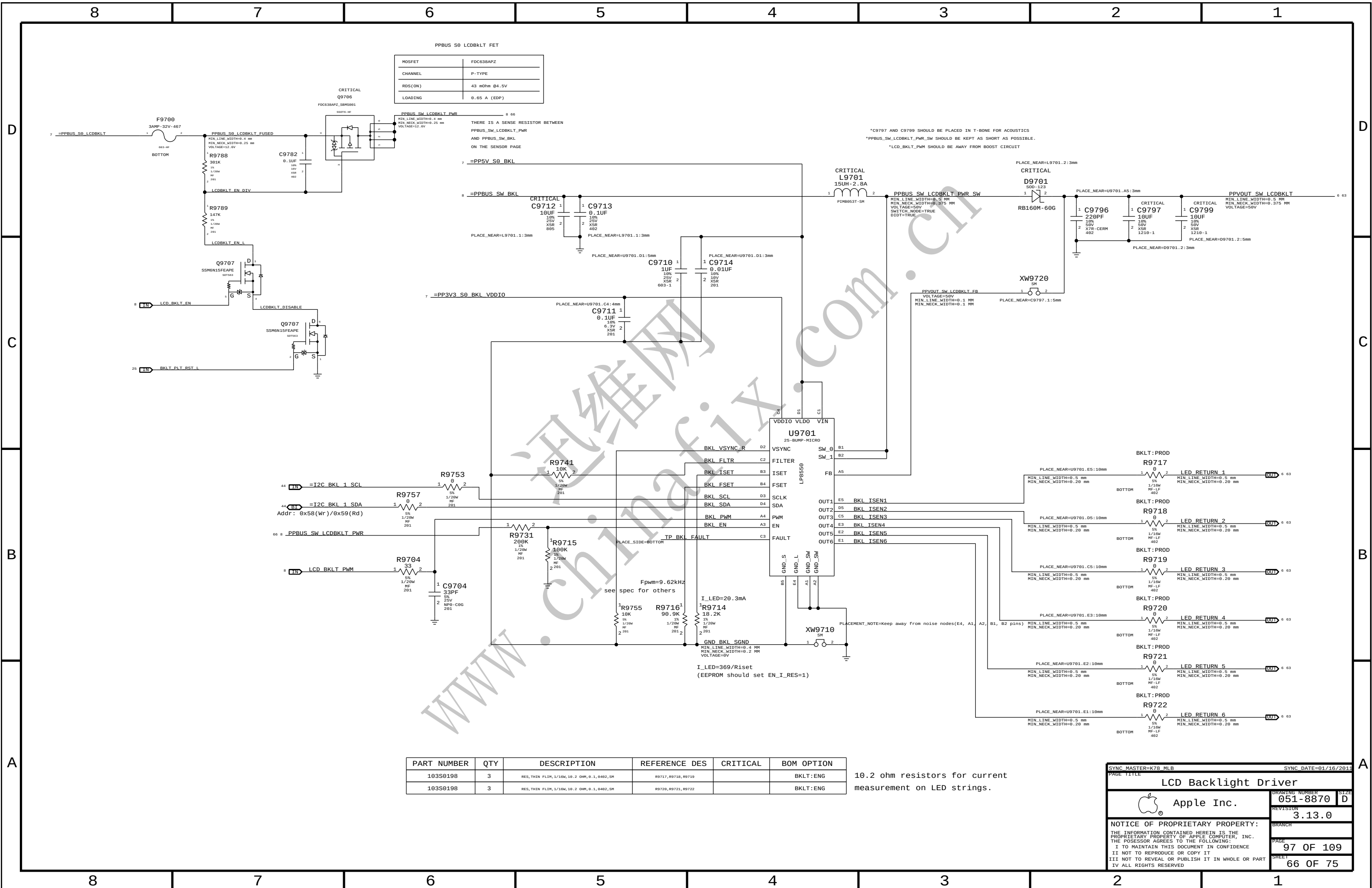
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PAGE TITLE			
DisplayPort/T29 A Connector			
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		SIZE	D
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES, THIN FILM, 1/10W, 10.2 OHM, 0.1, 0402, SM	R9717, R9718, R9719		BKLT:ENG
103S0198	3	RES, THIN FILM, 1/10W, 10.2 OHM, 0.1, 0402, SM	R9720, R9721, R9722		BKLT:ENG

10.2 ohm resistors for current measurement on LED strings.

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PAGE TITLE			
LCD Backlight Driver			
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CPU Constraints

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Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<5..0>	8 11 27 28 32
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<5..0>	8 11 27 28 32
MEM_A_CTRL	MEM_55S	MEM_CTRL	MEM A CKE<3..0>	8 11 27 28 32
MEM_A_CTRL	MEM_55S	MEM_CTRL	MEM A CS L<3..0>	8 11 27 28 32
MEM_A_CTRL	MEM_55S	MEM_CTRL	MEM A ODT<3..0>	8 11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A A<15..0>	8 11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A BA<2..0>	11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A RAS L	11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A CAS L	11 27 28 32
MEM_A_CMD	MEM_55S	MEM_CMD	MEM A WE L	11 27 28 32
MEM_A_DQ_BYTE0	MEM_50S	MEM_DATA	MEM A DQ<7..0>	11 27
MEM_A_DQ_BYTE1	MEM_50S	MEM_DATA	MEM A DQ<15..8>	11 27
MEM_A_DQ_BYTE2	MEM_50S	MEM_DATA	MEM A DQ<23..16>	11 27
MEM_A_DQ_BYTE3	MEM_50S	MEM_DATA	MEM A DQ<31..24>	11 27
MEM_A_DQ_BYTE4	MEM_50S	MEM_DATA	MEM A DQ<39..32>	11 28
MEM_A_DQ_BYTE5	MEM_50S	MEM_DATA	MEM A DQ<47..40>	11 28
MEM_A_DQ_BYTE6	MEM_50S	MEM_DATA	MEM A DQ<55..48>	11 28
MEM_A_DQ_BYTE7	MEM_50S	MEM_DATA	MEM A DQ<63..56>	11 28
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS P<0>	11 27
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS N<0>	11 27
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS P<1>	11 27
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS N<1>	11 27
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS P<2>	11 27
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS N<2>	11 27
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS P<3>	11 27
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS N<3>	11 27
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS P<4>	11 28
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS N<4>	11 28
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS P<5>	11 28
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS N<5>	11 28
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS P<6>	11 28
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS N<6>	11 28
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS P<7>	11 28
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS N<7>	11 28
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK P<5..0>	8 11 29 30 32
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK N<5..0>	8 11 29 30 32
MEM_B_CTRL	MEM_55S	MEM_CTRL	MEM B CKE<3..0>	8 11 29 30 32
MEM_B_CTRL	MEM_55S	MEM_CTRL	MEM B CS L<3..0>	8 11 29 30 32
MEM_B_CTRL	MEM_55S	MEM_CTRL	MEM B ODT<3..0>	8 11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B A<15..0>	8 11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B BA<2..0>	11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B RAS L	11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B CAS L	11 29 30 32
MEM_B_CMD	MEM_55S	MEM_CMD	MEM B WE L	11 29 30 32
MEM_B_DQ_BYTE0	MEM_50S	MEM_DATA	MEM B DQ<7..0>	11 29
MEM_B_DQ_BYTE1	MEM_50S	MEM_DATA	MEM B DQ<15..8>	11 29
MEM_B_DQ_BYTE2	MEM_50S	MEM_DATA	MEM B DQ<23..16>	11 29
MEM_B_DQ_BYTE3	MEM_50S	MEM_DATA	MEM B DQ<31..24>	11 29
MEM_B_DQ_BYTE4	MEM_50S	MEM_DATA	MEM B DQ<39..32>	11 30
MEM_B_DQ_BYTE5	MEM_50S	MEM_DATA	MEM B DQ<47..40>	11 30
MEM_B_DQ_BYTE6	MEM_50S	MEM_DATA	MEM B DQ<55..48>	11 30
MEM_B_DQ_BYTE7	MEM_50S	MEM_DATA	MEM B DQ<63..56>	11 30
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS P<0>	11 29
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS N<0>	11 29
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS P<1>	11 29
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS N<1>	11 29
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS P<2>	11 29
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS N<2>	11 29
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS P<3>	11 29
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS N<3>	11 29
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS P<4>	11 30
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS N<4>	11 30
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS P<5>	11 30
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS N<5>	11 30
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS P<6>	11 30
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS N<6>	11 30
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS P<7>	11 30
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS N<7>	11 30
		MEM_PWR	PP1V5 S3RS0	7
		MEM_PWR	PP1V5 S3	7
		MEM_PWR	PP0V75 S3 MEM VREFCA A	27 28 29 30 31
		MEM_PWR	PP0V75 S3 MEM VREFDQ A	9 27 28 29 30 31

Need to support MEM_*-style wildcards!

DDR3: Sandybridge SFF 2C when routed on Type-3 (Through hole) should follow rPGA guidelines per Huron River SFF D6 rev1.0 (#438297).

DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.

DQ to DQS matching per byte lane should be within 0.127mm.

DQS to clock matching should be within [CLK-63.5mm] and [CLK+38.1mm].

CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.9508mm.

CONTROL signals should be matched within [CLK-2.54mm] to [CLK+0.6mm] of CLK pairs.

A/BA/CMD signals should be matched within [CLK-12.7mm] to [CLK+12.7mm] of CLK pairs A/BA/CMD signals to each other should match within 5.08mm.

DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.

Maximum length of any signal from die pad to SODIMM pad is 119.83mm, from processor ball to SODIMM pad is 88.9mm.

SOURCE: Huron River Platform DG, Rev 1.01 (#436735), Section 2.5

Memory Constraints

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: Calpella Platform Design Guide for Ixex Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SM8_S05	*	=50_OHR_SE	=50_OHR_SE	=50_OHR_SE	=50_OHR_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ixex Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OH1_SE	=55_OH1_SE	=55_OH1_SE	=55_OH1_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPT_555	*	=55_OHR_SE	=55_OHR_SE	=55_OHR_SE	=55_OHR_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

DisplayPort Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_BSD	*	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?	DISPLAYPORT	TOP, BOTTOM	=4x_DIELECTRIC	?

PCI-Express Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_BSD	*	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_25M_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.

PCH Net Properties

[illegible]

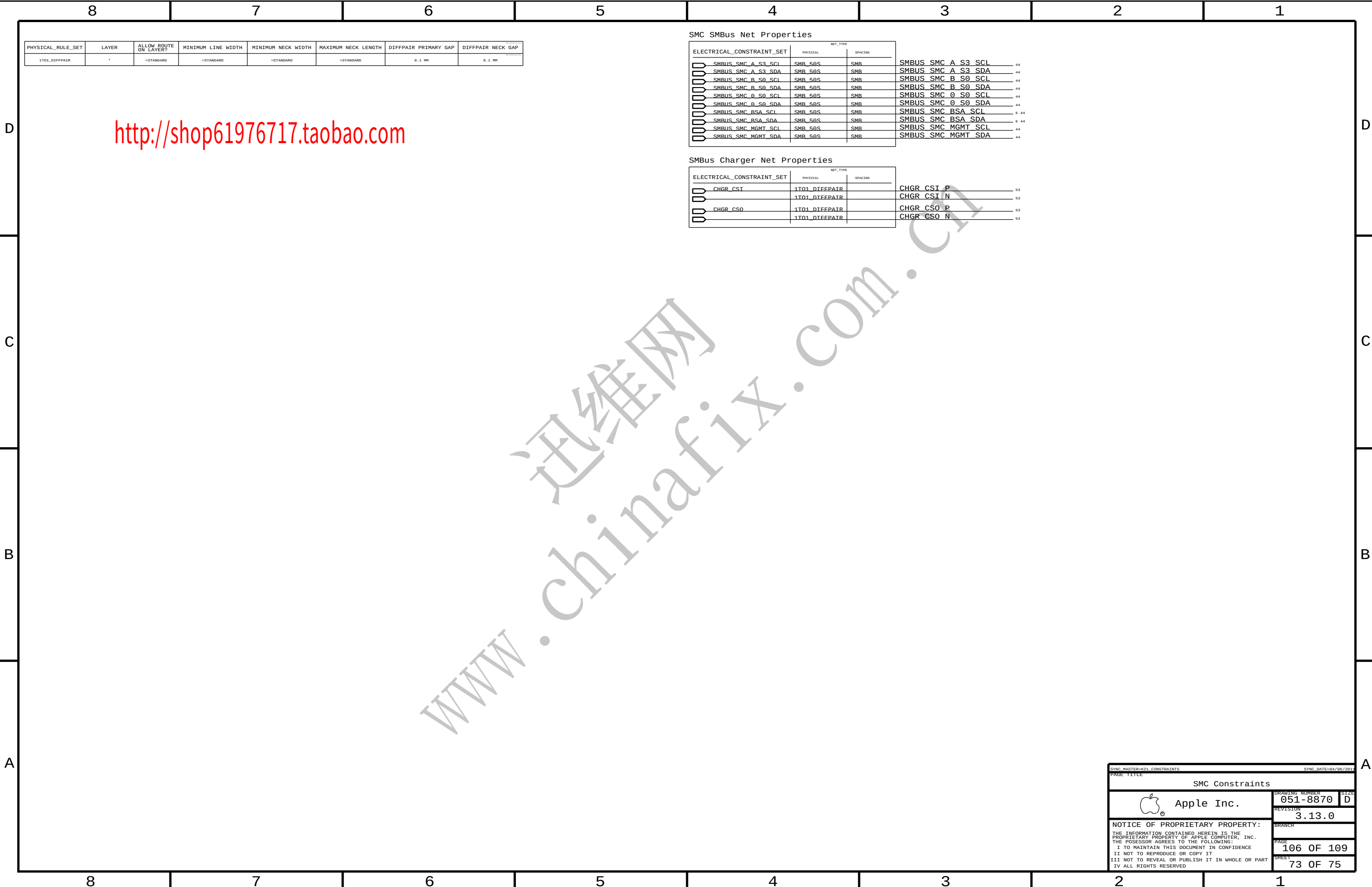
Chipset Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	DP_EXT_A ML	DP_85D	DISPLAYPORT	DP_EXT_A ML C P<3..0> 8 64
	DP_EXT_A ML	DP_85D	DISPLAYPORT	DP_EXT_A ML C N<3..0> 8 64
	DP_EXT_A ML	DP_85D	DISPLAYPORT	DP_EXT_A ML P<3..0> 64
	DP_EXT_A ML	DP_85D	DISPLAYPORT	DP_EXT_A ML N<3..0> 64
	DP_EXT_AUXCH	DP_85D	DISPLAYPORT	DP_EXT_AUXCH C P 8 64
	DP_EXT_AUXCH	DP_85D	DISPLAYPORT	DP_EXT_AUXCH C N 8 64
	DP_EXT_AUXCH	DP_85D	DISPLAYPORT	DP_EXT_AUXCH P 64
	DP_EXT_AUXCH	DP_85D	DISPLAYPORT	DP_EXT_AUXCH N 64
	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML C P<3..0> 63
	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML C N<3..0> 63
	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML P<3..0> 9 63
	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML N<3..0> 9 63
	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML F P<3..0> 6 63
	DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML F N<3..0> 6 63
	DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_CH C P 6 63
	DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_CH C N 6 63
	DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_CH P 9 63
	DP_INT_AUXCH	DP_85D	DISPLAYPORT	DP_INT_AUX_CH N 9 63
		PCI_E_85D	PCI_E	PCI_E T29 R2D C P<3..0> 8 34
		PCI_E_85D	PCI_E	PCI_E T29 R2D C N<3..0> 8 34
	PCI_E_PEG_D2R_LANE3	PCI_E_85D	PCI_E	PCI_E T29 D2R P<3> 8 34
	PCI_E_PEG_D2R_LANE2	PCI_E_85D	PCI_E	PCI_E T29 D2R P<2> 8
	PCI_E_PEG_D2R_LANE1	PCI_E_85D	PCI_E	PCI_E T29 D2R P<1> 8 34
	PCI_E_PEG_D2R_LANE0	PCI_E_85D	PCI_E	PCI_E T29 D2R P<0> 8
	PCI_E_PEG_D2R_LANE3	PCI_E_85D	PCI_E	PCI_E T29 D2R N<3> 8 34
	PCI_E_PEG_D2R_LANE2	PCI_E_85D	PCI_E	PCI_E T29 D2R N<2> 8 34
	PCI_E_PEG_D2R_LANE1	PCI_E_85D	PCI_E	PCI_E T29 D2R N<1> 8 34
	PCI_E_PEG_D2R_LANE0	PCI_E_85D	PCI_E	PCI_E T29 D2R N<0> 8 34
	PCI_E_PEG_R2D_LANE3	PCI_E_85D	PCI_E	PCI_E T29 R2D P<3> 34
	PCI_E_PEG_R2D_LANE2	PCI_E_85D	PCI_E	PCI_E T29 R2D P<2> 34
	PCI_E_PEG_R2D_LANE1	PCI_E_85D	PCI_E	PCI_E T29 R2D P<1> 34
	PCI_E_PEG_R2D_LANE0	PCI_E_85D	PCI_E	PCI_E T29 R2D P<0> 34
	PCI_E_PEG_R2D_LANE3	PCI_E_85D	PCI_E	PCI_E T29 R2D N<3> 34
	PCI_E_PEG_R2D_LANE2	PCI_E_85D	PCI_E	PCI_E T29 R2D N<2> 34
	PCI_E_PEG_R2D_LANE1	PCI_E_85D	PCI_E	PCI_E T29 R2D N<1> 34
	PCI_E_PEG_R2D_LANE0	PCI_E_85D	PCI_E	PCI_E T29 R2D N<0> 34
		PCI_E_85D	PCI_E	PCI_E T29 D2R C P<3..0> 34
		PCI_E_85D	PCI_E	PCI_E T29 D2R C N<3..0> 34
	PCI_E_CLK100M_T29	CLK_PCTE_90D	CLK_PCTE	PCI_E_CLK100M_T29 P 10 34
	PCI_E_CLK100M_T29	CLK_PCTE_90D	CLK_PCTE	PCI_E_CLK100M_T29 N 10 34

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	SYSCLK_CLK32K_RTC	CLK_SLOW_55S	CLK_SLOW	SYSCLK_CLK32K_RTC	10 25
	SYSCLK_CLK25M_SB	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_SB	10 25
		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_SB_R	10
		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_ENET	
		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_ENET_R	
	SYSCLK_CLK25M_T29	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29	25 34
		CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R	34

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PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
1T01_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

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SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
SMBUS_SMC_A_S3_SCL	SMB_50S	SMB	SMBUS_SMC_A_S3_SCL 44
SMBUS_SMC_A_S3_SDA	SMB_50S	SMB	SMBUS_SMC_A_S3_SDA 44
SMBUS_SMC_B_S0_SCL	SMB_50S	SMB	SMBUS_SMC_B_S0_SCL 44
SMBUS_SMC_B_S0_SDA	SMB_50S	SMB	SMBUS_SMC_B_S0_SDA 44
SMBUS_SMC_0_S0_SCL	SMB_50S	SMB	SMBUS_SMC_0_S0_SCL 44
SMBUS_SMC_0_S0_SDA	SMB_50S	SMB	SMBUS_SMC_0_S0_SDA 44
SMBUS_SMC_BSA_SCL	SMB_50S	SMB	SMBUS_SMC_BSA_SCL 6 44
SMBUS_SMC_BSA_SDA	SMB_50S	SMB	SMBUS_SMC_BSA_SDA 6 44
SMBUS_SMC_MGMT_SCL	SMB_50S	SMB	SMBUS_SMC_MGMT_SCL 44
SMBUS_SMC_MGMT_SDA	SMB_50S	SMB	SMBUS_SMC_MGMT_SDA 44

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
CHGR_CSI	1T01_DIEFPATR		CHGR_CSI_P 53
	1T01_DIEFPATR		CHGR_CSI_N 53
CHGR_CS0	1T01_DIEFPATR		CHGR_CS0_P 53
	1T01_DIEFPATR		CHGR_CS0_N 53

SYNC_MASTER=K21_CONSTRAINTS

SYNC_DATE=04/09/2013

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SMC Constraints

 Apple Inc.

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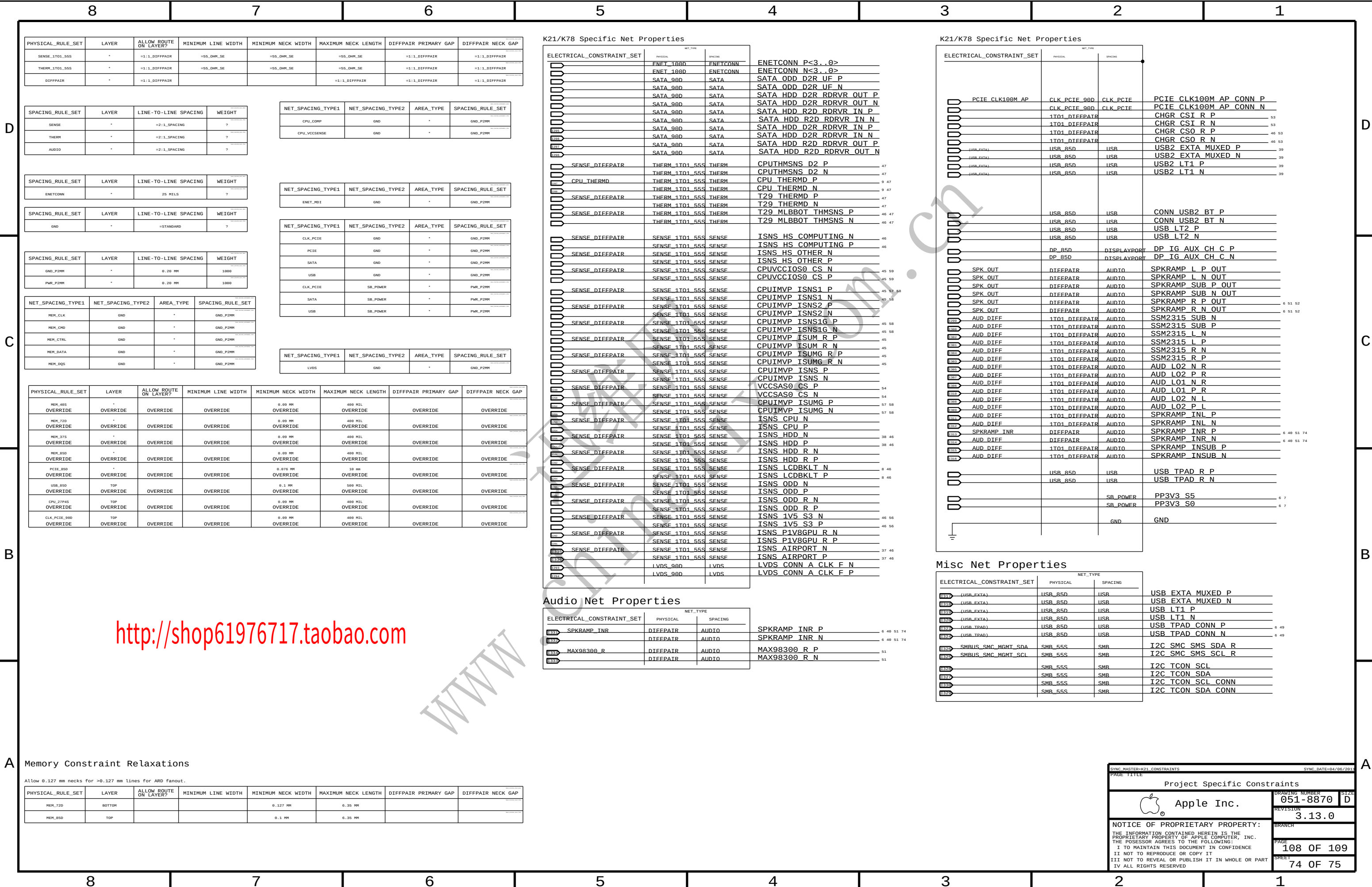
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A Memory Constraint Relaxations

Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MEM_720	BOTTOM			0.127 MM	0.35 MM		
MEM_850	TOP			0.1 MM	0.35 MM		

K21/K78 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
ENET_100D	ENET_100D	ENETCONN	ENETCONN P<3..0>
ENET_100D	ENET_100D	ENETCONN	ENETCONN N<3..0>
SATA_90D	SATA	SATA	SATA ODD D2R UF P
SATA_90D	SATA	SATA	SATA ODD D2R UF N
SATA_90D	SATA	SATA	SATA HDD D2R RDRVR OUT P
SATA_90D	SATA	SATA	SATA HDD D2R RDRVR OUT N
SATA_90D	SATA	SATA	SATA HDD R2D RDRVR IN P
SATA_90D	SATA	SATA	SATA HDD R2D RDRVR IN N
SATA_90D	SATA	SATA	SATA HDD D2R RDRVR IN P
SATA_90D	SATA	SATA	SATA HDD D2R RDRVR IN N
SATA_90D	SATA	SATA	SATA HDD R2D RDRVR OUT P
SATA_90D	SATA	SATA	SATA HDD R2D RDRVR OUT N
SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	CPUTHMSNS D2 P
SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	CPUTHMSNS D2 N
CPU_THERMD	THERM_1T01_55S	THERM	CPU_THERMD P
CPU_THERMD	THERM_1T01_55S	THERM	CPU_THERMD N
SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	T29_THERMD P
SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	T29_THERMD N
SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	T29_MLBBOT_THMSNS P
SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	T29_MLBBOT_THMSNS N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_HS_COMPUTING N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_HS_COMPUTING P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_HS_OTHER N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_HS_OTHER P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUVCCIOS0_CS N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUVCCIOS0_CS P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISNS1 P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISNS1 N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISNS2 P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISNS2 N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISNS1G P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISNS1G N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISUM_R P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISUM_R N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISUMG_R P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISUMG_R N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISNS P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISNS N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	VCCSAS0_CS P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	VCCSAS0_CS N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISUMG_P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUIMVP_ISUMG_N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_CPU_P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_CPU_N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_HDD_P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_HDD_R_N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_HDD_R_P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_LCDBKLT_N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_LCDBKLT_P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_ODD_N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_ODD_P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_ODD_R_N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_ODD_R_P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_1V5_S3_N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_1V5_S3_P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_P1V8GPU_R_N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_P1V8GPU_R_P
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_AIRPORT_N
SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_AIRPORT_P
LVDS_90D	LVDS	LVDS	LVDS_CONN_A_CLK_F_N
LVDS_90D	LVDS	LVDS	LVDS_CONN_A_CLK_F_P

Audio Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
SPKRAMP_INR	DIEFFPAIR	AUDIO	SPKRAMP_INR P
SPKRAMP_INR	DIEFFPAIR	AUDIO	SPKRAMP_INR N
MAX98300_R	DIEFFPAIR	AUDIO	MAX98300_R_P
MAX98300_R	DIEFFPAIR	AUDIO	MAX98300_R_N

K21/K78 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
PCIE_CLK100M_AP	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_CONN_P
PCIE_CLK100M_AP	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_CONN_N
1T01_DIEFFPAIR	1T01_DIEFFPAIR	CHGR_CSI_R_P	CHGR_CSI_R_P
1T01_DIEFFPAIR	1T01_DIEFFPAIR	CHGR_CSI_R_N	CHGR_CSI_R_N
1T01_DIEFFPAIR	1T01_DIEFFPAIR	CHGR_CS0_R_P	CHGR_CS0_R_P
1T01_DIEFFPAIR	1T01_DIEFFPAIR	CHGR_CS0_R_N	CHGR_CS0_R_N
USB_EXTN	USB_85D	USB	USB2_EXTN_MUXED_P
USB_EXTN	USB_85D	USB	USB2_EXTN_MUXED_N
USB_EXTN	USB_85D	USB	USB2_LT1_P
USB_EXTN	USB_85D	USB	USB2_LT1_N
USB_85D	USB	CONNN_USB2_BT_P	CONNN_USB2_BT_P
USB_85D	USB	CONNN_USB2_BT_N	CONNN_USB2_BT_N
USB_85D	USB	USB_LT2_P	USB_LT2_P
USB_85D	USB	USB_LT2_N	USB_LT2_N
DP_85D	DISPLAYPORT	DP_IG_AUX_CH_C_P	DP_IG_AUX_CH_C_P
DP_85D	DISPLAYPORT	DP_IG_AUX_CH_C_N	DP_IG_AUX_CH_C_N
SPK_OUT	DIEFFPAIR	AUDIO	SPKRAMP_L_P_OUT
SPK_OUT	DIEFFPAIR	AUDIO	SPKRAMP_L_N_OUT
SPK_OUT	DIEFFPAIR	AUDIO	SPKRAMP_SUB_P_OUT
SPK_OUT	DIEFFPAIR	AUDIO	SPKRAMP_SUB_N_OUT
SPK_OUT	DIEFFPAIR	AUDIO	SPKRAMP_R_P_OUT
SPK_OUT	DIEFFPAIR	AUDIO	SPKRAMP_R_N_OUT
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SSM2315_SUB_N
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SSM2315_SUB_P
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SSM2315_L_N
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SSM2315_L_P
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SSM2315_R_N
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SSM2315_R_P
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	AUD_L02_N_R
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	AUD_L02_P_R
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	AUD_L01_N_R
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	AUD_L01_P_R
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	AUD_L02_N_L
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	AUD_L02_P_L
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SPKRAMP_INL_P
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SPKRAMP_INL_N
SPKRAMP_INR	DIEFFPAIR	AUDIO	SPKRAMP_INR_P
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SPKRAMP_INR_N
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SPKRAMP_INSUB_P
AUD_DIFF	1T01_DIEFFPAIR	AUDIO	SPKRAMP_INSUB_N
USB_85D	USB	USB_TPAD_R_P	USB_TPAD_R_P
USB_85D	USB	USB_TPAD_R_N	USB_TPAD_R_N
SB_POWER	SB_POWER	PP3V3_S5	PP3V3_S5
SB_POWER	SB_POWER	PP3V3_S0	PP3V3_S0
GND	GND	GND	GND

Misc Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
USB_EXTN	USB_85D	USB	USB_EXTN_MUXED_P
USB_EXTN	USB_85D	USB	USB_EXTN_MUXED_N
USB_EXTN	USB_85D	USB	USB_LT1_P
USB_EXTN	USB_85D	USB	USB_LT1_N
USB_TPAD	USB_85D	USB	USB_TPAD_CONN_P
USB_TPAD	USB_85D	USB	USB_TPAD_CONN_N
SMBUS_SMC_MGMT_SDA	SMB_55S	SMB	I2C_SMC_SMS_SDA_R
SMBUS_SMC_MGMT_SCL	SMB_55S	SMB	I2C_SMC_SMS_SCL_R
SMB_55S	SMB	I2C_TCON_SCL	I2C_TCON_SCL
SMB_55S	SMB	I2C_TCON_SDA	I2C_TCON_SDA
SMB_55S	SMB	I2C_TCON_SCL_CONN	I2C_TCON_SCL_CONN
SMB_55S	SMB	I2C_TCON_SDA_CONN	I2C_TCON_SDA_CONN

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